

ANNUAL ENERGY AUDIT REPORT



Designated Consumer



TATA POWER-DDL

**TATA POWER DELHI DISTRIBUTION
LIMITED**

NDPL House, Hudson Lines,
Kingsway Camp, Delhi-110009

FY 2024 -25

Conducted by



A-Z Energy Engineers Private Limited

PLOT NO. 12, 4860-62, HARBANS SINGH STREET, KOTHI NO. -24,
WARD NO.-II, DARYA GANJ, NEW DELHI-110002
Tel: 0129-4046120, Mob. 9811402040

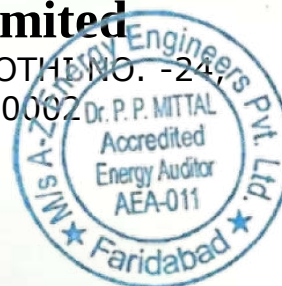


Table of Contents

List of Abbreviations	5
Acknowledgement	7
1. Executive Summary	8
1.1. Goals and Objectives	8
1.2. About Energy Audit firm	9
1.3. AT&C losses for FY 2024-25	9
2. Background	10
2.1. Regulations and role of BEE	10
2.2. Purpose of audit and accounting Report	12
2.3. Period of Energy Auditing and accounting	12
3. DISCOM Introduction and Overview	14
3.1. Name and address of DISCOM	14
3.2. Name and contact details of energy manager (BEE Certified, if any) and Authorized signatory of DISCOM (Nodal Officer)	14
3.3. Summary profile of DISCOM	14
3.4. Electrical infrastructure and assets voltage wise	19
3.5. Energy Conservation measures	24
4. Energy flow analysis	28
4.1. Energy flow across 5 Service Levels	28
4.2. Validation of metered data	29
4.3. Validation of energy flow data and losses	36
5. Loss and subsidy computation	37
5.1. Energy account analysis for previous year	37
5.2. Energy accounts analysis and performance in current year (based on quarterly data)	39
5.3. Subsidy computation and analysis (based on quarterly data)	54
5.4. Trend analysis and identification of key exceptions	56
6. Energy Audit findings	58
6.1. Review of capacity of DISCOM's energy accounting and audit wing	58
6.2. Critical Analysis	59
6.3. Revised findings based on data validation and field verification	66
6.4. Inclusions and Exclusions	66
6.5. Summary of critical analysis and way forward proposed by Energy Auditor	67
6.6. Summary of key findings – energy balance and losses	67
6.7. Recommendations and best practices	68
6.8. Action plan for line loss reduction	71
6.9. Action plan for monitoring and reporting	72
6.10. Action plan for automated energy accounting	72
Annexures	74
Annexure I - Introduction of Verification Firm	74
Annexure II - Minutes of Meeting with the DISCOM team	76
Annexure III - Check List prepared by auditing Firm	77
Annexure IV - Brief Approach, Scope & Methodology for audit	78
Annexure V - Infrastructure Details	79
Annexure VI - Electrical Distribution System	80
Annexure VII - Power Purchase Details	81
Annexure VIII - Single Line Diagram (SLD)	84
Annexure IX - Category of service details (With Consumer and voltage-wise)	85
Annexure X - Field Verification data and reports	86
Annexure XI - List of documents verified with each parameter	89
Annexure XII - Brief Description of Unit	90

Annexure XIII - List of Parameters arrived through calculation or formulae with list of documents as source of data	93
Annexure XIV - Detailed Formats	94

List of Tables

Table 1: Energy Balance & Losses for FY 2024-25	9
Table 2: Period of Energy Auditing and Accounting	13
Table 3: Name and Address of DISCOM	14
Table 4: Details of Energy Manager and Authorized signatory of DISCOM	14
Table 5: Details of Energy Manager and Authorized signatory of DISCOM	15
Table 6: Administrative hierarchy structure in Tata Power-DDL	16
Table 7: Administrative hierarchy in Tata Power-DDL up to section level	16
Table 8: Customer Profile for FY 2024-25	17
Table 9: Network Infrastructure details	19
Table 10: Input Energy & Infrastructure details	19
Table 11: Voltage wise Meter Consumers	20
Table 12: Numbers of Distribution Transformers and Metering	20
Table 13: Voltage wise numbers of Feeders and Metering	21
Table 14: Length of Distribution Lines	21
Table 15: Energy Flow details	21
Table 16: Consumer category wise consumption pattern	22
Table 17: Data of 10% 11kV feeder Meters	29
Table 18: 1% Input Energy Points	34
Table 19: Data of Rohini Division	34
Table 20: Data of Mangolpuri Division	35
Table 21: Data of Pitampura Division	35
Table 22: Data of Keshav Puram Division	36
Table 23: AT&C losses of previous three years	37
Table 24: Division wise Input and Billed Energy FY 2022-23 & 2023-24	38
Table 25: Division wise T&D and AT&C losses in FY 2022-23 & FY 2023-24	38
Table 26: Circle wise Comparison for FY 2024-25	39
Table 27: Division wise input energy & billed energy for FY 2024-25	39
Table 28: Energy Input and AT&C Losses for FY 2024-25	41
Table 29: Voltage-wise AT&C Losses for FY 2024-25	41
Table 30: Circle wise consumers, connected load, Input and Billed Energy in FY 2024-25	42
Table 31: Circle wise T&D losses, Collection Efficiency and AT&C losses for FY 2024-25	43
Table 32: Division wise T&D losses, Collection Efficiency and AT&C losses for FY 2024-25	44
Table 33: Divisions wise connections & energy consumptions	47
Table 34: Divisions wise AT&C losses (%) & T&D losses (%)	47
Table 35: High Loss Feeders	49
Table 36: Category wise subsidy Division wise subsidy detail	55
Table 37: Comparison of AT&C and T&D losses for last two Years	56
Table 38: AT&C losses Trend for last three Years	57
Table 39: Details of energy manager and Authorized signatory of DISCOM	58
Table 40: Compliance status w.r.t Timelines and Pre-requisites	59
Table 41: Summary of Data gaps	64
Table 42: Comments by Energy Auditor and responses of DISCOM management	65
Table 43: Energy balance and losses	67
Table 44: Infrastructure details	79
Table 45: Metering details	79
Table 46: Category of service details	85
Table 47: Field inspection details	86
Table 48: List of documents verified with each parameter	89
Table 49: Formulas used to derive the parameters	93

List of Figures

Figure 1: Regulatory framework for Energy Accounting & Audit.....	11
Figure 2: Organogram of Energy accounting in the DISCOM	15
Figure 3: Category wise consumer share.....	18
Figure 4: Category wise Load share.....	18
Figure 5: Category wise Billed Units	18
Figure 6: Circle wise Energy Input (MU) and Energy Billed (MU) Pattern	22
Figure 7: Core Values of TATA Power-DDL.....	23
Figure 8: Energy Flow Diagram.....	28
Figure 9: Comparison of Losses – Last 3 FYs – 21-22/22-23/23-24.....	37
Figure 10: Division wise Input Energy Share	40
Figure 11: Division wise Input Energy Share	40
Figure 12: Last two years AT&C loss trend.....	56
Figure 13: Last two years T&D loss trend	57
Figure 14: Organogram of Energy accounting in the DISCOM	58
Figure 15: Single Line Diagram (SLD) of Tata Power DDL.....	84

List of Abbreviations

ADMS	Advanced Distribution Management System
ADR	Automated Demand Response
AMI	Advanced Metering Infrastructure
AMR	Automated Meter Reading
AMRUT	Atal Mission for Rejuvenation and Urban Transformation
AT&C	Aggregate Technical and Commercial
BDR	Behavioural Demand Response
BEE	Bureau of Energy Efficiency
BLDC	Brushless DC
CT	Current Transformer
DC	Designated Consumer
DEEP	Discovery of Efficient Electricity Price
DISCOM	Electricity Distribution Company
DMS	Distribution Management System
DT	Distribution Transformer
EA	Energy Auditor
EHT	Extra High Tension
EHV	Extra High Voltage
EM	Energy Manager
ERP	Enterprise Resource Planning
FFA	Field Force Automation
FY	Financial Year
GIS	Geographic Information System
GIUNC	Global Intelligent Utility Network Coalition
HT	High Tension
HVDS	High Voltage Distribution System
IPDS	Integrated Power Development Scheme
kVA	Kilo Volt Ampere
LT	Low Tension
MoP	Ministry of Power
MU	Million Units
M&V	Monitoring & Verification
MW	Mega Watt
NO	Nodal Officer
OA	Open Access
OMS	Outage Management System
PAT	Perform, Achieve and Trade
POC	Point of Connection
PSS	Packaged sub-station
PT	Potential Transformer
PX	Power Exchange
R-APDRP	Re-structured Accelerated Power Development and Reforms Programme
RDSS	Revamped Distribution Sector Scheme
RE	Renewable Energy
RLDC	Regional Load Dispatch Centre
RLB	Rural Local Bodies

RPO	Renewable Purchase Obligation
SCADA	Supervisory Control and Data Acquisition
SDA	State Designated Agency
SLD	Single Line Diagram
SLDC	State Load Dispatch Centre
T&D	Transmission and Distribution
ULB	Urban Local Bodies

Acknowledgement

We would like to express our heartfelt gratitude to Tata Power Delhi Distribution Limited, Delhi for providing us with the opportunity to conduct Energy Audit of the DISCOM for FY 2024-25, in accordance with Bureau of Energy Efficiency (Manner and Intervals for Conduct of Energy Audit in electricity distribution companies) Regulations, 2022 and its Amendments.

We are immensely grateful to the management of Tata Power Delhi Distribution Limited, Delhi, for their invaluable cooperation and providing us with all the relevant information necessary for the successful completion of the Annual Energy Audit FY 2024-25.

We also extend our sincere thanks to the entire working group comprising of:

Mr. Dwijadas Basak – CEO

Mr. H C Sharma– General Manager (Nodal Officer)

Mr. Pushpendra Kumar- Addl. General Manager, Head of Department, Energy Audit

Mr. Banwari Lal Gupta – AGM (Energy Manager)

Mr. Hari Om Sharma – Head of Group, Energy Audit Group

Mr. Neeraj Singh – Sr. Manager, Energy Audit

Mr. Sahil Yadav – Executive, Energy Audit

for their immense support and assistance throughout the audit process.

We look forward to a continued partnership with Tata Power Delhi Distribution Limited, Delhi, and we express our gratitude for their continued support in all our future endeavours.



Signature

Name: Dr. P.P Mittal

Designation: Director

Registered No: EmAeA 0024

Firm: A-Z Energy Engineers Pvt Ltd

1. Executive Summary

This section presents a brief summary of the results of the Annual Energy Audit carried out during May & June 2025. The study covers mainly verification process for monitoring of input energy and consumption pattern at various voltage levels, identification of area of energy leakage, wastage or inefficient use, identification of high loss-making areas and networks, identification of overloaded segments of the network for necessary capacity additions, highlighting the strengths and weaknesses of the Tata Power as DISCOM in the management of energy and energy resources with a focus mainly on proposals and recommendations on Energy Conservation.

Tata Power-DDL, a joint venture between Tata Power Company and the Government of NCT of Delhi with the majority stake being held by Tata Power (51%). It distributes electricity in North & Northwest parts of Delhi. The company started its operations on July 1, 2002, post the unbundling of erstwhile Delhi Vidyut Board to distribute electricity from Delhi Transco Limited end to the end consumers.

The Input energy purchased (from Generation source), Net Input energy (Discom periphery after adjusting for transmission losses and traded energy) & Total Energy billed by Tata Power-DDL for their customers is 13094MU, 11436.67MU & 10795.16MU respectively, for FY 2024-25, and the average monthly consumption stands at 427.08kWhr/per consumer/month.

Tata Power-DDL caters to network area spread in 5 circles, 12 Divisions across an area of 510sq.kms.

1.1. Goals and Objectives

Tata Power-DDL is a designated consumer in Discom sector. Being a designated Consumer, Tata Power-DDL ought to have Annual energy audit (Accounting) of their facilities as per BEE notification No. 18/1/BEE/Discom/2021 dated 6th October 2021.

The Annual Energy Audit & Accounting at Tata Power-DDL is conducted with the following Objectives:

- Verification of existing pattern of energy distribution across periphery of DISCOM.
- Verification of accounted energy flow submitted by DISCOM at all applicable voltage levels of the distribution network.
- Verification of the accuracy of the data collected, analysis of data with respect to consistency, improvement in accounting and reducing losses of DISCOM.
- Verification of the information submitted by DC to the SDA/BEE about status of Energy Input, Output and Losses for the previous two/three years.
- Access the past performance of the establishment.
- Quantification of Energy Losses and Energy Saving Potential.

1.2. About Energy Audit firm

A-Z Energy Engineers Pvt. Ltd., is an Accredited Energy Auditor from BEE and an ISO 9001:2015 certified company that aims to assist all stakeholders in implementing energy efficiency and creating awareness about the merits of implementing energy efficiency and safety practices. They are empaneled by BEE for PAT M & V Audits and Mandatory Energy Audit Projects and have completed more than 1800 projects, including 75 PAT projects.

The founder Director, Shri. Dr PP Mittal, has received several awards and recognitions for his services in the field of energy. We have a pool of experienced BEE Accredited & Certified Energy Auditors, Electrical Engineers, Mechanical Engineers and Technicians having experience of more than 32 years. The Energy Audits are being carried out with sophisticated instruments namely Power-Analyzer, Flue Gas Analyzer, Ultra-sonic flow meter, Techo-meter, Anemometer, Hego-Meter, Digital Thermometer, Thermographic Camera's, Lux Meter, Leak detectors. Laser gun etc.

1.3. AT&C losses for FY 2024-25

The AT&C loss for FY 2024-25 for TATA Power-DDL stands at 5.55%, which is approximately same as the T&D (Transmission and Distribution) losses as the collection efficiency has been close to 100.06%. The AT&C losses for the FY 2024-25 are shown in the table below:

Table 1: Energy Balance & Losses for FY 2024-25

Energy Input Details	Formula	UoM	Value
Input Energy Purchase (From Generation Source)	A	MU	13094.00
Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	B	MU	11436.67
Total Energy billed (is the Net energy billed, adjusted for energy traded)	C	MU	10795.16
Transmission and Distribution (T&D) loss Details	D	MU	641.51
	$E = D/B \times 100$	%	5.61%
Collection Efficiency	F	%	100.06%
Billing Efficiency	G	%	94.39%
Aggregate Technical & Commercial Loss	$H = 1 - \left\{ \frac{G}{F} \right\} \times \text{Min}(F, 100\%)$	%	5.55%

2. Background

2.1. Regulations and role of BEE

2.1.1. Objectives of BEE

- To develop policies and programs on efficient use of energy and its conservation with the involvement of stakeholders.
- To plan, manage and implement energy conservation programs as envisaged in the Energy Conservation Act, 2001
- To assume leadership and provide policy framework and direction to national energy efficiency and conservation efforts and programs.
- To demonstrate energy efficiency delivery mechanisms, as envisaged in the Energy Conservation Act, 2001, through Public-Private Partnership (PPP).
- To establish systems and procedures to measure, monitor and verify energy efficiency results in individual sectors as well as at the national level.
- To leverage multi-lateral, bi-lateral and private sector support in implementation of programs and projects on efficient use of energy and its conservation.
- To promote awareness of energy savings and energy conservation.

2.1.2. Role of BEE

- BEE coordinates with designated agencies, designated consumers and other organizations working in the field of energy conservation/efficiency to recognize and utilize the existing resources and infrastructure in performing the functions assigned to the Bureau under the Energy Conservation Act.
- The Act provides regulatory mandate for: standards & labeling of equipment and appliances; energy conservation building code for commercial buildings; and energy consumption norms for energy intensive industries.
- The EC Act was amended in 2010 to incorporate few additional provisions required to better equip BEE to manage ever evolving sphere of energy efficiency in the country.
- Create awareness and disseminate information on energy efficiency and conservation.
- Arrange and organize training of personnel and specialists in the techniques for efficient use of energy and its conservation.
- Strengthen consultancy services in the field of Energy Efficiency.
- Promote research and development.
- Develop testing and certification procedures and promote testing facilities.
- Formulate and facilitate implementation of pilot projects and demonstration projects.

- Promote use of energy efficient processes, equipment, devices and systems.
- Take steps to encourage preferential treatment for use of energy efficient equipment or appliances.
- Promote innovative financing of energy efficiency projects.
- Give financial assistance to institutions for promoting efficient use of energy and its conservation.
- Prepare educational curriculum on efficient use of energy and its conservation.
- Implement international co-operation programs relating to efficient use of energy and its conservation.

2.1.3. Regulatory framework for Energy Accounting & Audit

The Energy Conservation Act 2001 (hereafter referred to as EC Act 2001) was enacted on 29th September 2001. The EC Act 2001 empowers BEE to notify regulations regarding energy conservation and efficiency improvement. In accordance with the EC Act 2001, BEE notified the Bureau of Energy Efficiency (Manner and Intervals for Conduct of Energy Audit) Regulations, 2021, on 6th October 2021. BEE subsequently amended these regulations with the Bureau of Energy Efficiency (Manner and Intervals for Conduct of Energy Audit) (Amendment) Regulations, 2022. The Ministry of Power (MoP) issued guidelines on 17th January 2023, for energy accounting and auditing of distribution companies, in line with the BEE regulations. Distribution companies and energy audit firms must comply with this regulatory framework when preparing energy accounts and audit reports. The regulatory framework for Energy Accounting and Energy Auditing is shown in the below figure:

Figure 1: Regulatory framework for Energy Accounting & Audit



Key highlights of the Regulatory framework are listed below:

- Bureau of Energy Efficiency (BEE) through Ministry of Power, Government of India issued regulations for Conduct of Mandatory Annual Energy Audit and Periodic Energy Accounting in DISCOMs. As per the regulation, all Electricity Distribution Companies are mandated to conduct annual energy audit and periodic energy accounting on quarterly basis.
- Owing to the impact of energy auditing on the entire distribution and retail supply business and absence of an existing framework with dedicated focus on the same, it was imperative to develop a set of comprehensive guidelines that all Distribution utilities across India can follow and adhere to.
- Accordingly, Regulations on Manner and Intervals for Conduct of Energy Audit and Accounting in Electricity Distribution Companies has been framed. Energy Accounting means accounting of all energy inflows at various voltage levels in the distribution periphery of the network, including renewable energy generation and open access consumers, and energy consumption by the end consumers. Energy accounting and a consequent annual energy audit would help to identify areas of high loss and pilferage, and thereafter focus efforts to take corrective action.
- These Regulations for Energy audit in Electricity Distribution Companies provides broad framework for conduct of Annual and Quarterly Periodic Energy Accounting with necessary pre-requisites and reporting requirements to be met.

2.2. Purpose of audit and accounting Report

Tata Power-DDL is a designated consumer in Discom sector. Being a designated Consumer, Tata Power-DDL need to have Annual and Quarterly Energy Audit and Accounting of its facilities as per BEE notification No 18/1/BEE/Discom/2021 dated 6thOctober 2021.

The energy intensity of India is higher with respect to GDP growth and there is an urgent need to address these issues on priority through integrated and comprehensive approach and by adopting latest techniques and technologies with active participation of all stakeholders.

Annual Energy Audit and Accounting not only help in reducing losses in system, but it also helps DISCOM in sustainable growth. The objective of this energy audit is to reduce T&D loss and AT&C loss of the DISCOM through identification of commercially viable and implementable schemes for reduction of technical and commercial loss in the DISCOM thus leading to sustainable energy cost reductions.

2.3. Period of Energy Auditing and accounting

Energy audit activity was started with a meeting with Nodal Officer of Tata Power-DDL in the month of May '2025. Based on the requirement, visits were carried out to Division,

Subdivision and Grid etc. for data collection and technical discussion. The period of study was from April 2024 to March 2025.

Table 2: Period of Energy Auditing and Accounting

Particulars	Energy Accounting				Energy Audit
	Q1	Q2	Q3	Q4	FY 2024-25
Applicable period	01-Apr-24 to 30-Jun-24	01-Jul-24 to 30-Sep-24	01-Oct-24 to 31-Dec-24	01-Jan-25 to 31-Mar-25	01-Apr-24 to 31-Mar-25
Date of Commencement	01-July-24	01-Oct-24	01-Jan-25	01-Apr-25	May-25
Date of Publishing	24-Sep-24	19-dec-24	28-Mar-25	10-Jun-25	-
Officer In charge	Mr Hari Om Sharma (Head of Group, Energy Audit)				Dr. P P Mittal [AEA 0011] Registration No: EmAEA-0024



[Handwritten signature]

3. DISCOM Introduction and Overview

3.1. Name and address of DISCOM

Tata Power-DDL is having its corporate office at NDPL House, Hudson Lines, Kingsway Camp, Delhi – 110009

Table 3: Name and Address of DISCOM

Particulars	Details
Name of DC	Tata Power Delhi Distribution Limited
Address	NDPL House, Hudson Lines, Kingsway Camp, Delhi-09, P.O. GTB Nagar, Delhi

3.2. Name and contact details of energy manager (BEE Certified, if any) and Authorized signatory of DISCOM (Nodal Officer)

Energy Accounting/Audit wing is headed by Mr. Dwijadas Basak (CEO). BEE certified Energy Manager, Mr. Banwari Lal Gupta (AGM) is leading the energy accounting activities in Tata Power-DDL. Mr. H C Sharma (GM) is Authorized Signatory/ Nodal Officer.

The details of DISCOM's energy manager and authorized signatory for this report are shown below:

Table 4: Details of Energy Manager and Authorized signatory of DISCOM

Particulars	Details
Energy Manager	Mr. Banwari Lal Gupta (AGM) EA-2317 Mobile: 9971392990 Email: bl.gupta@tatapower-ddl.com
Authorized Signatory	Mr. H C Sharma (General Manager) Ph: 91-1166050595 Email: hc.sharma@tatapower-ddl.com

3.3. Summary profile of DISCOM

3.3.1. Jurisdiction of DISCOM

Tata Power Delhi Distribution Limited [Tata Power-DDL] is a joint venture between Tata Power and the Government of NCT of Delhi with the majority stake being held by Tata Power Company (51%). Tata Power-DDL is acknowledged for its consumer-friendly practices. Since privatization, the Aggregate Technical & Commercial (AT&C) losses in Tata Power-DDL area have shown a record decline. Tata Power-DDL Strives to Serve Utilities across the globe to improve their standards of performance and create long term value with their strategic partners. Its distribution network is spread over an area of 510 sq. km covering 5 circles, 12 Divisions, catering to a customer base of over 2 million.

3.3.2. Energy Accounting/Audit wing in the DISCOM:

The Energy Accounting/Audit wing in the DISCOM has been established on 11-Feb-22. The Organogram of the DISCOM wrt Energy Auditing and Accounting is as shown below:

Figure 2: Organogram of Energy accounting in the DISCOM

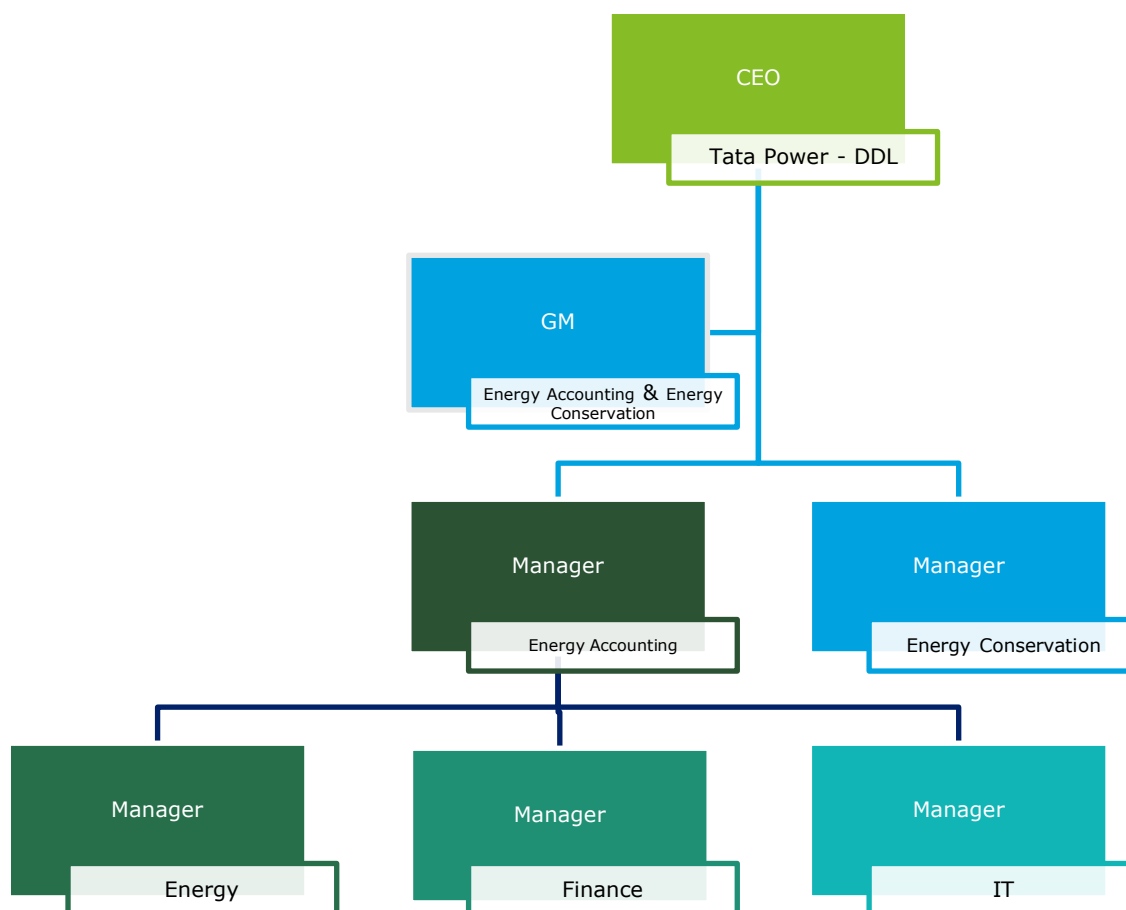


Table 5: Details of Energy Manager and Authorized signatory of DISCOM

Designation	No. of Officers
CEO	Mr. Dwijadas Basak
General Manager	Mr. HC Sharma
AGM (Energy Manager)	Mr. Banwari Lal Gupta
Head of Department	Mr. Pushpendra Kumar
Head of Group, Energy Audit	Mr. Hari Om Sharma
Sr. Manager, Energy Audit	Mr. Neeraj Singh
Executive, Energy Audit	Mr. Sahil Yadav
Finance Manager	Mr. Sachin Gupta
IT Manager	Ms. Kamakshi Oberoi

3.3.3. Administrative hierarchy

Tata Power-DDL is having its corporate office at NDPL House, Hudson Lines, Kingsway Camp, Delhi-09 and has 5 Circles. These Circles are further divided into Divisions and Sub-Divisions as shown in the below tables.

Table 6: Administrative hierarchy structure in Tata Power-DDL

Parameters	Value
Number of circles	5
Number of divisions	12
Number of sub-divisions	31
Number of sections	-

The Circles are further divided into Divisions and Sub-Divisions as shown in the table below:

Table 7: Administrative hierarchy in Tata Power-DDL up to section level

Circle	District	Zone	Zone code
Urban	Badli	Samaya Pur	507
		Prahladpur	516
		Holambi	572
Sub- Urban	Bawana	Bawana	512
		Pooth Khurd	521
		Kanjhawala	533
Town	Civil Lines	Vivekanand Puri + Tibya College+Kamla Nagar	421+423+417+424
		Civil Lines+Timar Pur+Vijay Nagar	411+418+416
Town	Keshav Puram	Keshav Puram + Jai Mata Market+Shahzada Bagh	501+520+422
		W.Pur Indl. Area A & B-Blk. +Shastri Nagar	502+509+425
Metro	Mangolpuri	Poothkalan	519
		Mangolpuri+Sultanpuri	515+518
Town	Model Town	Jahangir Puri	505
		Gujrawalan Town + Adarsh Nagar+Model Town +Mukherjee	402+415+412+413
City	Moti Nagar	Rama Road + Lakkad Mandi	1301+1303
		Naraina + Pusa	1302+1304

Circle	District	Zone	Zone code
Sub- Urban	Narela	Ali Pur	511
		Narela	514
		Bakhtawarpur	517
		DSIDC	522
Metro	Pitam Pura	Saraswati Vihar + Rani Bagh	504+510
		Pitam Pura + Lok Vihar	508+530
Urban	Rohini	Avantika	551
		Rithala	561
		Rohini+Badli Residential	571+581
Metro	Kirari	Kirari	523
		Karala	513
Urban	Shalimar Bagh	Burari	414
		Bhalswa	503
		Shalimar Bagh	506+531
		Sant Nagar	532

3.3.4. Consumer Details

Energy consumption with type of customer is given in the table:

Table 8: Customer Profile for FY 2024-25

Category	No. of Connections		Connected Load		Billed Energy		Billed Amount in Rs. Crore	Collected Amount in Rs. Crore
	Nos	%	MW	%	MU	%		
Residential	1792580	85.25%	3882.14	55.6%	5451.54	50.5%	3878.90	3887.54
Agricultural	4471	0.21%	34.519	0.5%	16.14	0.1%	12.63	12.49
Commercial/ Industrial-LT	295912	14.07%	2418.31	34.7%	3937.02	36.5%	6183.75	6185.86
Commercial/ Industrial-HT	783	0.04%	333.144	4.8%	638.11	5.9%	949.02	949.61
Others	9091	0.43%	308.46	4.4%	752.35	7.0%	788.15	784.64
Total	2102837	100%	6976.57	100%	10795.16	100%	11812.46	11820.14

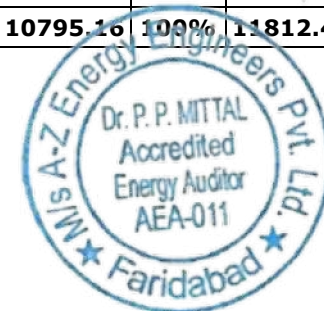


Figure 3: Category wise consumer share

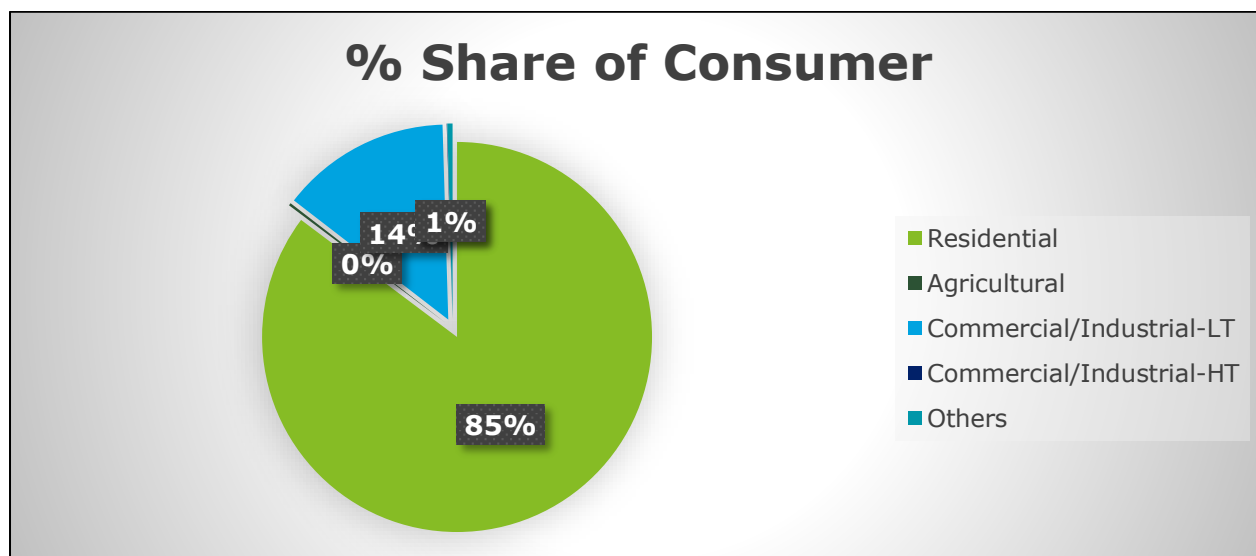


Figure 4: Category wise Load share

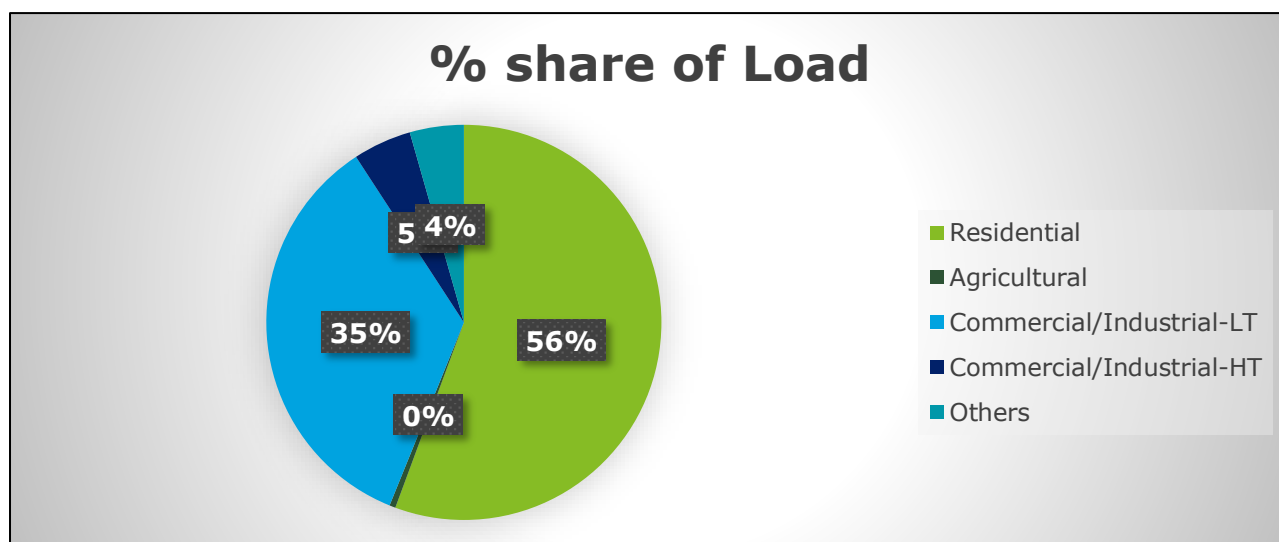
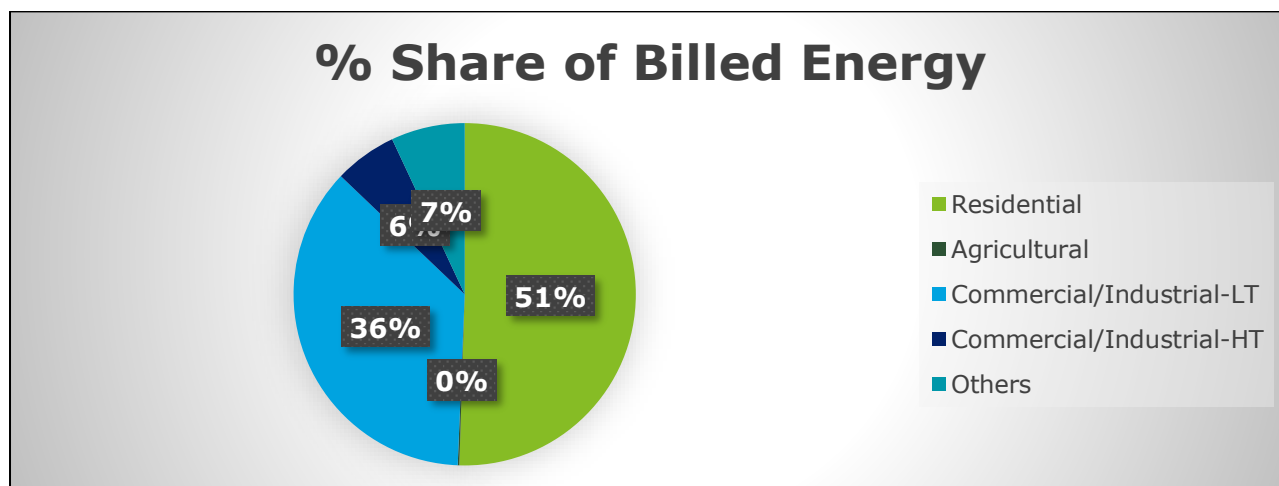


Figure 5: Category wise Billed Units



3.4. Electrical infrastructure and assets voltage wise

The following table provides the details of network infrastructure owned by Tata Power-DDL:

Table 9: Network Infrastructure details

Asset	Particulars	Unit	FY22-23	FY 23-24	FY 24-25
66 kV and above	66/ 11 kV Sub station	Nos	NA	NA	NA
	66 kV Feeders	Nos	137	142	149
	66 kV Line	Ckt. km	554.01	567	598.5
33 kV	33/ 11 kV Sub station	Nos	NA	NA	NA
	33 kV Feeders	Nos	111	115	119
	33 kV Line	Ckt. km	482.19	519.65	540.9
11 kV	11 kV Feeders	Nos	1286	1312	1394
	11 kV Overhead Line	Ckt. km	5236.9	2209	2204.6
	11 kV Underground Line	Ckt. km	3028	3154	3264.6
LT	LT Line	Ckt. km	7588	7809.1	8035.9
PT	Power Transformer	Nos	219	222	219
	Power Transformer Capacity	MVA	5143	5267	5256
DT	Distribution Transformer	Nos	Above 250kVA-4986 nos., equal to and less than 250kVA-3171 nos.	Above 250kVA-5210nos., Equal to and less than 250kVA-3481	Above 250KVA-5548 nos., Equal to and less than 250KVA-3237
	Distribution Transformer Capacity	MVA	3421.1	3575.4	3804.9

The Input energy, consumption, transmission losses and key infrastructure details of Tata Power-DDL are summarized in table below:

Table 10: Input Energy & Infrastructure details

Parameters	FY 2024-25
Input Energy purchased (MU)	13094
Transmission loss (%)	4%
Transmission loss (MU)	548.6386
Energy sold outside the periphery (MU)	1140
Open access sale (MU)	96.968653
EHT sale	273.88
Net input energy (received at DISCOM periphery or at distribution point)-(MU)	11527.40
Is 100% metering available at 66/33 kV (Select yes or no from list)	Yes
Is 100% metering available at 11 kV (Select yes or no from list)	Yes
% of metering available at DT	77%
% of metering available at consumer end	100%

Parameters	FY 2024-25
No of feeders at 66kV voltage level	149
No of feeders at 33kV voltage level	119
No of feeders at 11kV voltage level	1394
No of LT feeders level	18341
Line length (ckt. km) at 66kV voltage level	598.5
Line length (ckt. km) at 33kV voltage level	540.9
Line length (ckt. km) at 11kV voltage level	5469.2
Line length (km) at LT level	8035.9
Length of Aerial Bunched Cables	5934.3
Length of Underground Cables	6869.2
HT/LT ratio	0.822384549

3.4.1. Metering details

The status of meters installed in Tata Power-DDL as on 31-03-25 are given in the below tables:

Table 11: Voltage wise Meter Consumers

Parameters	66kV and above	33kV	11/22kV	LT
Number of conventional metered consumers	0	0	0	1541229
Number of consumers with 'smart' meters	1	2	552	529083
Number of consumers with 'smart prepaid' meters	0	0	0	10094
Number of consumers with 'AMR' meters	7	4	412	19241
Number of consumers with 'non-smart prepaid' meters	0	0	0	3190
Number of unmetered consumers	0	0	0	0
Number of total consumers	8	6	964	2102837

3.4.2. Distribution Transformer (DT) details

The details of distribution transformers in Tata Power-DDL as on 31-03-25 are given in the below tables:

Table 12: Numbers of Distribution Transformers and Metering

Parameters	66kV and above	33kV	11/22kV	LT
Number of conventionally metered Distribution Transformers	0	0	0	154
Number of DTs with communicable meters	0	0	0	6567
Number of unmetered DTs	0	0	0	2064
Number of total Transformers	0	0	0	8785

3.4.3. Feeder details

The details of feeders in Tata Power-DDL as on 31-03-25 are given in the below tables:

Table 13: Voltage wise numbers of Feeders and Metering

Parameters	66kV and above	33kV	11/22kV	LT
Number of metered feeders	149	119	1394	0
Number of feeders with communicable meters	149	119	1394	0
Number of unmetered feeders	0	0	0	18341
Number of total feeders	149	119	1394	18341

3.4.4. Distribution Line details

The details of distribution lines in Tata Power-DDL as on 31-03-25 are given in the below tables:

Table 14: Length of Distribution Lines

Particulars	Value(km)
Line length (ckt km)	1841
Length of Aerial Bunched Cables	5934.3
Length of Underground Cables	6869.2

3.4.5. Energy Flow details

Energy flow details for FY 2024-25 are given in the below table:

Table 15: Energy Flow details

Parameters	Values
Input Energy purchased (MU)	13094
Transmission loss (%)	4%
Transmission loss (MU)	548.64
Energy sold outside the periphery (MU)	1140
Open access sale (MU)	96.97
EHT sale	273.88
Net input energy (received at DISCOM periphery or at distribution point)-(MU)	11527.4
Total Energy billed (MU)	10795.16
Transmission and Distribution (T&D) Loss (MU)	641.51
T&D Loss (%)	5.61%

3.4.6. Pattern of energy distribution

Power Purchase:

During the analyzed period, Tata Power-DDL purchased energy of 13094 million units (MUs) in FY 2024-25. The company's energy purchase varies considerably from month to month depending upon seasonal impact.

Energy Billed:

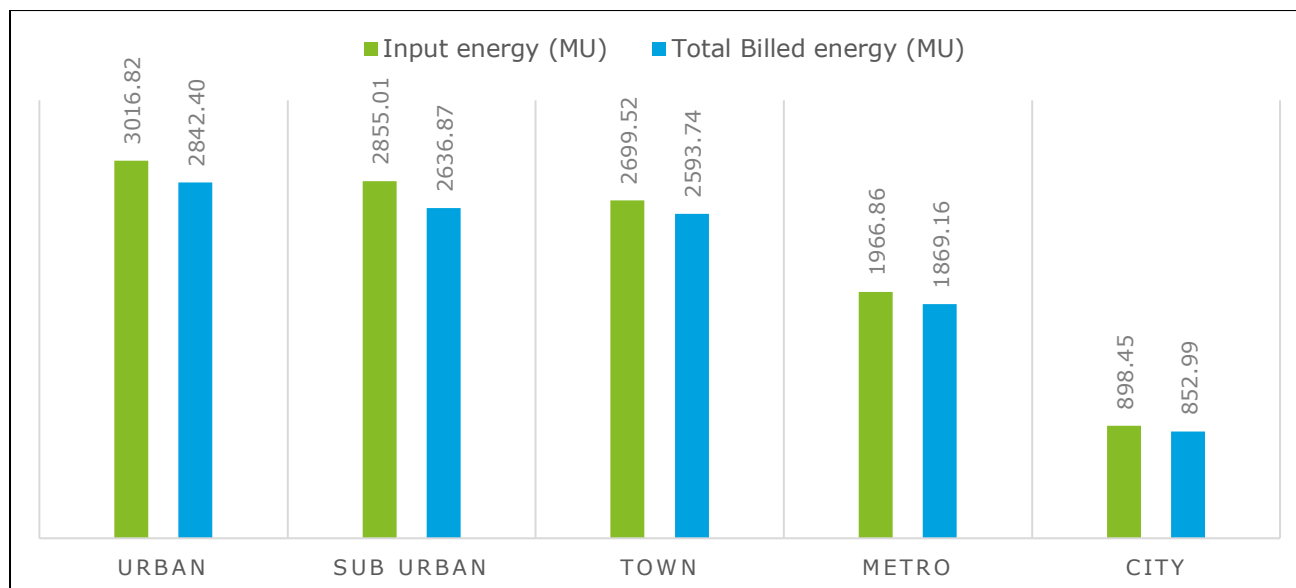
Category wise – consumer count and energy consumption has been provided in table below. Maximum energy consumption has been recorded against domestic consumers.

Table 16: Consumer category wise consumption pattern

Consumer Category	Consumers		Energy Consumption	
	No.	%Share	MUs	%Share
Residential	17,92,580	85.25%	5,451.54	50.50%
Commercial/Industrial	296695	14.11%	4575.13	42.38%
Agricultural	4471	0.21%	16.14	0.15%
Others-3 (specify in remarks)	9091	0.43%	752.35	6.97%
Total	21,02,837	100%	10,795.16	100%

Circle wise Energy Input and Energy Billed has been shown below:

Figure 6: Circle wise Energy Input (MU) and Energy Billed (MU) Pattern



3.4.7. Salient features

Tata Power-DDL main objectives are to achieve efficiency gains and make necessary changes to make the company commercially viable, progressively self-sustainable, and less dependent on the government while balancing the interests of consumers with regards to quality of service and economical tariffs.

a) Vision

Be a benchmark power utility providing sustainable, affordable and innovative energy solutions.

b) Mission

- Keep the customer at the centre of what we do.
- Embed safety and sustainability in every aspect of our business.
- Create new benchmarks through leadership in technology, innovation and continuous improvement.
- Empower teams that are driven by passion and purpose.
- Support the development of communities we work in.

c) Core Values

Tata Power-DDL is a value-driven organisation and there are Five Core values as an integral part of its work culture.

Figure 7: Core Values of TATA Power-DDL



3.5. Energy Conservation measures

Energy conservation is a critical issue in today's world, as the demand for energy continues to increase while the resources available to produce it are finite. Energy conservation measures that have already been taken and proposed for the future are explained below.

1. Energy Conservation Measures taken by Tata Power DDL for Network strengthening and Loss Reduction:

1. **Sub-Transmission Network Optimization:** Tata Power-DDL has mesh network at 33 & 66 kV level which is operated in a radial manner. The NOPs (Normal Open Points) are decided after conducting thorough loss studies and the most optimal configuration is adopted for minimizing loss while meeting other operational & reliability requirements. The exercise is conducted annually to incorporate benefits from addition of new circuits. Also a separate configuration is adopted for the winter months (Nov to Mar) to take advantage of lower loading levels. The winter reconfiguration result is expected saving of approximately 0.6 GWh annually.
2. **11 kV Network Optimization:** The 11kV network is also a mesh network operated in a ring main manner. TATA Power-DDL has initiated an internal project in FY24-25 to reconfigure the network for minimizing loss while meeting other operational & reliability requirements. This is expected to contribute to reduction in technical loss by over 0.5 GWh annually.
3. **Change in Cable Specifications:** Due to operational requirements, the standard rating of service cable catering to single phase consumers with sanctioned load of 6 to 10 kW has been revised from 2x10 sq.mm. Al XLPE to 2x25 sq.mm. Al XLPE. The reduction in resistivity has resulted in reduction of technical losses.
4. **Energy Management Systems:** The Company has implemented energy management systems to monitor and control energy consumptions in grids, office buildings, to identify areas of energy waste, and optimize energy usage. TATA Power-DDL has already been certified with EnMS Standard ISO 50001:2018.
5. **Procurement of Renewable Energy:** The Company has increased the use of renewable energy sources, such as solar and wind power to reduce dependency on fossil fuels and contribute to reduce carbon emissions.
6. **Periodic and Annual Energy Audits:** Regular energy audits are conducted to identify energy waste and implement measures to reduce it. Also 3rd party ISO 50001 Audit is being conducted annually prior to re-certification of ISO standard.
7. **Use of Co-axial co-extruded Cables in 11 kV network:** With use of co-extruded cables in 11 kV network, has resulted in increased life span of cables and efficiency improvement in terms of power reliability and delivery.
8. **Smart DT Meters with DIDO (Digital Input/Output):** Smart DT meters installed for Distribution transformers, are being installed in network with added feature of DI/DO (digital input and digital output) port. These digital ports in meters can be connected to float sensors installed inside conservation oil tank for monitoring of oil levels of DT. When oil level dips below required level, smart digital

ports connected to breaker would trip off the power supply to DT and would prevent from any overheating/ damage to DT. This has been included by CEA "Guidelines and best practices for Operation and Maintenance of Distribution Transformers".

9. **Li-ion battery with BMS (battery management system):** For providing DC supply in grid sub-stations, conventional batteries used were of lead acid type. Gradually these types of batteries are now replaced with Li-ion batteries. Benefits of Li-ion over conventional are 90% of depth of discharge than 60%, 95% efficiency, non-hazardous disposal, less charging time, high specific energy output etc.
10. **Energy efficient Distribution Transformers:** Loss levels and efficiency levels for distribution and power transformers have been already defined and set in our technical specifications for OEMs (original equipment manufacturers) to comply with.
11. **Mineral oil vs. Ester oil:** Conventionally, petroleum based mineral oil is being used as dielectric medium in transformers. Ester oil has been introduced now with features of High flash point of 330 deg C providing high level of fire safety, Chemical properties of ester oil enhances DT insulation performance and life expectancy, Insulation system of distribution transformer can be run at 20°C warmer conditions, thus increasing the loading capability of transformer for the same life expectancy. Transformer with ester oil can operate better in areas with high ambient temperature.
12. **RF based street light management system:** Radio frequency based remote monitoring and operations of LED street light projects are being implemented. Main features include 'Dimming' of lights during non-peak hours and 'ON/OFF' feature based on Astro timer switch. Energy management through Street light controller would reduce energy consumption and CO2 emission. Unnecessary over-lighting can be prevented by target dimming of sectors of the city, roads or individual luminaires. Thus, the SLC (street light controller) can be used to make effective and consistent energy savings –approximately up to 50% per year.
13. **Sub-station re-designing for space constrained areas:** To cater to ever increasing load requirements, few occurrences of space constraints are part of network designing. For such space constrained areas, new designs of sub-stations are proposed. Few new designs are Vertical double decker sub-station, wherein two DTs would be placed up and down in double tier arrangement; DT installation of single spun pole with LT and HT switches beneath the mounted DT; Drain top based sub-station where no space is available etc.

TATA Power-DDL has always been frontrunner in choosing energy efficiency practices for building a sustainable greener tomorrow. The DISCOM has already undertaken sustainable initiative in scheme named 'Urja Arpan'.

- **Demand Side Management/ behavioural Demand Response Program:** Raising awareness among individuals and organizations about responsible and

efficient electricity consumption. Till now, 1.26 lac + enrolled customers have participated and successfully achieved a cumulative load curtailment of 452 MW.

- **Energy Efficient Lighting and Fans:** Energy efficient BLDC fan scheme: Tata Power-DDL is offering Energy Efficient Products (EEPs) such as LED Bulbs and LED Tube Lights which have low maintenance and usage costs. These are durable and use up to 75% less energy than any other incandescent bulbs resulting in the reduction of monthly energy bills.
In addition to lighting, option for BLDC fans at discounted rates is available. BLDC fans provide high cooling efficiency with lower energy consumption and also savings in electricity bills. BLDC motors do not have brushes so they are more reliable, high life expectancy and energy savings as compared to conventional ceiling fans. With the added convenience of remote control, BLDC fan is a financially and environmentally smart choice for customers as it consumes half electrical load as compared to ordinary fans.
- **Launch of Clean Cooling Program with discount based Energy efficient ACs Scheme for Customers:** Tata Power-DDL in association with Voltas, Blue Star, General, Godrej, Hitachi is providing a discount-based scheme with or without Replacement of Old AC for all the customers.
- **Home Automation:** Tata Power-DDL is offering an IoT based solution which allows to operate, schedule, monitor appliances like lighting, fans, ACs, washing machines, geysers, TVs, refrigerators, etc. with phone or with voice control via Google assistant or Alexa.
- **Home Automation (Sensors & Sensor Lights):** With Motion sensors, lights and appliances can be switched on during presence of a person in the area. In absence of motion, automatically turn off the lights. When motion is detected, sensor lights automatically illuminate at full brightness. In the absence of motion automatically dim the area lights to 20%.
- **Air Purifier Scheme:** Tata Power-DDL in association with Voltas has initiated the Air Purifiers scheme at special discounted rates. The Air Purifiers are powered with 6-stage filtration process and removes up to 99.90% airborne pollutants, eliminates impurities with ease.
- **Washing Machine Scheme:** Tata Power-DDL has launched a 5-star rated washing machine scheme at special discounted rates in collaboration with Voltas-Beko.
- **Digitization/ Adoption of digital lifestyle:** By reducing paper consumption, and exploring digital alternatives, TATA Power-DDL contributes to environmental conservation and decreased carbon footprints, thus fostering a greener and eco-friendly workplace. E-billing is being adopted for consumer billing.
- **Supporting green initiatives:** Participation in local environmental programs, such as tree planting and clean-up drives.

- **Celebration of Earth Day on 22nd April every year:** As a collective responsibility to nurture and protect Mother Earth, DISCOM is urging consumers to shut down their power for 1 hour on Earth Day.

Some other Energy Conservation measures planned for future by TATA Power-DDL are mentioned below:

- 1) Solid insulated switchgears
- 2) Dry type Transformers
- 3) Plastic packaging prohibition
- 4) Cobalt free silica gels in transformer breathers
- 5) Water harvesting in grid sub-stations
- 6) Meter plastic box re-usage
- 7) GIS: Apps for route optimization and hence reduction in cable/ line lengths and hence technical losses
- 8) Recycled aggregate made out of construction and demolition waste in concrete mixtures
- 9) Direct delivery of material at site, thus saving in additional transportation requirements, contributing to lowering carbon emissions
- 10) Recycling used carton boxes and plastics through NGO
- 11) Scrapping of meters and then segregation of scrap for re-usage.



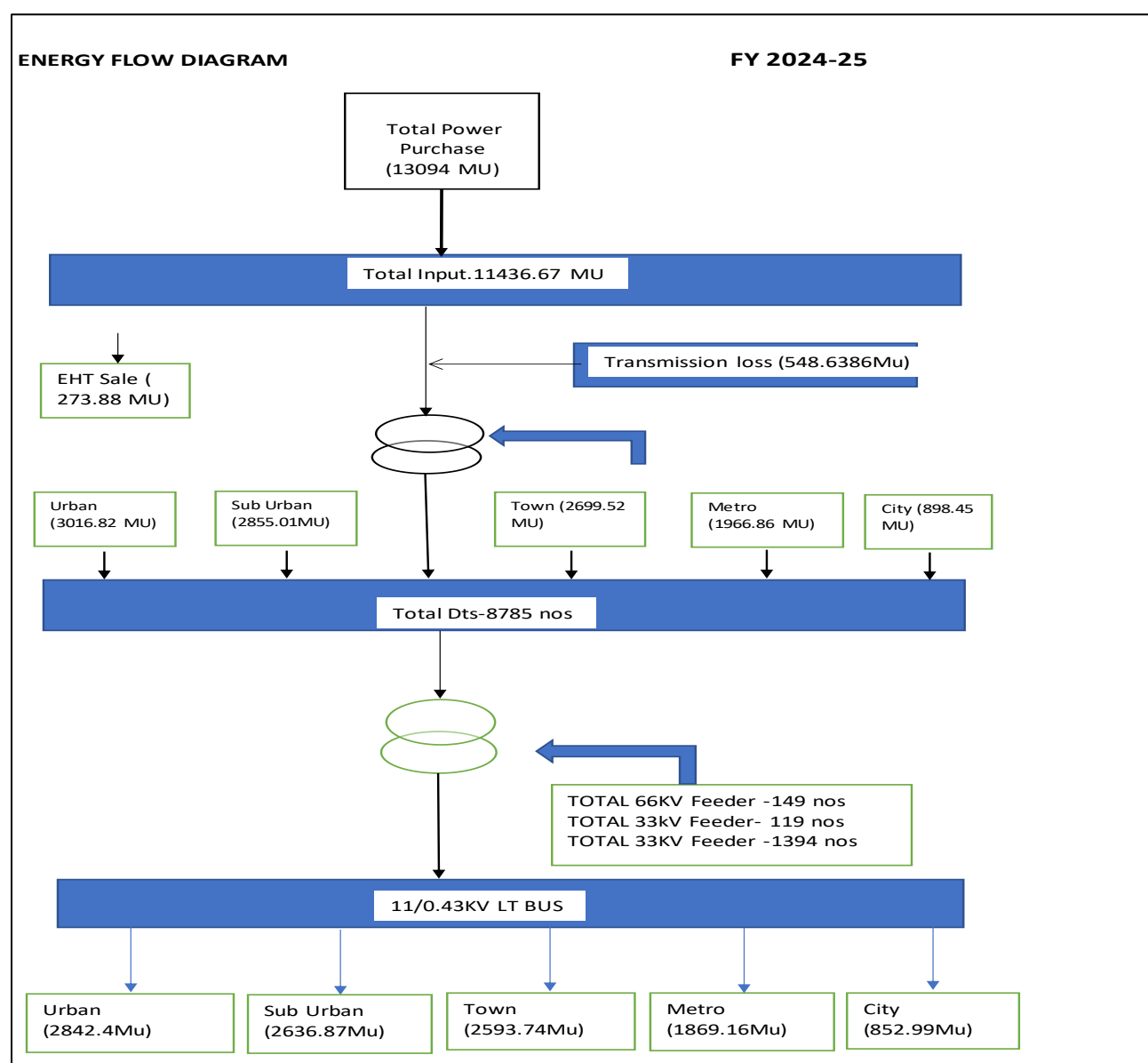
A handwritten signature in blue ink, appearing to be "P.P. Mittal", written over two horizontal lines.

4. Energy flow analysis

4.1. Energy flow across 5 Service Levels

- Energy flow between DTL 220KV Grid Substations to Tata Power DDL 66kV/33kV Grid Substation
- Energy flow between DTL 220 KV Grid Substations and Tata Power DDL 11kV/6.6 kV Substations
- Energy flow between 11kV/6.6kV feeders and distribution transformers, or high voltage distribution system
- Energy flow between distribution transformer, or high voltage distribution system to end consumer
- Energy flow between Feeders to end-consumer

Figure 8: Energy Flow Diagram



4.2. Validation of metered data

- a) Validation of feeder data:** Based on data available for 11 kV Feeder meters at substation, for a sample size of 10% for which documentary evidence has been captured in the audit report.

Data of 10% of 11 kV Feeder meters:

Table 17: Data of 10% 11kV feeder Meters

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
1	412	AZADPUR GRID	AZADPUR GRID TO AAKASH CINEMA S/STN	3.06	1.02
2	505	AZADPUR GRID	AZADPUR GRID TO INTERCONNECTOR 1 & 2 TO JAHANGIRPURI GRID S/S	2.16	2.83
3	506	SMB - KHOSLA GRID	SHALIMAR BAGH GRID TO 13 B S/S	1.44	1.35
4	507	BADLI GRID	BADLI GRID TO GALI NO-9 SAMAYPUR	2.80	1.19
5	516	BADLI GRID	BADLI GRID (NW) TO B.S.CHEMICAL	2.79	2.09
6	517	AIR KHAM PUR GRID	AIR KHAM PUR GRID TO BAKHTAWAR PUR - 1	2.48	2.27
7	522	DSIDC NARELA-2 GRID	DSIDC-2 GRID TO S/S NO-9 A-BLOCK	4.16	0.78
8	514	BAWANA CLEAR WATER GRID	BAWANA CLEAR WATER GRID TO GHOGHA (K) FDR NO-1	6.34	6.73
9	512	BAWANA CLEAR WATER GRID	BAWANA CLEAR WATER GRID TO DARYAPUR NANGAL S/S (O/D)	9.59	8.97
10	512	POOTH KHURD GRID	POOTH KHURD GRID TO SOS BAWANA	5.97	6.81
11	521	POOTH KHURD GRID	POOTH KHURD GRID TO POOTH KHURD VILLAGE FDR	2.76	3.39
12	551	ROHINI - 2 GRID	ROHINI - 2 GRID TO DDA FLAT S/S NO.1 FDR NO-1	1.68	4.60
13	519	ROHINI - 2 GRID	ROHINI - 2 GRID TO RMU OUTDOOR S/S	4.89	2.51
14	515	MANGOL PURI - 1 GRID	MANGOL PURI - 1 GRID TO S/S NO 15 FDR NO.1+ B BLK MGP IND.AREA PH-1 S/S	3.05	4.65
15	515	MANGOL PURI - 1 GRID	NDDDB	2.50	1.04
16	510	P. P. - 2 GRID	P. P. - 2 GRID TO LSC MKT.SAINIK VIHAR	1.54	1.61
17	515	P. P. - 2 GRID	P. P - 2 GRID TO N-BLOCK FDR-2	1.87	3.77
18	571	ROHINI - 3 GRID	ROHINI - 3 GRID TO RMU OPPOSITE CRPF SCHOOL S/S	1.27	0.82
19	504	RANI BAGH GRID	RANI BAGH GRID TO KAILAH ENCLAVE (Z-504)+JHULA LAL NO-1	4.36	5.07
20	530	P.P.-3 GRID	P.P.-3 GRID TO MERA BABA FEEDER-2	1.90	2.28
21	510	P.P.-3 GRID	P.P.-3 GRID TO RMU INSIDE	1.84	4.37

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
			PP-3 GRID		
22	501	RAMPURA GRID	RAMPURA GRID TO INTERCONNECTOR	1.88	2.58
23	1303	SARASWATI GARDEN GRID	SARASWATI GARDEN GRID TO RAMESH NGR	1.85	1.87
24	1302	INDERPURI GRID	INDERPURI GRID TO KRISHI KUNJ NO.-1	1.93	0.75
25	1301	SB MILL GRID	51 RAMA ROAD	1.52	1.44
26	416	HUDSON LINE GRID	HUDSON LINE GRID TO HUDSON LINE-1	1.93	1.35
27	418	CIVIL LINE GRID	CIVIL LINE GRID TO PRESS ROAD OUTDOOR S/S FDR NO-2	2.08	3.36
28	411	CIVIL LINE GRID	CIVIL LINE GRID TO 6.6KV CHANDRAWAL WATER WORKS S/S NO-2 FDR NO.1	5.13	4.71
29	411	CIVIL LINE GRID	CIVIL LINE GRID TO MAZARWALA FEEDER NO.-1	3.21	2.07
30	501	WAZIRPUR - 2 GRID	WAZIRPUR - 2 GRID TO PRESS COMPLEX-2	3.20	1.08
31	502	WAZIRPUR - 1 GRID	WAZIRPUR - 1 GRID TO PETROL PUMP WPIA	2.88	2.40
32	417	SHAKTI NAGAR (GIS) GRID	SHAKTI NAGAR GRID TO TAXI STAND	1.35	0.58
33	425	GULABI BAGH GRID	GULABI BAGH GRID TO NIMRI VILLAGE	1.44	1.28
34	507	BADLI GRID	BADLI GRID TO TEACHERS COLONY	3.99	0.88
35	1302	PAYAL GRID	PAYAL GRID TO NIA PH-1, MARUTI O/D S/S	1.90	0.68
36	581	HAIDER PUR GRID	HAIDER PUR GRID TO 448 SFS NO-1	1.87	1.70
37	581	HAIDER PUR GRID	HAIDER PUR GRID TO 1/15 S/S	3.72	4.31
38	501	TRINAGAR GRID	TRINAGAR GRID TO BOOSTER PUMP FDR NO-2	1.68	1.71
39	418	DIFR GRID	DIFR GRID TO CEFEES	4.92	4.86
40	418	DIFR GRID	DIFR GRID TO INMASS	3.84	4.46
41	521	BAWANA-7 GRID	BAWANA GRID -7 TO SEC-5 BNO BLOCK	2.13	1.96
42	572	BAWANA-7 GRID	BAWANA-7 GRID TO HOLAMBI KHURD O/H CKT.	4.36	4.72
43	521	BAWANA-7 GRID	BAWANA GRID -7 TO RAW WATER WITH T-OFFS	4.64	0.62
44	572	BAWANA-7 GRID	BAWANA-7 GRID TO RADHA KRISHNA RICE MILL FDR-2	3.09	2.67
45	572	BAWANA-7 GRID	BAWANA-7 GRID TO KHEDA KHURD O/H CKT.	5.63	6.26
46	515	MANGOL PURI - 1 GRID	MANGOL PURI - 1 GRID TO S-BLOCK	3.11	1.78
47	413	INDRA VIHAR GRID	INDIRA VIHAR GRID TO MUKHERJEE NAGAR-3 S/S	1.81	0.83
48	516	BADLI GRID	BADLI GRID (NW) TO AMAR JYOTI	1.93	2.28
49	516	BADLI GRID	BADLI GRID (NW) TO BIA S/S NO-7 FDR NO-2	1.94	1.80

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
50	571	ROHINI - 5 GRID	ROHINI - 5 GRID TO 3/9 S/S	2.37	0.96
51	533	GHEVRA SAVDA GRID	GHEVRA GRID TO GARI RINDHALA FEEDER	3.68	1.45
52	533	GHEVRA SAVDA GRID	GHEVRA GRID TO IGL PUMP CNG STATION NO. 2 SAWDA	1.73	1.99
53	532	WAZIRABAD GRID	WAZIRABAD GRID TO JAGATPUR EXT (K)	4.41	0.86
54	1303	REWARI LINE GRID	REWARI LINE GRID TO D-BLOCK MS GARDEN	1.36	1.26
55	561	ROHINI - 1 GRID	ROHINI - 1 GRID TO S/S - 1/7	1.64	2.31
56	532	WAZIRABAD GRID	WAZIRABAD GRID TO HANUMAN CHOWK O/D S/S	1.92	2.87
57	501	RAMPURA GRID	RAMPURA GRID TO STRIPLEX	2.25	1.20
58	422	SHAHZADA BAGH GRID	SHAHZADA BAGH GRID TO SHAHZADA BAGH NO-5	3.64	1.87
59	422	SHAHZADA BAGH GRID	SHAHZADA BAGH GRID TO ANAND NAGAR (K)	1.53	2.34
60	561	ROHINI DC-1	ROHINI DC-1 GRID TO CITY CENTER MALL	1.49	1.82
61	1301	D.M.S. GRID	D.M.S. GRID TO PHILLIPS S/S MOTI NAGAR	1.38	0.65
62	517	TIGI PUR GRID	TIGIPUR GRID TO JHANGOLA VILLAGE	1.57	1.61
63	511	TIGI PUR GRID	TIGIPUR GRID TO HIRANKI FEEDER	5.37	5.67
64	581	ROHINI - 6 GRID	ROHINI - 6 GRID TO 13/15 (I/D) S/S	5.88	5.55
65	522	A-7 NARELA GRID	A-7 NARELA GRID TO S/S NO-31 & B G SHIRKE	4.26	4.10
66	1304	PUSA GRID	NPL	2.23	1.05
67	551	ROHINI - 1 GRID	ROHINI - 1 GRID TO 3/4 S/S	1.48	1.76
68	511	BHALSWA GRID	BHALSWA GRID TO NEW KAUSHIK ENCLAVE NO-1 S/S	2.04	0.76
69	511	BHALSWA GRID	BHALSWA GRID TO SWROOP VIHAR	3.35	0.64
70	414	BHALSWA GRID	BHALSWA GRID TO KATARIA FARM	1.31	0.74
71	505	JAHANGIR PURI GRID	JAHANGIRPURI GRID TO A BLK PARK WALA S/STN	1.48	3.53
72	517	TIGI PUR GRID	TIGI PUR GRID TO TIGI PUR GRID WALA S/S	1.41	1.56
73	517	TIGI PUR GRID	TIGIPUR GRID TO TIGIPUR VILLAGE (MCD PARK)	1.54	1.17
74	581	220KV SHALIMARBAGH GRID	220KV SMB GRID TO SEWAGE PUMPING STATION SEC-19 + CNG PUMP	1.92	0.93
75	561	220KV ROHINI GRID	220 ROHINI GRID TO DELHI ADMN.	4.07	1.59
76	516	220KV ROHINI GRID	220 ROHINI GRID TO SEC-26 NALA (O/D)	2.12	1.06
77	522	220KV NARELA GRID	220KV NARELA GRID TO S/S NO-2 A5 PKT-6	1.22	0.74
78	533	220KV KANJHAWLA	220KV KANJHAWALA GRID	3.68	2.82

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
		GRID	TO U/G KHANJHAWLA CHOWK		
79	533	220KV KANJHAWLA GRID	220KV KANJHAWLA GRID TO GHEVRA	4.90	4.28
80	533	220KV KANJHAWLA GRID	220KV KANJHAWLA GRID TO SOS BAWANA (MAJARA)	4.75	5.55
81	521	POOTH KHURD GRID	POOTH KHURD GRID TO OLD ZONAL OFFICE	4.63	1.77
82	503	SMB FACILITY CENTER GRID	SMB FC GRID TO INTERCONNECTOR 1&2 TO JAHANGIRPURI GRID S/S	1.45	2.69
83	422	SHAHZADA BAGH GRID	SHAHZADA BAGH GRID TO INDERLOK S/STN.	3.53	2.71
84	572	DSIDC NARELA-2 GRID	DSIDC NARELA-2 GRID TO ALIPUR -4	1.43	3.12
85	572	ROHINI- 28 GRID	RG 28 GRID TO JINDAL NIRYAT	2.88	1.58
86	572	ROHINI- 28 GRID	RG-28 TO DJB-1 KHERA KHURD	1.87	1.75
87	516	ROHINI- 28 GRID	RG 28 GRID TO DDA COMMAND TANK S/S	1.48	3.56
88	412	MODEL TOWN GRID	MODEL TOWN GRID TO OLD TAGORE PARK	2.47	3.17
89	502	ASHOK VIHAR GRID	ASHOK VIHAR GRID TO JA-BLOCK MURGA MARKET FEEDER NO-2	2.06	1.37
90	521	BAWANA-7 GRID	BAWANA GRID -7 TO D BLOCK SEC-5	1.41	1.03
91	571	PRASHANT VIHAR (GIS) GRID	PRASHANT VIHAR GRID TO 4/7 S/S FDR NO-1	2.95	2.21
92	414	BHALSWA GRID	BHALSWA GRID TO JINDAL FARM	2.00	1.93
93	503	BHALSWA GRID	BHALSWA GRID TO GURUNANAK DEV COLONY S/S	2.10	0.92
94	503	BHALSWA GRID	BHALSWA GRID TO GALI NO-12 SWAROOP NAGAR (O/D)	1.49	1.19
95	416	G.T.K. GRID	G.T.K. ROAD GRID TO RUB SANGAM PARK S/S	1.27	0.91
96	1303	SUDARSHAN PARK GRID	SUDARSHAN PARK GRID TO GADARIA	1.67	1.63
97	511	SGTN-2 (GIS) GRID	SGTN-2 GRID TO KADIPUR	1.38	2.92
98	507	SGTN-2 (GIS) GRID	SGTN-2 GRID TO MCD OFFICE	2.80	1.83
99	503	SGTN-2 (GIS) GRID	SGTN-2 GRID TO A-BLK SWAROOP NAGAR O/D S/S	2.66	2.69
100	511	SIRASPUR GRID	SIRASPUR GRID TO SDM OFFICE ALIPUR	2.55	1.82
101	507	SIRASPUR GRID	SIRASPUR GRID TO ESS-1	2.67	1.94
102	572	SIRASPUR GRID	SIRASPUR GRID TO NEHRU ENCLAVE	1.70	1.16
103	512	NARELA SUB CITY GRID	NARELA SUB CITY GRID TO MBS BAWANA S/S (O/D)	4.20	4.54
104	521	NARELA SUB CITY GRID	NARELA SUB CITY GRID TO L-BLOCK SEC-5	2.34	1.91

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
105	581	ROHINI - 6 GRID	ROHINI - 6 GRID TO 5/16 S/S FDR-1	3.03	2.40
106	516	ROHINI 30 GRID	RG 30 ROHINI GRID TO PRAHLADPUR CHOWK S/S NO-2	2.07	1.10
107	581	ROHINI 30 GRID	INTER CONNECTOR 3&4 RG-30 TO RG-6	3.61	2.89
108	510	RANI BAGH CC GRID	RANI BAGH CC GRID TO PURI BUILDER S/S	1.63	1.30
109	425	GULABI BAGH GRID	GULABI BAGH GRID TO STATUE PARK (K)	1.41	1.30
110	532	BURARI GRID	DJB BURARI GRID TO GHANTESHWAR MAHADEV S/S SANT NAGAR BURARI	2.10	1.01
111	533	KARALA (GIS) GRID	KARALA GRID TO CHANDPUR MAJARA S/S	2.75	2.66
112	533	KARALA (GIS) GRID	KARALA GRID TO LADPUR S/S	4.22	3.93
113	513	KARALA (GIS) GRID	KARALA GRID TO 11 KV O/G BHAGYA VIHAR FDR	2.76	1.58
114	513	KARALA (GIS) GRID	KARALA GRID TO ZONAL OFFICE KARALA FEEDER NO.1	4.15	2.45
115	518	ROHINI-20 (GIS) GRID	ROHINI-20 GRID TO P3 SULTANPURI	1.39	1.41
116	518	ROHINI-20 (GIS) GRID	ROHINI-20 GRID TO WORK CENTRE-2 SULTANPURI	1.88	8.87
117	519	ROHINI-20 (GIS) GRID	ROHINI-20 GRID TO PKT-16 SEC-20 S/S NO-4 (O/D)	1.41	0.94
118	581	ROHINI - 6 GRID	ROHINI-6 GRID TO 1/17 S/S	3.21	1.92
119	417	SWISS APARTMENT (GIS) GRID	SWISS APARTMENT GRID TO NARMADA SWITCHING S/S (O/D)	1.40	1.43
120	418	SWISS APARTMENT (GIS) GRID	SWISS APARTMENT GRID TO NEW ST. STEPHENS HOSPITAL	1.50	1.33
121	521	BAWANA-4 (GIS) GRID	BAWANA-4 GRID TO MANGALAM HOSPITAL	5.62	1.10
122	521	BAWANA-4 (GIS) GRID	BAWANA-4 GRID TO I BLOCK SEC-4	1.49	0.59
123	503	DHEERPUR GRID	DHEERPUR GRID TO INDERPAHALWAN NO-1 S/S	1.29	1.45
124	532	DHEERPUR GRID	DHEERPUR GRID TO PANCHSHEEL ASHRAM NO.-4 FDR-2	1.50	0.76
125	512	BAWANA-1 66kV	BAWANA-1 GRID TO DSIIDC INFRA OFFICE S/S (O/D)	2.30	1.58
126	505	MODEL TOWN GRID	MODEL TOWN GRID TO E BLOCK MKT S/S	2.11	2.59
127	561	ROHINI - 4 GRID	ROHINI - 4 GRID TO NAALEWALA	2.36	1.66
128	1302	PAYAL GRID	PAYAL GRID TO C-142 NIA PH-1 S/S	1.32	1.60
129	519	ROHINI - 23 GRID	ROHINI 23 GRID TO MUSKAN HOSPITAL O/D S/S	3.26	0.83
130	519	ROHINI - 23 GRID	ROHINI-23 GRID TO 4 WAY RMU NEAR SHAMSHAAN	3.04	0.59

SI No.	Zone	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
			GHAT		
131	523	ROHINI - 22 GRID	ROHINI - 22 GRID TO NITHARI	2.50	3.14
132	519	ROHINI - 22 GRID	ROHINI - 22 GRID TO RMU NEAR RG-22 GRID	1.39	3.82
133	513	ROHINI - 22 GRID	ROHINI - 22 GRID TO AGAR NAGAR FEEDER	1.71	3.57

b) Validation of energy flow data and losses: Based on field survey and as per following sample size:

- **Checking functional and communication status of consumers meters:** Verification of metering and connection status of min. of 10 or 2% consumers of the Division (whichever is higher) of various category of consumers – Agriculture (Metered and Un-metered), Govt. category connection (and LT Industrial etc.

1% of input energy points at (T-D) interface

Table 18: 1% Input Energy Points

Exchange point meters	Consumption (FY 23-24)	Consumption (FY 24-25)	MF
Narela T-1	302.25	328.37	1000
Narela T-2	329.70	359.51	1000
Narela T-3	195.80	212.78	1000
BWN T-2	174.43	184.45	1000
BWN T-3	236.06	255.28	333.33

Data of at-least 1/3rd of Divisions: -

Min 10 nos. of input energy points

Table 19: Data of Rohini Division

SI No.	Zone	District	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
1	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 1/16 (I/D) S/S	0.41	0.43
2	561	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO SWARN JYANTI PARK STREET LIGHT NO.1	0.46	0.33
3	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 17/16 S/S	1.09	0.57
4	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 320 LIG FLATS SEC-17	0.46	0.56
5	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 21/16 S/S	0.64	1.50
6	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 13/15 (I/D) S/S	5.88	5.55
7	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 11/16 S/S	0.27	0.76
8	581	Rohini	ROHINI - 6 GRID	ROHINI - 6 GRID TO 5/16 S/S FDR-1	3.03	2.40
9	581	Rohini	ROHINI - 6 GRID	ROHINI-6 GRID TO 1/17 S/S	3.21	1.92
10	571	Rohini	ROHINI - 3 GRID	ROHINI - 3 GRID TO 2/9 S/S FDR NO-2	1.23	0.29

Table 20: Data of Mangolpuri Division

SI No.	Zone	District	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
1	519	Mangolpuri	ROHINI - 23 GRID	ROHINI - 23 GRID TO PKT-16 SEC-20 S/S NO-1	1.19	1.67
2	519	Mangolpuri	ROHINI - 2 GRID	ROHINI - 2 GRID TO RMU OUTDOOR S/S	4.89	2.51
3	515	Mangolpuri	MANGOL PURI - 1 GRID	MANGOL PURI - 1 GRID TO S/S NO 15 FDR NO.1+ B BLK MGP IND.AREA PH-1 S/S	3.05	4.65
4	515	Mangolpuri	MANGOL PURI - 1 GRID	NDDDB	2.50	1.04
5	515	Mangolpuri	P. P. - 2 GRID	P. P - 2 GRID TO N-BLOCK FDR-2	1.87	3.77
6	515	Mangolpuri	MANGOL PURI - 1 GRID	MANGOL PURI - 1 GRID TO S-BLOCK	3.11	1.78
7	519	Mangolpuri	ROHINI - 24 GRID	ROHINI - 24 GRID TO RMU NEAR RG-24 S/S	7.60	0.38
8	519	Mangolpuri	ROHINI - 23 GRID	ROHINI 23 GRID TO MUSKAN HOSPITAL O/D S/S	3.26	0.83
9	519	Mangolpuri	ROHINI - 23 GRID	ROHINI - 23 GRID TO BEGUMPUR BUS STAND (K)	1.67	0.18
10	519	Mangolpuri	ROHINI - 23 GRID	ROHINI-23 GRID TO 4 WAY RMU NEAR SHAMSHAAN GHAT	3.04	0.59

Table 21: Data of Pitampura Division

SI No.	Zone	District	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
1	510	Pitam Pura	P. P. - 2 GRID	P. P. - 2 GRID TO LSC MKT.SAINIK VIHAR	1.54	1.61
2	504	Pitam Pura	P. P. - 2 GRID	P. P. - 2 GRID TO U.P. SAMAJ FDR NO- 1	1.14	1.81
3	504	Pitam Pura	ROHINI - 3 GRID	ROHINI - 3 GRID TO DUDIYAL APPT.	1.37	0.29
4	504	Pitam Pura	RANI BAGH GRID	RANI BAGH GRID TO KAILAH ENCLAVE (Z-504)+JHULA LAL NO-1	4.36	5.07
5	530	Pitam Pura	P.P.-3 GRID	P.P.-3 GRID TO MERA BABA FEEDER-2	1.90	2.28
6	510	Pitam Pura	P.P.-3 GRID	P.P.-3 GRID TO RMU INSIDE PP-3 GRID	1.84	4.37
7	508	Pitam Pura	WAZIRPUR - 2 GRID	WAZIRPUR - 2 GRID TO KASTURBA POLYTECHNIC	1.01	1.70
8	530	Pitam Pura	P P - 1 GRID	P P - 1 GRID TO KOHAT ENCLAVE	1.68	0.27
9	508	Pitam Pura	220KV SHALIMARBAG H GRID	220KV SMB GRID TO PETROL PUMP	1.21	0.26
10	504	Pitam Pura	MANGOL PURI - 2 GRID	MANGOL PURI - 2 GRID TO A 40 MIA PH-2	3.27	0.26

Table 22: Data of Keshav Puram Division

SI No.	Zone	District	Name of the Station	Feeder Name	FY 23-24 T&D losses (MU)	FY 24-25 T&D losses (MU)
1	520	Keshav Puram	RAMPURA GRID	RAMPURA GRID TO ONKAR NAGAR-B	1.85	1.70
2	501	Keshav Puram	RAMPURA GRID	RAMPURA GRID TO INTERCONNECTOR	1.88	2.58
3	501	Keshav Puram	WAZIRPUR - 2 GRID	WAZIRPUR - 2 GRID TO PRESS COMPLEX-2	3.20	1.08
4	502	Keshav Puram	WAZIRPUR - 1 GRID	WAZIRPUR - 1 GRID TO PETROL PUMP WPIA	2.88	2.40
5	502	Keshav Puram	ASHOK VIHAR GRID	ASHOK VIHAR GRID TO J.J. COLONY (F BLK)	1.28	0.38
6	502	Keshav Puram	ASHOK VIHAR GRID	ASHOK VIHAR GRID TO RAMLILA GROUND ASHOK VIHAR	1.09	1.30
7	502	Keshav Puram	ASHOK VIHAR GRID	ASHOK VIHAR GRID TO MOTHER DAIRY JJ COLONY	2.08	0.26
8	501	Keshav Puram	TRINAGAR GRID	TRINAGAR GRID TO BOOSTER PUMP FDR NO-2	1.68	1.71
9	501	Keshav Puram	RAMPURA GRID	RAMPURA GRID TO STRIPLEX	2.25	1.20
10	509	Keshav Puram	WAZIRPUR - 1 GRID	WAZIRPUR - 1 GRID TO C - 33 WPIA	1.04	2.17

c) Field verification report of the activities undertaken in a) and b) above to be included as an annexure to the energy audit report.

- Field visit of Substations, DTs and commercial building was done during the audit time.
- It was also observed that 100% of the consumers are metered.
- At the time of field survey, it was found that HT feeder meters are communicable.
- It was also observed that average power factor was found in the range of 0.96 to 0.99.
- It was also observed that hourly load parameters are monitored and energy consumption is monitored at every 24hrs.

4.3. Validation of energy flow data and losses

Voltage wise power purchase details are not available, as the invoice of power purchase contains only units purchased and details of power plant voltage not available.

Voltage wise energy flow data is not available, Inputs at different voltage levels are not available.

Majority of feeders are common to LT & HT. So, input energy supplied is inseparable.

Cumulated EHT sales MUs for 33kV and above are maintained and therefore bifurcation of same into 66 kV and 33 kV is not available.

5. Loss and subsidy computation

5.1. Energy account analysis for previous year

Performance comparison is done for last 3 years FY 21-22, FY 22-23 & FY23-24 to find, where the DISCOM has improved and where it needs improvement, etc.

a) Summary of AT&C losses for previous years

The AT&C losses for the FY 2021-22, FY 2022-23, FY 2023-24 are as shown below:

Table 23: AT&C losses of previous three years

Technical Details	UoM	FY 2021-22	FY 2022-23	FY 2023-24
Input Energy Purchase (From Generation Source)	Million kWh	11737.11	12741.47	12686.10
Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	Million kWh	9424.75	10622.62	10660.43
Total Energy billed (is the Net energy billed, adjusted for energy traded))	Million kWh	8752.21	9946.33	10028.15
Transmission and Distribution (T&D) loss Details	Million kWh	672.55	676.29	632.28
	%	7.14%	6.37%	5.93%
Collection Efficiency	%	100.36%	100%	100.01%
Aggregate Technical & Commercial Loss	%	6.80%	6.36%	5.92%

Figure 9: Comparison of Losses – Last 3 FYs – 21-22/22-23/23-24

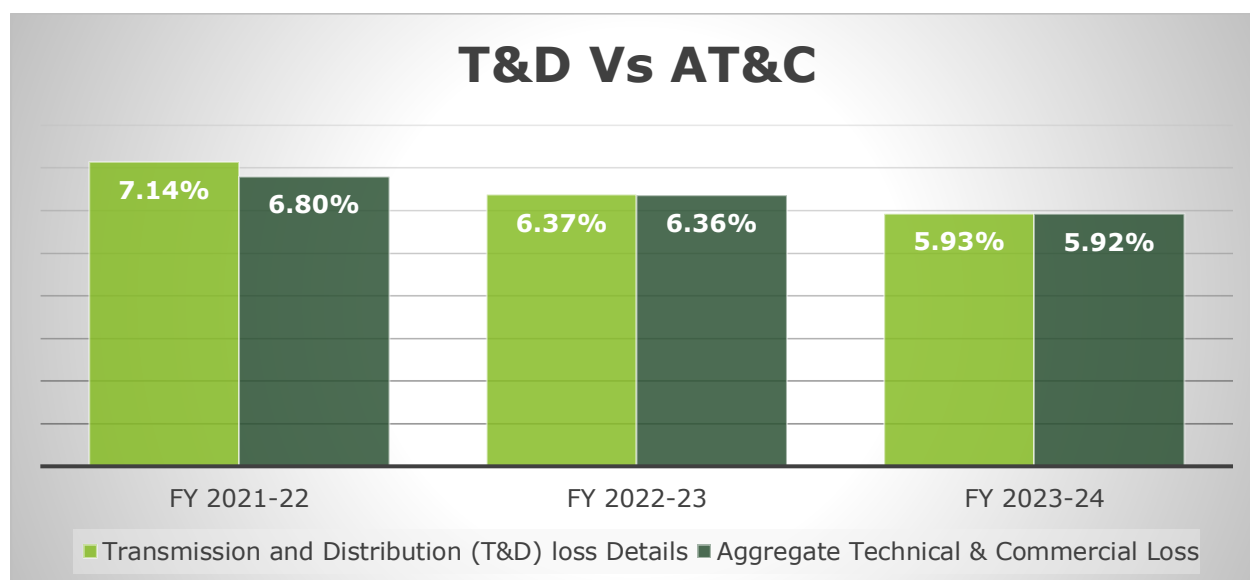


Table shows trend of T&D loss, Collection efficiency and AT&C loss for last 3 years. T&D and AT&C losses have reduced as compare to previous year.

b) Division Wise Comparison

Division wise input Energy, Billed Energy, T&D losses and AT&C losses is given in following table:

Table 24: Division wise Input and Billed Energy FY 2022-23 & 2023-24

Name of Division	FY-2022-23		FY-2023-24	
	Input Energy (MUs)	Billed Energy (MUs)	Input Energy (MUs)	Billed Energy (MUs)
Badli	697.77	640.87	720.57	663.52
Bawana	1457.23	1323.2	1466.37	1352.51
Civil Lines	903.66	889.91	890.25	871.21
Keshav Puram	886.18	838.01	873.94	832.66
Mangolpuri	698.77	645.64	695.58	656.23
Model Town	755.87	717.99	749.44	709.17
Moti Nagar	847.33	803.81	831.61	788.4
Narela	1231.63	1128.6	1274.31	1176.54
Pitam Pura	677.68	645.99	662.42	629.08
Rohini	1161.64	1112.3	1153.93	1103.4
Kirari	437.49	384.56	446.88	400.3
Shalimar Bagh	867.37	815.48	895.11	845.13
Total	10622.62	9946.33	10660.43	10028.15

Table 25: Division wise T&D and AT&C losses in FY 2022-23 & FY 2023-24

Name of Division	FY-2022-23		FY-2023-24	
	T&D LOSS (%)	AT & C LOSS (%)	T&D LOSS (%)	AT & C LOSS (%)
Badli	8.15%	7.80%	7.92%	8.59%
Bawana	9.20%	8.90%	7.76%	7.66%
Civil Lines	1.52%	2.47%	2.14%	2.33%
Keshav Puram	5.44%	6.35%	4.72%	4.31%
Mangolpuri	7.60%	7.21%	5.66%	5.58%
Model Town	5.01%	5.03%	5.37%	5.42%
Moti Nagar	5.14%	4.91%	5.20%	5.39%
Narela	8.36%	8.34%	7.67%	7.82%
Pitam Pura	4.68%	4.47%	5.03%	5.13%
Rohini	4.25%	4.08%	4.38%	4.22%
Kirari	12.10%	11.83%	10.42%	8.24%
Shalimar Bagh	5.98%	5.83%	5.58%	5.86%
At DISCOM level	6.37%	6.36%	5.93%	5.92%

5.2. Energy accounts analysis and performance in current year (based on quarterly data)

5.2.1. Circle wise Consumer Count, Connected Load, Input Energy, Billed Energy, T&D losses & AT&C loss%

Table 26: Circle wise Comparison for FY 2024-25

Name of circle	Total Number of connections (Nos)	Total Connected Load (MW)	Input Energy (MU)	Total Billed energy (MU)	T&D loss (MU)	T&D loss (%)	AT & C loss (%)
URBAN CIRCLE	684028	1903.39	3016.82	2842.40	174.422	5.78%	5.73%
SUB-URBAN CIRCLE	256491	1434.31	2855.01	2636.87	218.145	7.64%	7.60%
TOWN CIRCLE	476395	1698.85	2699.52	2593.74	105.779	3.92%	3.86%
METRO CIRCLE	532277	1310.44	1966.86	1869.16	97.705	4.97%	4.99%
CITY CIRCLE	153646	629.588	898.45	852.99	45.45	5.06%	4.75%
At DISCOM level	2102837	6976.57	11436.7	10795.16	641.51	5.61%	5.55%

5.2.2. Division wise Input energy and billed energy details

The Division wise input energy & billed energy for FY 2024-25 is shown in below table.

Table 27: Division wise input energy & billed energy for FY 2024-25

S. No	Name of Division	Input energy (MU)	% of Input energy consumption	Total Billed energy (MUs)	% of energy consumption
1	Badli	768.72	6.7%	702.19	6.5%
2	Bawana	1527.02	13.4%	1408.36	13.0%
3	Civil lines	937.55	8.2%	917.40	8.5%
4	Keshavpuram	957.20	8.4%	912.38	8.5%
5	Kirari	502.40	4.4%	469.04	4.3%
6	Mangolpuri	755.68	6.6%	720.09	6.7%
7	Model town	804.77	7.0%	763.96	7.1%
8	Moti nagar	898.45	7.9%	852.99	7.9%
9	Narela	1327.99	11.6%	1228.51	11.4%
10	Pitam pura	708.78	6.2%	680.03	6.3%
11	Rohini	1252.40	11.0%	1203.15	11.1%
12	Shalimar bagh	995.71	8.7%	937.06	8.7%
		11436.67	100%	10795.16	100%

Figure 10: Division wise Input Energy Share

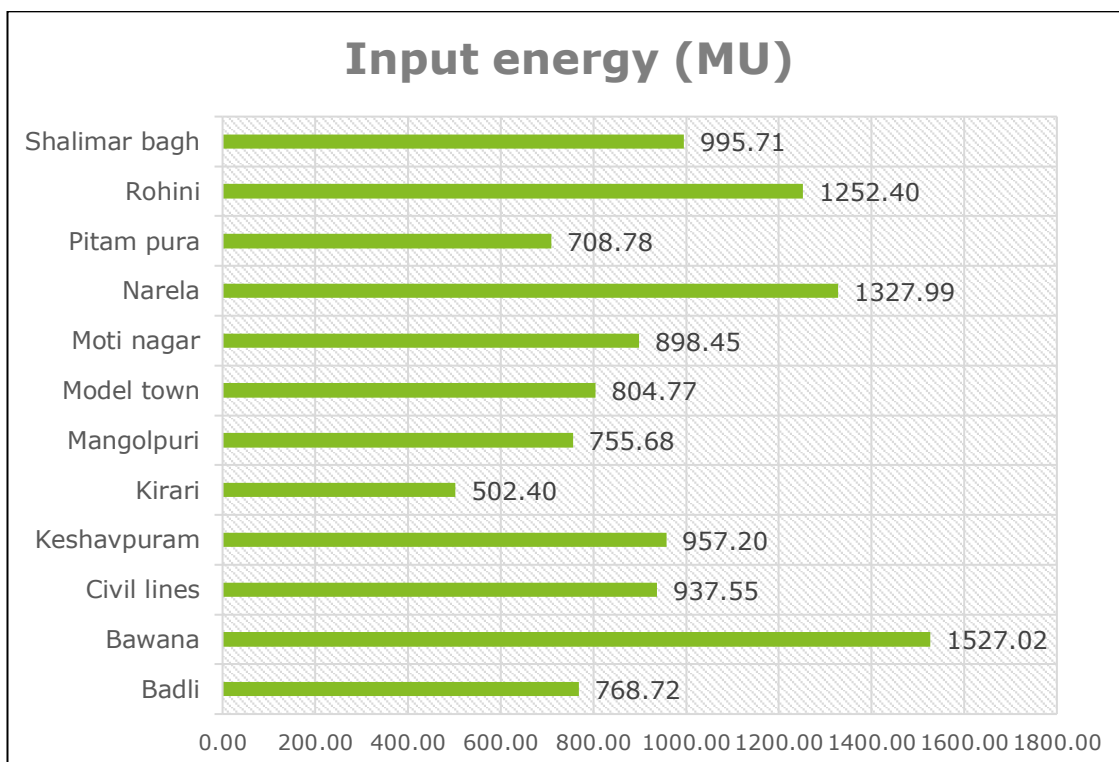
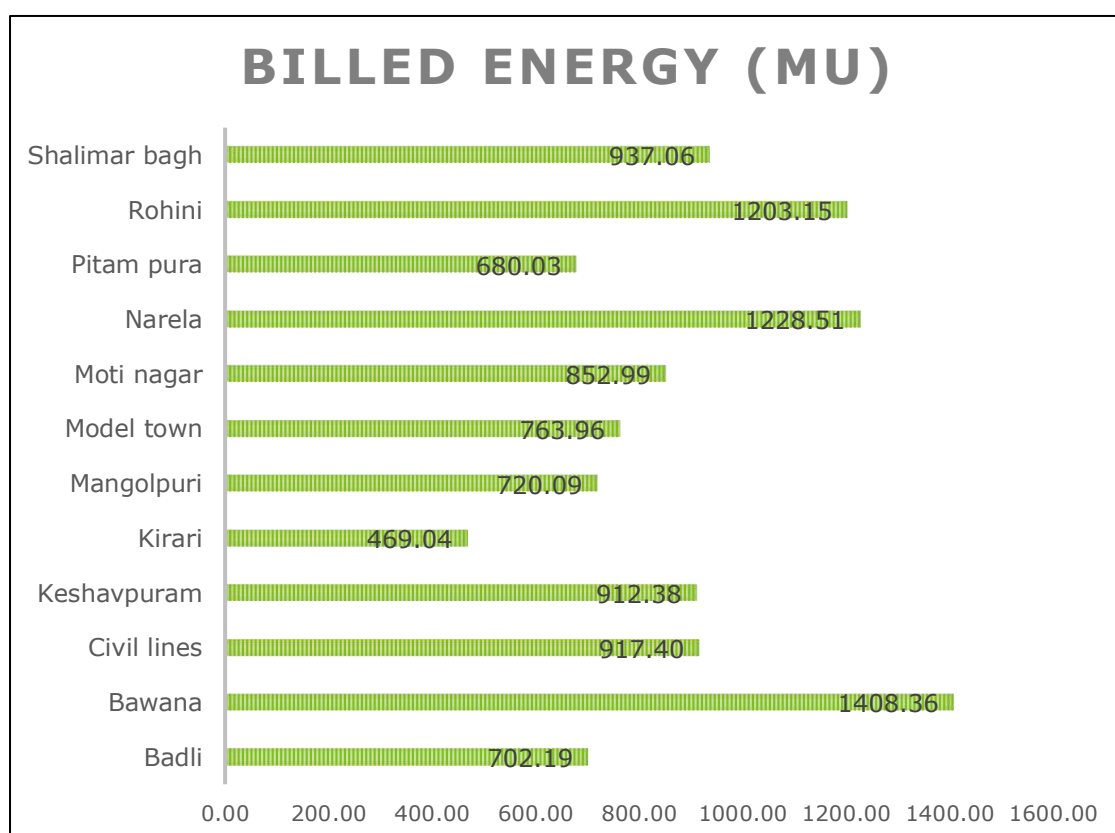


Figure 11: Division wise Input Energy Share



5.2.3. Quarterly and annual AT&C losses

The Quarter wise and annual AT&C losses for FY 2024-25 are shown in below table.

Table 28: Energy Input and AT&C Losses for FY 2024-25

Energy Input Details	Formula	UoM	Quarterly				Annual
			Q1	Q2	Q3	Q4	FY 24-25
Input Energy Purchase (From Generation Source)	A	MU	3875.7	3954.89	2641.14	2640.99	13094.00
Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	B	MU	3360.9	3476.33	2417.59	2181.85	11436.67
Total Energy billed (is the Net energy billed, adjusted for energy traded)	C	MU	2929.5	3300.18	2466.18	2099.18	10795.16
Transmission and Distribution (T&D) loss Details	D	MU	431.40	176.15	-48.59	82.67	641.51
	$E = D/B \times 100$	%	12.84%	5.07%	-2.01%	3.79%	5.61%
Collection Efficiency	F	%	88.96%	104.11%	105.65%	103.00%	100.06%
Aggregate Technical & Commercial Loss	$G = 1 - \{(1-E) \times \text{Min}(F, 100\%)\}$	%	22.46%	1.17%	-7.77%	0.90%	5.55%

5.2.4. Voltage wise AT&C losses

Voltage wise power purchase details are not available, as the invoice of power purchase contains only units purchased and details of power plant voltage not available.

Majority of feeders are common to LT & HT. So input energy supplied is inseparable.

Cumulated EHT sales MUs for 33kV and above are maintained and therefore bifurcations of same are not available.

The voltage wise AT&C losses of Tata Power-DDL for FY 2024-25 are as shown in the below table:

Table 29: Voltage-wise AT&C Losses for FY 2024-25

S. No.	Particulars	Units	Values
1	Losses in 132 KV System and Connected Equipment		
1.a.	Total Energy delivered into 132 KV Distribution System from EHT SSs	MUs	-
1.b.	Energy consumed by HT consumers at 132KV (Sales + Third Party)	MUs	-
1.c.	Energy Delivered to lower voltage	MUs	-
1.d.	Losses (132 kV System)	MUs	-
1.e.	% Losses (132 kV System)	%	-
2	Losses in 33 KV System and Connected Equipment		-
2.a.	Total Energy delivered into 33 KV Distribution System from EHT SSs	MUs	-
2.b.	Energy consumed by HT consumers at 33KV (Sales + Third Party)	MUs	-

S. No.	Particulars	Units	Values
2.c.	Energy Delivered into 11 KV and LT System from 33/11 KV SSs	MUs	-
2.d.	Losses (33 kV System)	MUs	-
2.e.	% Losses (33 kV System)	%	-
3	Losses in 11 KV System and Connected Equipment		-
3.a.	Total Energy delivered into 11 KV and LT Distribution System	MUs	-
3.b.	Energy consumed by HT consumers at 11KV (Sales + Third Party)	MUs	-
3.c.	Total Output from 11kV to LT	MUs	-
3.d.	Losses (11kV System)	MUs	-
3.e.	% Losses (11kV System)	%	-
4	Losses in LT system and connected equipment		-
4.a.	Energy delivered to LT system from 11/400 V DTRs	MUs	-
4.b.	Energy sold at LT level	MUs	-
4.c.	Losses (LT System)	MUs	-
4.d.	% Losses (LT System)	%	-
5	Total losses in the Distribution System		-
5.a.	Total Input to the distribution system	MUs	11436.67
5.b.	Total Output from the Distribution System	MUs	10795.16
5.c.	Distribution System Losses	%	641.51
5.d.	% Distribution System Losses	%	5.61%

5.2.5. Circle wise AT&C losses analysis

1. Circle wise connections & energy consumptions for FY 2024-25

The circle wise connections, load, input energy & Billed energy with percentage share for different circles is given below.

"Urban Circle" having maximum numbers of consumers and "City Circle" having minimum numbers of consumers. "Urban Circle" has maximum input energy as well as billed units and "City Circle" have minimum input energy as well as billed units as shown in table:

Table 30: Circle wise consumers, connected load, Input and Billed Energy in FY 2024-25

Circle	Total Number of connections		Total Connected Load		Input energy		Billed energy	
	Nos.	% Share	MW	% Share	MU	% Share	MU	% Share
URBAN CIRCLE	684028	33%	1903.39	27%	3016.82	26%	2842.40	26%
SUB-URBAN CIRCLE	256491	12%	1434.31	21%	2855.01	25%	2636.87	24%
TOWN CIRCLE	476395	23%	1698.85	24%	2699.52	24%	2593.74	24%
METRO CIRCLE	532277	25%	1310.44	19%	1966.86	17%	1869.16	17%

Circle	Total Number of connections		Total Connected Load		Input energy		Billed energy	
	Nos.	% Share	MW	% Share	MU	% Share	MU	% Share
CITY CIRCLE	153646	7%	629.588	9%	898.45	8%	852.99	8%
Total	2102837	100%	6976.57	100%	11436.67	100%	10795.16	100%

2. Circle-wise AT&C losses

The circle wise AT&C losses are shown in the table below:

Table 31: Circle wise T&D losses, Collection Efficiency and AT&C losses for FY 2024-25

Name of Circle	Input energy (MU)	Billed energy (MU)	T&D loss		Collection Efficiency (%)	AT& C loss (%)
			(MU)	(%)		
URBAN CIRCLE	3016.82	2842.40	174.422	5.78%	100.06%	5.73%
SUB-URBAN CIRCLE	2855.01	2636.87	218.145	7.64%	100.04%	7.60%
TOWN CIRCLE	2699.52	2593.74	105.779	3.92%	100.06%	3.86%
METRO CIRCLE	1966.86	1869.16	97.705	4.97%	99.97%	4.99%
CITY CIRCLE	898.45	852.99	45.45	5.06%	100.33%	4.75%
Total	11436.67	10795.16	641.51	5.61%	100.06%	5.55%

Note: AT&C Loss's calculation as per BEE proforma the calculation is as per actual Collection efficiency.



Division wise AT&C losses analysis

3. Division-wise AT&C losses

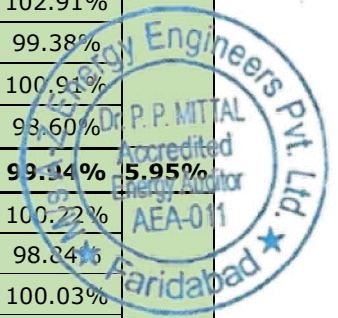
The Division wise AT&C losses are shown in the table below:

Table 32: Division wise T&D losses, Collection Efficiency and AT&C losses for FY 2024-25

S. No	Name of circle	Name of Division	Consumer category	Total Number of connections (Nos)	Total Connected Load (MW)	Input energy (MU)	Total Billed energy (MU)	T&D loss (MU)	T&D loss (%)	Billed Amount in Rs. Crore	Collected Amount in Rs. Crore	Collection Efficiency	AT & C loss (%)
1	Urban	Badli	Residential	119488	203.114	768.7	300.46	66.53	8.65%	198.46	198.87	100.21%	
			Agricultural	475	3.991		2.06			1.52	1.51	99.44%	
			Commercial/Industrial-LT	23334	222.39		362.52			578.88	579.03	100.03%	
			Commercial/Industrial-HT	45	11.407		22.55			33.76	33.81	100.13%	
			Others	673	9.119		14.60			18.20	18.03	99.02%	
				144015	450.021	768.7	702.19	66.53	8.65%	830.83	831.25	100.05%	8.61%
2	Sub- Urban	Bawana	Residential	76324	133.707	1527.0	185.76	118.66	7.77%	121.23	121.80	100.47%	
			Agricultural	1701	10.473		3.60			3.47	3.49	100.54%	
			Commercial/Industrial-LT	23087	593.371		1170.71			1772.82	1772.46	99.98%	
			Commercial/Industrial-HT	38	9.892		17.20			25.39	25.26	99.49%	
			Others	425	10.19		31.10			34.83	35.02	100.53%	
				101575	757.633	1527.0	1408.36	118.66	7.77%	1957.74	1958.03	100.01%	7.76%
3	Town	Civil lines	Residential	114931	316.495	937.5	444.95	20.14	2.15%	350.95	352.06	100.32%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	24955	112.363		140.15			236.82	236.61	99.91%	
			Commercial/Industrial-HT	73	56.115		94.61			151.58	151.72	100.09%	
			Others	946	81.792		237.68			251.58	248.19	98.65%	
				140905	566.765	937.5	917.40	20.14	2.15%	990.92	988.58	99.76%	2.38%
4	Town	Keshavpuram	Residential	127833	304.102	957.2	419.24	44.82	4.68%	309.48	309.93	100.15%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	28302	238.037		370.74			587.55	588.46	100.15%	
			Commercial/Industrial-HT	100	33.617		69.92			97.57	97.29	99.71%	
			Others	678	23.574		52.47			40.45	42.00	103.82%	
				156913	599.33	957.2	912.38	44.82	4.68%	1035.06	1037.68	100.25%	4.44%
5	Metro	Kirari	Residential	154326	219.15	502.4	351.99	33.36	6.64%	207.76	207.31	99.78%	

S. No	Name of circle	Name of Division	Consumer category	Total Number of connections (Nos)	Total Connected Load (MW)	Input energy (MU)	Total Billed energy (MU)	T&D loss (MU)	T&D loss (%)	Billed Amount in Rs. Crore	Collected Amount in Rs. Crore	Collection Efficiency	AT & C loss (%)
			Agricultural	58	0.33		0.20			0.13	0.13	102.10%	
			Commercial/Industrial-LT	18071	54.351		72.22			119.81	119.34	99.61%	
			Commercial/Industrial-HT	8	2.489		3.73			7.20	7.20	99.89%	
			Others	454	19.274		40.90			38.47	36.84	95.75%	
				172917	295.594	502.4	469.04	33.36	6.64%	373.37	370.82	99.32%	7.28%
6	Metro	Mangolpuri	Residential	207618	348.769		526.06			326.27	326.87	100.18%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	24409	109.321	755.7	153.82	35.60	4.71%	245.71	244.95	99.69%	
			Commercial/Industrial-HT	24	9.684		22.64			31.21	31.02	99.38%	
			Others	665	7.985		17.57			21.68	21.42	98.77%	
				232716	475.759	755.7	720.09	35.60	4.71%	624.88	624.25	99.90%	4.81%
7	Town	Model Town	Residential	155018	364.445		509.38			380.43	381.97	100.41%	
			Agricultural	3	0.007		0.01			0.00	0.00	107.89%	
			Commercial/Industrial-LT	22667	111.635	804.8	139.83	40.81	5.07%	234.27	234.66	100.17%	
			Commercial/Industrial-HT	44	13.398		24.71			40.60	41.23	101.55%	
			Others	845	43.268		90.03			97.74	96.59	98.83%	
				178577	532.753	804.8	763.96	40.81	5.07%	753.03	754.45	100.19%	4.89%
8	City	Moti Nagar	Residential	122887	328.649		448.52			344.66	345.45	100.23%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	29819	232.575	898.4	297.77	45.45	5.06%	506.07	507.92	100.37%	
			Commercial/Industrial-HT	116	50.942		83.42			129.04	128.32	99.44%	
			Others	824	17.422		23.29			25.02	26.40	105.49%	
				153646	629.588	898.4	852.99	45.45	5.06%	1004.79	1008.09	100.33%	4.75%
9	Sub- Urban	Narela	Residential	131134	228.041		331.24			218.80	218.70	99.95%	
			Agricultural	2164	19.234		9.84			7.30	7.13	97.73%	
			Commercial/Industrial-LT	20493	377.512	1328.0	767.11	99.48	7.49%	1136.92	1137.95	100.09%	
			Commercial/Industrial-HT	131	33.154		90.03			126.46	126.57	100.09%	

S. No	Name of circle	Name of Division	Consumer category	Total Number of connections (Nos)	Total Connected Load (MW)	Input energy (MU)	Total Billed energy (MU)	T&D loss (MU)	T&D loss (%)	Billed Amount in Rs. Crore	Collected Amount in Rs. Crore	Collection Efficiency	AT & C loss (%)
			Others	994	18.735		30.28			40.39	40.78	100.95%	
				154916	676.676	1328.0	1228.51	99.48	7.49%	1529.87	1531.13	100.08%	7.41%
10	Metro	Pitam Pura	Residential	107690	377.413	708.8	459.98	28.75	4.06%	380.00	381.40	100.37%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	18223	115.811		145.25			241.58	242.20	100.26%	
			Commercial/Industrial-HT	78	37.788		61.64			91.29	91.14	99.84%	
			Others	653	8.074		13.16			14.60	15.47	105.93%	
				126644	539.086	708.8	680.03	28.75	4.06%	727.47	730.21	100.38%	3.69%
11	Urban	Rohini	Residential	214196	576.378	1252.4	780.86	49.25	3.93%	584.96	586.43	100.25%	
			Agricultural	0	0		0.00			0.00	0.00	0.00%	
			Commercial/Industrial-LT	32368	137.111		163.14			270.73	271.25	100.19%	
			Commercial/Industrial-HT	68	47.382		93.46			137.76	138.20	100.32%	
			Others	1088	50.574		165.71			169.71	168.95	99.55%	
				247720	811.445	1252.4	1203.15	49.25	3.93%	1163.16	1164.82	100.14%	3.79%
12	Urban	Shalimar bagh	Residential	261135	481.872	995.7	693.09	58.65	5.89%	455.90	456.74	100.18%	
			Agricultural	70	0.484		0.43			0.21	0.22	102.91%	
			Commercial/Industrial-LT	30184	113.836		153.78			252.59	251.02	99.38%	
			Commercial/Industrial-HT	58	27.276		54.20			77.16	77.86	100.91%	
			Others	846	18.452		35.56			35.46	34.97	98.60%	
				292293	641.92	995.7	937.06	58.65	5.89%	821.33	820.81	99.94%	5.95%
			Residential	1792580	3882.135	11436.7	5451.54	641.51	5.61%	3878.90	3887.54	100.22%	
			Agricultural	4471	34.519		16.14			12.63	12.49	98.84%	
			Commercial/Industrial-LT	295912	2418.313		3937.02			6183.75	6185.86	100.03%	
			Commercial/Industrial-HT	783	333.144		638.11			949.02	949.61	100.06%	
			Others	9091	308.459		752.35			788.15	784.64	99.55%	
	Total			2102837	6976.57	11436.7	10795.16	641.51	5.61%	11812.46	11820.14	100.06%	5.55%



4. Division wise AT&C loss

1. Division wise connections & energy consumptions for FY 2024-25

Division wise connections, connected load, input energy & Billed energy with percentage share in different division is given below. "Shalimar Bagh" Division having maximum numbers of consumers and "Bawana" Division having minimum numbers of consumers. "Bawana" has maximum input energy as well as billed units and "Kirari" has minimum input energy as well as billed units as shown in table:

Table 33: Divisions wise connections & energy consumptions

Name of Division	Total Number of connections		Total Connected Load		Input energy		Total Billed energy	
	Number	%	Load (MW)	%	MU	%	MU	%
Badli	144015	6.8%	450.021	6.5%	768.72	6.7%	702.19	6.5%
Bawana	101575	4.8%	757.633	10.9%	1527.02	13.4%	1408.36	13.0%
Civil lines	140905	6.7%	566.765	8.1%	937.55	8.2%	917.40	8.5%
Keshavpuram	156913	7.5%	599.33	8.6%	957.20	8.4%	912.38	8.5%
Kirari	172917	8.2%	295.594	4.2%	502.40	4.4%	469.04	4.3%
Mangolpuri	232716	11.1%	475.759	6.8%	755.68	6.6%	720.09	6.7%
Model town	178577	8.5%	532.753	7.6%	804.77	7.0%	763.96	7.1%
Moti nagar	153646	7.3%	629.588	9.0%	898.45	7.9%	852.99	7.9%
Narela	154916	7.4%	676.676	9.7%	1327.99	11.6%	1228.51	11.4%
Pitam pura	126644	6.0%	539.086	7.7%	708.78	6.2%	680.03	6.3%
Rohini	247720	11.8%	811.445	11.6%	1252.40	11.0%	1203.15	11.1%
Shalimar bagh	292293	13.9%	641.92	9.2%	995.71	8.7%	937.06	8.7%
Total	2102837	100%	6976.57	100%	11436.67	100%	10795.16	100%

Table 34: Divisions wise AT&C losses (%) & T&D losses (%)

S.No.	Division	T&D loss (%)	AT&C loss (%)
1	Badli	8.65%	8.61%
2	Bawana	7.77%	7.76%
3	Civil lines	2.15%	2.38%
4	Keshavpuram	4.68%	4.44%
5	Kirari	6.64%	7.28%
6	Mangolpuri	4.71%	4.81%
7	Model town	5.07%	4.89%
8	Moti nagar	5.06%	4.75%
9	Narela	7.49%	7.41%
10	Pitam pura	4.06%	3.69%
11	Rohini	3.93%	3.79%
12	Shalimar bagh	5.89%	5.95%

Further, it was observed that the collection efficiency is nearly equal to 100% in all the Divisions as shown below: The collection efficiency recorded greater than 100% due to seasonal variation and legacy cases.

Figure 11: Division wise Billed Collection efficiency (%)



5.2.6. Feeder wise AT&C losses analysis

1. Feeder wise AT&C losses

Tata Power-DDL has ring main power distribution network. To provide uninterrupted power, back feeding points/normal operating points (NOP) are usually changed. Due to this dynamic nature of the network, connected consumers to a feeder at a particular point of time gets changed as the NOP changes. Tata Power-DDL is working to implement technological solutions to capture these dynamic changes.

2. High loss feeders

If we compare input energy of feeder and related consumption, in many cases, input is low as compared to consumption. Vis-a- Vis, in some feeders, gap is high on positive side.

Though critical parameters are monitored on monthly basis, substantial difference in the data will not be reflected on quarterly basis. Due to spread over of meter reading and billing in multiple cycle, rollover of consumers takes place from preceding quarter to current quarter and vice-a-versa. Hence, in some of the feeders, the gap between input and consumption may be high on positive side and in some of the feeders it is negative. As the period of consideration on rolling basis, variation gets normalized.

Considering the following feeders individually, it gives an impression that these feeders are having high MU loss. But considering them with the feeders with which these feeders share input energy and consumption, the MU losses get reduced drastically.

Table 35: High Loss Feeders

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
BAWANA-1 66kV	22030879	BAWANA-1 GRID TO I-19 SEC-2 NEAR GOL CHAKKAR	8.89	1.53	7.36
BAWANA-6 GRID	230463820	BAWANA GRID -6 TO L-216 SEC-3 DSIIDC BWN	6.50	1.39	5.12
BAWANA CLEAR WATER GRID	22000108	BAWANA CLEAR WATER GRID TO AUCHANDI FDR	13.15	2.81	10.34
AIR KHAM PUR GRID	21964064	AIR KHAMPUR GRID TO NIT FEEDER NO-2	10.23	2.44	7.79
220KV DSIDC BAWANA GRID	22019724	220 KV BAWANA GRID TO G-BLOCK SEC-2	10.37	3.11	7.25
BAWANA CLEAR WATER GRID	21979376	BAWANA CLEAR WATER GRID TO GHOGHA (K) FDR NO-2	10.96	3.32	7.64
220KV KANJHAWLA GRID	22019624	220KV KANJHAWALA GRID TO RANI KHERA O/H FEEDER	9.04	2.75	6.29
BAWANA CLEAR WATER GRID	21979426	BAWANA CLEAR WATER GRID TO GHOGHA (K) FDR NO-1	9.74	3.01	6.73
ROHINI - 24 GRID	21998443	ROHINI - 24 GRID TO PKT-4&5 SEC-23 S/S NO-7	12.55	4.16	8.40
CIVIL LINE GRID	21986170	CIVIL LINE GRID TO PRESS ROAD OUTDOOR S/S FDR NO-2	5.15	1.79	3.36
BAWANA CLEAR WATER GRID	21979466	BAWANA CLEAR WATER GRID TO DARYAPUR NANGAL S/S (O/D)	14.41	5.44	8.97
RAMPURA GRID	21983960	RAMPURA GRID TO B - 38	14.45	5.95	8.50
BAWANA-1 66kV	22030939	BAWANA-1 GRID TO K-1 BLK.SEC-1	18.74	7.78	10.97
220KV KANJHAWLA GRID	22019594	220KV KANJHAWALA GRID TO J.J COLONY SWADA NO-1 (BULLA COLONY)	12.50	5.28	7.22
POOTH KHURD GRID	21979866	POOTH KHURD GRID TO SOS BAWANA	12.71	5.90	6.81
ROHINI - 1 GRID	21982733	ROHINI - 1 GRID TO POLICE STATION SOUTH ROHINI	2.12	1.00	1.12
BAWANA-7 GRID	21998303	BAWANA-7 GRID TO KHEDA KHURD O/H CKT.	12.02	5.76	6.26
ROHINI - 1 GRID	22012612	ROHINI - 1 GRID TO SEVEN SEAS S/STN	2.43	1.20	1.23
220KV DSIDC BAWANA GRID	22032541	220 KV BAWANA GRID TO O-BLOCK SEC-2 S/S (O/D)	8.75	4.35	4.40
PAYAL GRID	21996003	PAYAL GRID TO A-BLK KHURANA S/S NIA PH-1 FEEDER NO-1	5.21	2.62	2.60
BAWANA CLEAR WATER GRID	22022804	BAWANA CLEAR WATER GRID TO NANGAL FDR	13.37	6.74	6.63
ROHINI - 34 GRID	22040171	RG-34 GRID TO SEWAGE PUMP SECTOR-35 S/S	9.33	4.81	4.52
220KV KANJHAWLA GRID	22019564	220KV KANJHAWALA GRID TO SOS BAWANA (MAJARA)	11.56	6.01	5.55
ROHINI-20 (GIS) GRID	75451284	ROHINI-20 GRID TO WORK CENTRE-2 SULTANPURI	18.56	9.69	8.87
TIGI PUR GRID	22011457	TIGIPUR GRID TO HIRANKI FEEDER	11.93	6.26	5.67
ROHINI - 6 GRID	22012144	ROHINI - 6 GRID TO 13/15 (I/D) S/S	11.84	6.30	5.55

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
A-7 NARELA GRID	70911529	A-7 NARELA GRID TO PARSHURAM CHOWK S/S (O/D)	9.28	4.94	4.34
BAWANA-7 GRID	21998263	BAWANA GRID -7 TO F G BLOCK	12.86	6.87	5.99
BAWANA-1 66kV	22031009	BAWANA-1 GRID TO LM BLOCK SEC-1 S/S	12.46	6.74	5.72
ROHINI- 28 GRID	22023724	RG 28 GRID TO ESS-5 1800 LIG SEC-29	2.90	1.58	1.32
BURARI GRID	58593177	DJB BURARI GRID TO BURARI GRID O/D SHIV KUNJ	17.07	9.37	7.70
GULABI BAGH GRID	21987717	GULABI BAGH GRID TO VIVEKANANDA PURI	11.12	6.12	4.99
A-7 NARELA GRID	21998773	A-7 NARELA GRID TO SINGHOLA	24.23	13.53	10.70
GHEVRA SAVDA GRID	22001285	GHEVRA GRID TO IGL PUMP CNG STATION NO. 2 SAWDA	4.54	2.55	1.99
KARALA (GIS) GRID	75009210	KARALA GRID TO LADPUR S/S	9.06	5.13	3.93
BHALSWA GRID	22013924	BHALSWA GRID TO BABA COLONY (95405073+95405731)	12.01	6.80	5.21
BAWANA-7 GRID	21998193	BAWANA-7 GRID TO HOLAMBI KHURD O/H CKT.	10.90	6.18	4.72
BAWANA CLEAR WATER GRID	21979456	BAWANA CLEAR WATER GRID TO GHOGHA DAIRY COLONY O/H CKT FDR-1	11.82	6.75	5.07
ROHINI - 2 GRID	21981526	ROHINI - 2 GRID TO DDA FLAT S/S NO.1 FDR NO-1	10.79	6.19	4.60
KARALA (GIS) GRID	75008937	KARALA GRID TO ROOPALI ENCLAVE S/S	8.62	4.99	3.63
KARALA (GIS) GRID	75009366	KARALA GRID TO PASHU AUSHADHALAYA S/S	15.12	8.76	6.37
KARALA (GIS) GRID	75009251	KARALA GRID TO ZONAL OFFICE KARALA FEEDER NO.2	12.07	6.99	5.08
ROHINI - 34 GRID	22039501	RG-34 GRID TO M BLOCK SEC-1 S/S	6.81	3.99	2.82
220KV KANJHAWLA GRID	22019544	220KV KANJHAWLA GRID TO GHEVRA	10.34	6.06	4.28
ROHINI - 23 GRID	21999758	ROHINI - 23 GRID TO RMU NEAR RG - 23 GRID	10.59	6.22	4.36
BAWANA CLEAR WATER GRID	22000148	BAWANA CLEAR WATER GRID TO A-BLOCK SEC.-2	5.50	3.27	2.22
BAWANA CLEAR WATER GRID	21979436	BAWANA CLEAR WATER GRID TO SANNOTH (K) FDR NO-1	10.85	6.46	4.38
220KV KANJHAWLA GRID	22019534	220KV KANJHAWLA GRID TO U/G KHANJHAWLA CHOWK	6.98	4.16	2.82
220KV SUBZI MANDI GRID	21949424	SUBZI MANDI GRID TO KABIR BASTI	7.46	4.47	3.00
RANI BAGH GRID	21983253	RANI BAGH GRID TO KAILAH ENCLAVE (Z-504)+JHULA LAL NO-1	12.78	7.71	5.07
ROHINI - 3 GRID	21983123	ROHINI - 3 GRID TO 4/7 S/S FDR NO-2	7.53	4.56	2.97
Bawana-1 Ph-2 66/11 KV	42202574	BAWANA -1 PH2 GRID TO SEC-L,DSIDC BHORGARH FEEDER-2	10.69	6.51	4.18

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
A-7 NARELA GRID	21998743	A-7 NARELA GRID TO GT ROAD AIR KHAMPUR S/S	11.46	6.99	4.47
A-7 NARELA GRID	39414674	A-7 NARELA GRID TO PURAN WALA S/S	14.50	8.88	5.62
BHALSWA GRID	75467399	BHALSWA GRID TO BHALSWA JHEEL NO-2 MANDIRWALA	14.49	8.90	5.59
WAZIRPUR - 2 GRID	22009897	WAZIRPUR - 2 GRID TO AD BLOCK PSS	14.60	8.97	5.62
SHAHZADA BAGH GRID	22033346	SHAHZADA BAGH GRID TO EAST MOTI BAGH	11.35	7.03	4.32
GULABI BAGH GRID	21987617	GULABI BAGH GRID TO CHUNA BHATTI FDR NO-1	6.86	4.26	2.60
ROHINI - 23 GRID	22031119	ROHINI - 23 GRID TO RMU INFRONT OF RG-23 POOTHKALAN NO.1 S/S	11.56	7.24	4.32
ROHINI- 28 GRID	22023784	RG-28 TO DJB-1 KHERA KHURD	4.81	3.06	1.75
DHEERPUR GRID	34516849	DHEERPUR GRID TO WAZIRABAD FLYOVER NEHRU VIHAR S/S	9.58	6.12	3.46
JAHANGIR PURI GRID	22014179	JAHANGIRPURI GRID TO A BLK PARK WALA S/STN	9.81	6.28	3.53
WAZIRABAD GRID	22003222	WAZIRABAD GRID TO JAGATPUR EXT.FEEDER NO-2	13.75	8.86	4.90
NARELA SUB CITY GRID	42584830	NARELA SUB CITY GRID TO LAMPUR O/D S/S	7.61	4.91	2.70
WAZIRPUR - 1 GRID	22009877	WAZIRPUR - 1 GRID TO C - 33 WPIA	6.11	3.94	2.17
NARELA SUB CITY GRID	42584810	NARELA SUB CITY GRID TO MBS BAWANA S/S (O/D)	12.97	8.43	4.54
POOTH KHURD GRID	21979846	POOTH KHURD GRID TO DISTRICT OFFICE BAWANA - 1	14.27	9.31	4.97
HUDSON LINE GRID	22008627	HUDSON LINE GRID TO ORBIT PLAZA MALL FDR	5.59	3.66	1.94
HAIDER PUR GRID	21996903	HAIDER PUR GRID TO 1/15 S/S	12.46	8.15	4.31
MANGOL PURI - 1 GRID	21982056	MANGOL PURI - 1 GRID TO S/S NO 15 FDR NO.1+ B BLK MGP IND.AREA PH-1 S/S	13.75	9.10	4.65
ROHINI - 23 GRID	21999778	ROHINI - 23 GRID TO RMU SUKHBIR NAGAR	11.36	7.53	3.83
ROHINI - 22 GRID	263445699	ROHINI - 22 GRID TO RMU NEAR RG-22 GRID	11.71	7.89	3.82
AZADPUR GRID	21959254	AZADPUR GRID TO A BLOCK GTK	6.53	4.41	2.12
ROHINI - 4 GRID	21980916	ROHINI - 4 GRID TO SPFA NO-3 S/S	15.09	10.21	4.88
PRASHANT VIHAR (GIS) GRID	22035301	PRASHANT VIHAR GRID TO 4/7 S/S FDR NO-1	6.84	4.63	2.21
220KV DSIDC BAWANA GRID	22019744	220 KV BAWANA GRID TO C-BLOCK SEC-2	18.00	12.24	5.76
BAWANA-4 (GIS) GRID	129186247	BAWANA-4 GRID TO C BLOCK SEC-3	17.43	11.86	5.57
INDERPURI GRID	21984559	INDERPURI GRID TO GENE BANK	8.56	5.84	2.72
G.T.K. GRID	22036936	G.T.K. ROAD GRID TO G-1 G.T.K S/S	3.48	2.38	1.10

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
ROHINI DC-1	22009977	ROHINI DC-1 GRID TO CITY CENTER MALL	5.81	3.99	1.82
KARALA (GIS) GRID	75009320	KARALA GRID TO ZONAL OFFICE KARALA FEEDER NO.1	7.85	5.40	2.45
G.T.K. GRID	22013462	G.T.K. GRID TO MULTI LEVEL CAR PARKING	3.73	2.57	1.16
WAZIRPUR - 1 GRID	21986957	WAZIRPUR - 1 GRID TO PETROL PUMP WPIA	7.80	5.39	2.40
POOTH KHURD GRID	21979876	POOTH KHURD GRID TO POOTH KHURD VILLAGE FDR	11.02	7.63	3.39
SWISS APARTMENT (GIS) GRID	114526545	SWISS APARTMENT GRID TO INDIRA MKT (O/D)	9.23	6.40	2.83
BAWANA-7 GRID	22030009	BAWANA GRID -7 TO I BLOCK SEC-5	13.86	9.63	4.23
BAWANA-7 GRID	22029999	BAWANA GRID -7 TO E-BLK SEC-5	17.87	12.41	5.46
BAWANA-1 66kV	177677382	BAWANA-1 GRID TO F BLOCK SEC-1	17.84	12.42	5.43
SUDARSHAN PARK GRID	22032631	SUDARSHAN PARK GRID TO ESI HOSPITAL NEW	4.61	3.21	1.40
A-21 NARIANA GRID	22028834	A-21 NARIANA GRID TO RANJIT NAGAR COMMUNITY CENTRE	6.27	4.38	1.89
BURARI GRID	58592839	DJB BURARI GRID TO BURARI PRADHAN ENCLAVE S/S	12.17	8.54	3.63
ROHINI- 28 GRID	22023834	RG 28 GRID TO DDA COMMAND TANK S/S	11.95	8.39	3.56
GULABI BAGH GRID	49690500	GULABI BAGH GRID TO SUBHADRA COLONY RAMLEELA GROUND	10.37	7.30	3.08
DHEERPUR GRID	165581076	DHEERPUR GRID TO DJB WWTP MAJLIS PARK S/S	3.88	2.73	1.15
G.T.K. GRID	21985183	G.T.K. GRID TO F.C.I.	4.38	3.11	1.27
SMB FACILITY CENTER GRID	22020644	SMB FC GRID TO INTERCONNECTOR 1&2 TO JAHANGIRPURI GRID S/S	9.28	6.60	2.69
P.P.-3 GRID	21983693	P.P.-3 GRID TO RMU INSIDE PP-3 GRID	15.20	10.84	4.37
ROHINI - 23 GRID	21999748	ROHINI - 23 GRID TO 4A/23 S/S	9.27	6.69	2.58
NARELA SUB CITY GRID	42585034	NARELA SUB CITY GRID TO L-BLOCK SEC-5	6.89	4.98	1.91
SWISS APARTMENT (GIS) GRID	114526519	SWISS APARTMENT GRID TO PARAG ICE FACTORY	12.62	9.17	3.46
RANI BAGH GRID	21983273	RANI BAGH GRID TO RANI BAGH NO-2 FDR -2	11.49	8.37	3.11
ROHINI - 1 GRID	21982713	ROHINI - 1 GRID TO MANGALAM PLACE S/S NO-5 FDR NO-2	10.78	7.86	2.92
ROHINI - 4 GRID	190786863	ROHINI-4 GRID TO 1/3 S/S	8.48	6.19	2.29
ROHINI- 28 GRID	22023754	RG 28 GRID TO JINDAL NIRYAT	5.87	4.30	1.58
ROHINI - 22 GRID	21981386	ROHINI - 22 GRID TO PKT-17 SEC-22 S/S NO.1	7.21	5.29	1.92
TRINAGAR GRID	21997413	TRINAGAR GRID TO BOOSTER PUMP FDR NO-2	6.46	4.74	1.71
A-7 NARELA GRID	22012284	A-7 NARELA GRID TO S/S NO-31 & B G SHIRKE	15.50	11.40	4.10

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
BAWANA CLEAR WATER GRID	22000158	BAWANA CLEAR WATER GRID TO HANUMAN MANDIR BWN S/S O/D	8.10	5.98	2.13
WAZIRPUR - 1 GRID	21987067	WAZIRPUR - 1 GRID TO RED LIGHT ASHOK VIHAR CROSSING	5.80	4.28	1.52
MODEL TOWN GRID	179429277	MODEL TOWN GRID TO E BLOCK MKT S/S	9.94	7.34	2.59
SARASWATI GARDEN GRID	21984239	SARASWATI GARDEN GRID TO E BLOCK KIRTI NAGAR	6.14	4.54	1.60
SGTN-2 (GIS) GRID	22039831	SGTN-2 GRID TO SDM OFFICE S/STN (O/D)	7.79	5.76	2.03
KARALA (GIS) GRID	75008981	KARALA GRID TO CHANDPUR MAJARA S/S	10.29	7.63	2.66
P. P. - 2 GRID	21982413	P. P. - 2 GRID TO PHOOLARAM NO-2	10.34	7.70	2.64
P. P. - 2 GRID	21982513	P. P - 2 GRID TO N-BLOCK FDR-2	14.77	11.00	3.77
GHEVRA SAVDA GRID	22001305	GHEVRA GRID TO 7620 FDR-3	3.65	2.72	0.93
PUSA GRID	21984769	PUSA GRID TO NASC FDR NO-2	6.70	5.00	1.69
SHAKTI NAGAR (GIS) GRID	21987117	SHAKTI NAGAR GRID TO CLOCK TOWER (K)	14.85	11.10	3.75
ROHINI - 3 GRID	21982963	ROHINI - 3 GRID TO 1/9 S/S FDR NO -1	8.70	6.51	2.19
MODEL TOWN GRID	22028644	MODEL TOWN GRID TO OLD TAGORE PARK	12.64	9.48	3.17
WAZIRPUR - 2 GRID	21986767	WAZIRPUR - 2 GRID TO KASTURBA POLYTECHNIC	6.78	5.09	1.70
BAWANA CLEAR WATER GRID	21979366	BAWANA CLEAR WATER GRID TO A BLK JJ COLONY+D BLK JJ COLONY	8.35	6.28	2.07
BAWANA-7 GRID	21998283	BAWANA-7 GRID TO RADHA KRISHNA RICE MILL FDR-2	10.78	8.11	2.67
220KV SHALIMARBAGH GRID	22018574	220KV SMB GRID TO TIKONA PARK FEEDER NO-2	13.02	9.80	3.22
KARALA (GIS) GRID	75009343	KARALA GRID TO ZONAL OFFICE KARALA FEEDER NO.3	5.36	4.05	1.31
PUSA GRID	21984779	PUSA GRID TO NCAP	8.00	6.04	1.96
JAHANGIR PURI GRID	21987887	JAHANGIR PURI GRID TO INTERCONNECTOR 1&2 TO SMB FC GRID S/S	8.96	6.77	2.19
DSIDC NARELA-2 GRID	22023584	DSIDC NARELA-2 GRID TO ALIPUR -4	12.79	9.67	3.12
ROHTAK ROAD GRID	21975575	ROHTAK ROAD GRID TO RAMPURA-3	9.45	7.17	2.29
A-21 NARIANA GRID	22028864	A-21 NARIANA GRID TO COMMUNITY CENTER NIA PH-2	4.70	3.57	1.13
ROHINI - 23 GRID	22031099	ROHINI - 23 GRID TO DHEERAJ VIHAR JAIN NAGAR	11.55	8.77	2.78
220KV ROHINI GRID	22019224	220 ROHINI GRID TO SEC-26 NALA (O/D)	4.44	3.38	1.06
RAMPURA GRID	21983910	RAMPURA GRID TO OLD MOTHER DAIRY RAMPURA	5.23	3.99	1.24
POOTH KHURD GRID	21979816	POOTH KHURD GRID TO MB HOSPITAL FDR-1	8.63	6.60	2.04

Name of the Station	Feeder Code/ID	Feeder Name	Received at Feeder (Final in MU)	Feeder Consumption (In MU)	T&D losses (Mus)
ASHOK VIHAR GRID	21987527	ASHOK VIHAR GRID TO RAMLILA GROUND ASHOK VIHAR	5.59	4.28	1.30
MODEL TOWN GRID	22028694	MODEL TOWN GRID TO H2 MKT S/S JAHANGIR PURI	15.69	12.03	3.65
AZADPUR GRID	21959284	AZADPUR GRID TO INTERCONNECTOR 1 & 2 TO JAHANGIRPURI GRID S/S	12.17	9.34	2.83
ROHINI - 2 GRID	21981686	ROHINI - 2 GRID TO RMU OUTDOOR S/S	10.82	8.31	2.51
SHAHZADA BAGH GRID	22023574	SHAHZADA BAGH GRID TO INDERLOK S/STN.	11.79	9.08	2.71
SIRASPUR GRID	35895320	SIRASPUR GRID TO ESS-1	8.44	6.50	1.94
220KV NARAINA GRID	22010217	NARAINA 220KV GRID TO KIRBI PLACE FDR NO-1	6.95	5.36	1.59
JAHANGIR PURI GRID	271239194	JAHANGIRPURI GRID TO C-BLK RELIANCE S/S	6.26	4.83	1.43
BAWANA-1 66kV	177677336	BAWANA-1 GRID TO DSIIDC INFRA OFFICE S/S (O/D)	6.95	5.37	1.58
BHALSWA GRID	75467274	BHALSWA GRID TO DUSIB ESS NO-6 S/S	4.60	3.56	1.04
G.T.K. GRID	21985233	G.T.K. GRID TO A-BLOCK DERAVAL KIOSK	13.53	10.48	3.05
DHEERPUR GRID	165581210	DHEERPUR GRID TO AUTHORITY (O/D) (Z-532)	10.44	8.09	2.35
WAZIRPUR - 2 GRID	21986457	WAZIRPUR - 2 GRID TO PRESS COMPLEX-2	4.79	3.71	1.08

5.2.7. Identify overloaded segments/ infrastructure

Count of Overload feeders in Tata Power DDL is Zero.

In TPDDL network planning of feeders/ sections is being done 2-3 years prior to implementation at site, for overcoming any chance of overloading crisis occurrence in future.

5.3. Subsidy computation and analysis (based on quarterly data)

The subsidy details during FY 2024-25 are shown in the table below:

Table 36: Category wise subsidy Division wise subsidy detail

Annexure - 1 : Proforma for Quarterly Consumer Category-wise Subsidy Billed/Received/Due for period 01/Apr/2024 to 31/March/2025																	
S.N.	Division Name	Consumer Category (Separate for each subsidized consumer category)	Billed Energy			Subsidized Billed Energy			Applicable rate of Subsidy as notified by State Govt.		Subsidy Due from State Govt.			Subsidy Actually Billed/claimed from State Govt. (As against col.12)	Subsidy Received from State Govt. (As against col.13)	Balance Subsidy yet to be Received from State Govt.	No.of benefitted consumers
			Metered	Un-metered*	Total	Metered (out of col.2)	Un-metered* (Out of col.3)	Total	Metered Energy**	Un-metered Energy**	Metered Energy	Un-metered Energy	Total	(In Rs. Cr.)	(In Rs. Cr.)	(In Rs. Cr.)	Count
			(In kwh)			(In kwh)			(In Rs/Kwh)		(In Rs. Cr.)			(In Rs. Cr.)	(In Rs. Cr.)	(In Rs. Cr.)	
1		1	2	3	4=2+3	5	6	7=5+6	8	9	10=5x8	11=6x9	12=10+11	13	14	15=13-14	16****
	Opening Balance (01/04/2024)													67.82			
	Residential	0-200 units pm	1113029922		1113029922	1113029922		1113029922	Entire current demand					1057.19	1035.30	98.06	
		201-400 units pm	1671576451		1671576451	#		#	Rs. 800 per month								
	Agriculture		15438419		15438419	##		##	Rs. 105 per kw/month					4.38			
	Commercial/Industrial-LT	Lawyer's Chamber	1977820		1977820	###		###	###								
	Commercial/Industrial-HT																
	Other (Specify) WW													3.97			
Total		2802022612.00	0.00	2802022612.00	1113029922.00	0.00	1113029922.00	0.00	0.00	0.00	0.00	0.00	0.00	1133.36	1035.30	98.06	
* Subsidy in 201-400 block is being given at flat rate of Rs. 800 per month, thus Subsidized Billed Energy cannot be ascertained.																	
** Subsidy in Agriculture category is being given at the flat of Rs. 105 per kw/month on Fixed charges																	
*** Subsidizing / Neutralizing non-domestic tariff 'Lawyer's chamber' to domestic tariff after subsidy																	

5.4. Trend analysis and identification of key exceptions

Details may be specified such as performance of DISCOM in FY 2023-24 Vs FY2024-25, annual comparison, circle wise analysis, where the DISCOM has improved and where needs improvement etc.

a) Summary Comparison of AT&C and T&D losses

Division Wise Comparison of T&D and AT&C losses For FY 2023-24 & FY 2024-25 are shown in below Table:

Table 37: Comparison of AT&C and T&D losses for last two Years

Name of Division	AT&C Loss		T&D Loss	
	FY 2023-24	FY 2024-25	FY 2023-24	FY 2024-25
Badli	8.04%	8.61%	7.92%	8.65%
Bawana	7.77%	7.76%	7.76%	7.77%
Civil Lines	2.33%	2.38%	2.14%	2.15%
Keshav Puram	4.35%	4.44%	4.72%	4.68%
Mangolpuri	5.59%	4.81%	5.66%	4.71%
Model Town	5.28%	4.89%	5.37%	5.07%
Moti Nagar	5.26%	4.75%	5.20%	5.06%
Narela	7.72%	7.41%	7.67%	7.49%
Pitam Pura	4.95%	3.69%	5.03%	4.06%
Rohini	4.18%	3.79%	4.38%	3.93%
Kirari	10.50%	7.28%	10.42%	6.64%
Shalimar Bagh	5.80	5.95%	5.58%	5.89%

Figure 12: Last two years AT&C loss trend

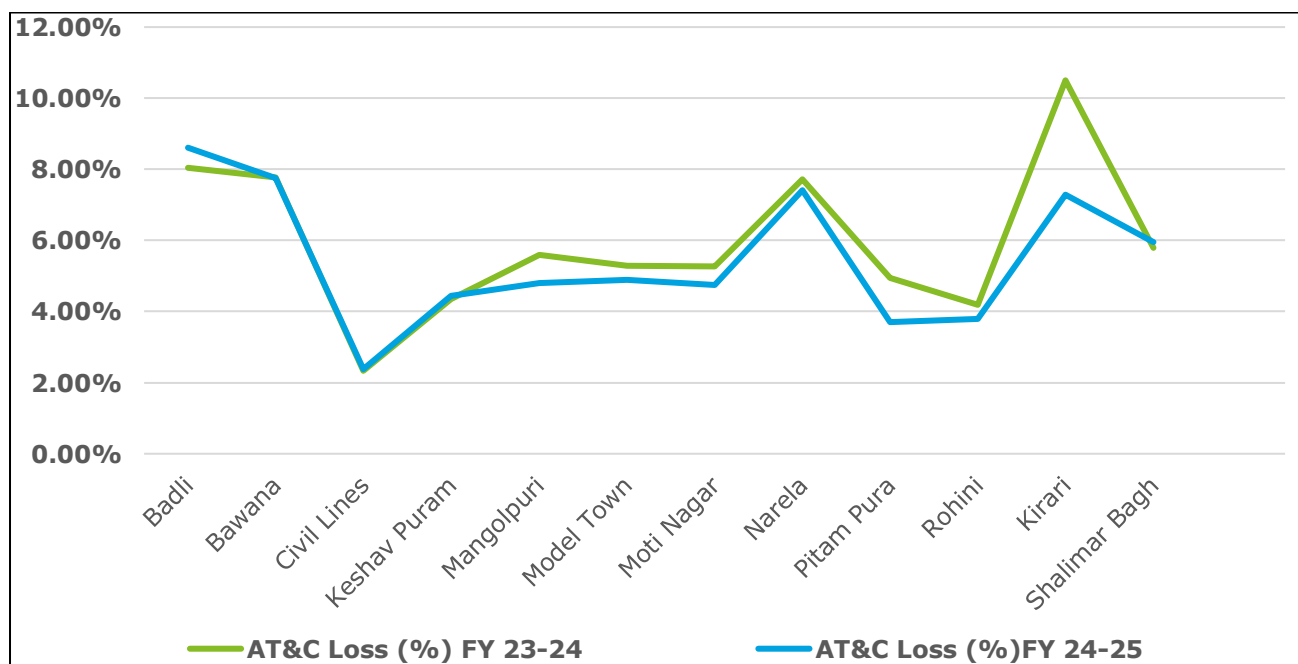
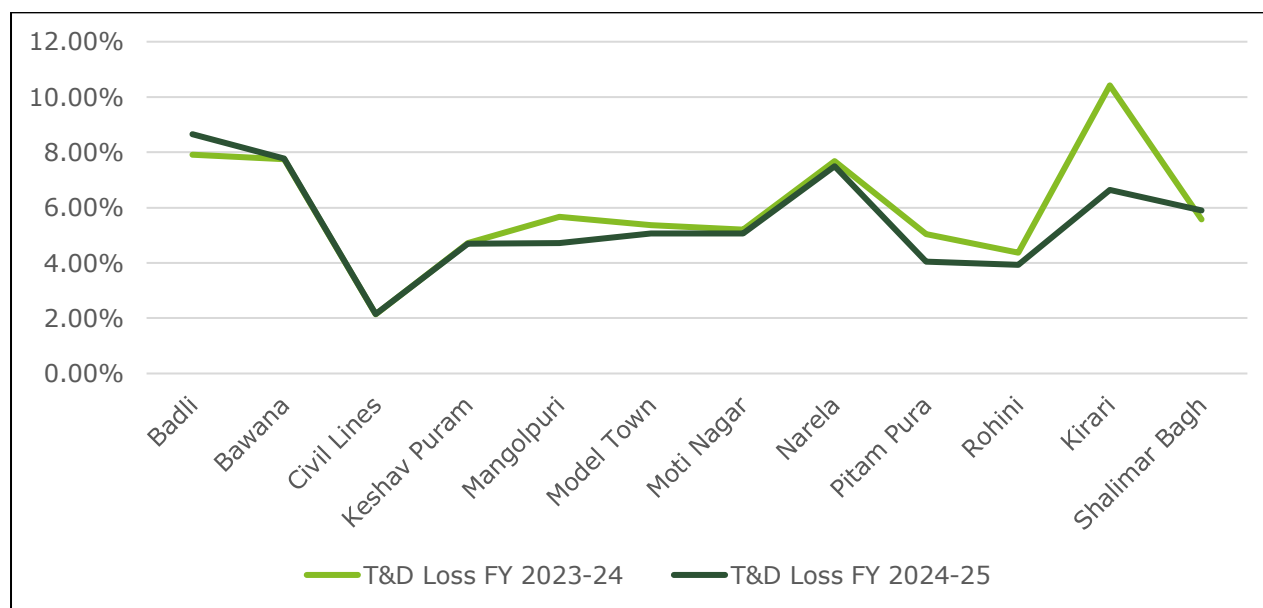


Figure 13: Last two years T&D loss trend



b) Summary of AT&C losses for last three Years

The AT&C losses for FY2022-23, FY 2023-24 & FY 2024-25 are as shown below:

Table 38: AT&C losses Trend for last three Years

Technical Details	UoM	FY 2022-23	FY 2023-24	FY 2024-25
Input Energy Purchase (From Generation Source)	Million kWh	12741.47	12686.10	13094.00
Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	Million kWh	10622.62	10660.43	11436.67
Total Energy billed (is the Net energy billed, adjusted for energy traded))	Million kWh	9946.33	10028.15	10795.16
Transmission and Distribution (T&D) loss Details	Million kWh	676.29	632.28	641.51
	%	6.37%	5.93%	5.61%
Collection Efficiency	%	100%	100.01%	100.06%
Aggregate Technical & Commercial Loss	%	6.36%	5.92%	5.55%

6. Energy Audit findings

6.1. Review of capacity of DISCOM's energy accounting and audit wing

The Energy Accounting/Audit wing in the DISCOM has been implemented from 11-Feb-22. The Organogram of the DISCOM is as shown below:

Figure 14: Organogram of Energy accounting in the DISCOM

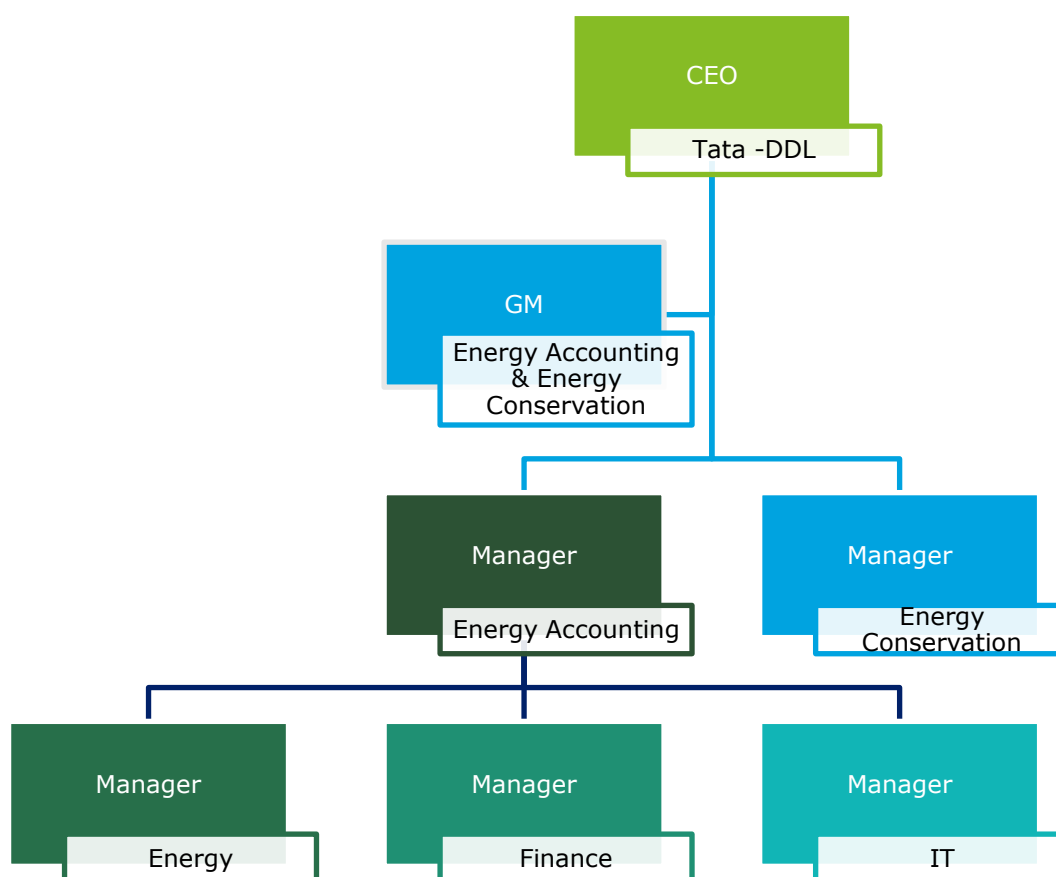


Table 39: Details of energy manager and Authorized signatory of DISCOM

Designation	No. of Officers
CEO	Mr. Dwijadas Basak
General Manager	Mr. H C Sharma
AGM (Energy Manager)	Mr. Banwari Lal Gupta
Head of Department, Energy Audit	Mr. Pushpendra Kumar
Head of Group, Energy Audit	Mr. Hari Om Sharma
Sr. Manager, Energy Audit	Mr. Neeraj Singh
Executive, Energy Audit	Mr. Sahil Yadav
Finance Manager	Mr. Sachin Gupta
IT Manager	Ms. Kamakshi Oberoi

6.2. Critical Analysis

- The monthly consumption per customer stands at 427.8kWh/Month. Tata Power-DDL caters to area spread in 5 circles, 12 Division, 32 numbers of sub-division, 20003 number of feeders, 8785 number of DTs and 2102837 number of consumers.
- Verified transmission losses, distribution (T&D) losses, collection efficiency & aggregate technical & commercial losses of Tata Power Delhi Distribution Limited for FY24-25 was 4.19%, 5.61 %, 100.06 % & 5.55 % respectively.
- The electrical energy is supplied by various interstate and intrastate generating stations and same is supplied to customers at 66 KV, 33 KV, 11 KV, 6.6KV, 400V and 230 V single phase.
- Tata Power-DDL has implemented several world-class technologies such as Advance Distribution Management system or ADMS which is designed to replace the conventional SCADA-DMS-OMS system with features like real-time integration of Smart Meter Data / Distributed Generation integration and single data model from GIS , Integrated Geographical Information System (GIS) for instant services, Advanced Metering Infrastructure (AMI), Automated Demand Response (ADR), Smart Street Light Management system, Field Force Automation, Upgraded Network etc.
- Tata Power DDL has implemented various energy conservation measures under DSM program i.e., AC Replacement Scheme, BLDC Ceiling Fan, LED Lighting Scheme, Behavioral Demand Response (BDR) etc.
- Tata Power DDL calculation methodology of AT&C Losses calculated as per BEE proforma the calculation is as per actual collection efficiency.
- Tata Power-DDL has 100% metering available at 11/33/66 KV system. However, there is 100% metering at consumer end and 77% metering available at DT.

6.2.1. Status and progress in compliance to pre-requisites to energy accounting

It was observed that Quarterly Accounts during all Quarters (i.e., Q1, Q2, Q3 & Q4 of FY 2024-25) with in given timeline. DISCOM needs to submit the Quarterly accounts within the timeframe stipulated in the Regulations. Further, the compliance with regards to Regulations and Pre-requisites are tabulated in the table below:

Table 40: Compliance status w.r.t Timelines and Pre-requisites

Clause	Details	Sub-Clause	Criteria	Compliance Status
3	Intervals of time for conduct of annual energy audit	A	Conducted an annual energy audit for every financial year and submitted the annual energy audit report to the Bureau and respective State Designated Agency and also made available on the	Yes

Clause	Details	Sub-Clause	Criteria	Compliance Status
			website of the electricity distribution company within a period of four months from the expiry of the relevant financial year	
4	Intervals of time for conduct of periodic energy accounting.	a	All feeder wise, circle wise and division wise periodic energy accounting is conducted by the energy manager of the electricity distribution company for each quarter of the financial year.	Yes
		b	All feeder wise, circle wise and division wise periodic energy accounting is conducted by the energy manager of the electricity distribution company for each quarter of the financial year.	Yes
		c	Electricity distribution company conducted its first periodic energy accounting, for the last quarter of the financial year immediately preceding the date of such commencement (i.e., 6th October 2021)	Yes
		d	Electricity distribution company conducted its subsequent periodic energy accounting for each quarter of the financial year for a period of two financial years from the date of such commencement and submits the periodic energy accounting report within sixty days from the date of periodic energy accounting.	Yes
5	Pre-requisites for annual energy audit and periodic energy accounting	a	Pre-requisites for annual energy audit and periodic energy accounting	Yes
		b	Identification and mapping of high tension and low-tension consumers	Yes
		c	Development and implementation of information technology enabled energy accounting and audit system, including associated software	Yes
		d	Electricity distribution company ensures the installation of functional meters for all consumers, transformers and	Yes

Clause	Details	Sub-Clause	Criteria	Compliance Status
			feeders. Meter installation is done in a phased manner within a period of three financial years from the date of the commencement of these regulations in accordance with the trajectory set out in the First Schedule d.1. 100% Communicable Feeder Metering integrated with AMI, by 31st December 2022 along with replacement of existing non-communicable feeder meters.	
			d.2. All Distribution Transformers (other than HVDS DT up to 25kVA and other DTs below 25 kVA) shall be metered with communicable meters. Communicable DT Metering for the following areas/ consumers to be completed by December 2023 and in balance areas by December 2025:	NA
			d.2.1. All Electricity Divisions of 500 AMRUT cities, with AT&C Losses > 15%	NA
			d.2.2. All Union Territories (for areas with technical difficulty, non-communicable meters may be installed)	NA
			d.2.3. All Industrial and Commercial consumers	Yes
			d.2.4. All Government offices at Block level and above	Yes
			d.2.5. Other high loss areas i.e., rural areas with losses more than 25% and urban areas with losses more than 15%	Yes
			d.3. Prepaid Smart Consumer Metering to be completed for all directly connected meters and AMR in case of other meters, by December 2023 in the following areas: d.3.1. All Electricity Divisions of 500 AMRUT cities, with AT&C Losses > 15%;	NA
			d.3.2. All Union Territories (for areas with technical difficulty, prepaid meters	NA

Clause	Details	Sub-Clause	Criteria	Compliance Status
			to be installed);	
			d.3.3. All Industrial and Commercial consumers;	NA
			d.3.4. All Government offices at Block level and above;	NA
			d.3.5. Other high loss areas i.e., rural areas with losses more than 25% and urban areas with losses more than 15%.	NA
			d.4. Consumer Metering: 98% by FY 2022-23 99% by FY 2023-24	Yes
			d.5. Targets for functional meters - Meter FY 22-23 FY 23-24 FY24-25 Feeder metering 98.5% 99.5% 99.5% DT metering 90% 95% 98% Consumer metering 93% 96% 98	Yes
		e	e.1. All distribution transformers (other than high voltage distribution system up to 25kVA and other distribution system below 25 kVA) is metered with communicable meters.	
			e.2. And existing non-communicable distribution transformer meters is replaced with communicable meters and integrated with advanced metering infrastructure.	Yes
		f	Electricity distribution company has established an information technology enabled system to create energy accounting reports without any manual interference and such systems may be within a period of three years from the date of the commencement of these regulations in case of urban and priority area consumers; and within five years from the date of the commencement of these regulations in case of rural consumers	Yes
		g	Electricity distribution company has a centralized energy accounting and audit	Yes

Clause	Details	Sub-Clause	Criteria	Compliance Status
			cell comprising of (i) a nodal officer, an energy manager and an information technology manager, having professional experience of not less than five years; and (ii) a financial manager having professional experience of not less than five years	
6	Reporting requirements for annual energy audit and periodic energy accounting	a	Electricity distribution company has a nodal officer, who is a full-time employee of the electricity distribution company in the rank of the Chief Engineer or above, for the purpose of reporting of the annual energy audit and periodic energy accounting and communicate the same to the Bureau	Yes
		b	Electricity distribution company ensures that the energy accounting data is generated from a metering system or till such time the metering system is not in place, by an agreed method of assumption as may be prescribed by the State Commission.	Yes
		c	Metering of distribution transformers at High Voltage Distribution System up to 25KVA is done on cluster meter installed by the electricity distribution company	NA
		d	The energy accounting and audit system and software is developed to create monthly, quarterly and yearly energy accounting reports.	Yes
		c	Electricity distribution company has provided the details of the information technology system in place as specified in clause (f) of regulation 5 that ensures minimal manual intervention in creating the energy accounting reports and any manual intervention of any nature, in	Yes

Clause	Details	Sub-Clause	Criteria	Compliance Status
			respect of the period specified therein, shall be clearly indicated in the periodic energy accounting report	

6.2.2. Data gaps

The Audit firm has raised the data gaps to the DISCOM. The summary of data gaps raised and response from DISCOM is summarized in the table below:

Table 41: Summary of Data gaps

S. No.	Data gaps raised by Energy Auditor	Response shared by DISCOM	Status of data submission by DISCOM
	Net Input Energy is 11436.67 MUs not matching the total energy available in Form Input 11527.40 MUs	Net Input Energy calculation: Net Input Energy received at DISCOM periphery – Open Access + Solar Generation	Submitted
	Voltage wise power purchase is not available as per BEE format as mentioned in Infrastructure sheet.	Voltage wise power purchase details are not available, as the invoice of power purchase contains only units purchased and details of power plant voltage not available.	
	Voltage wise input billed energy not available	Majority of feeders are common to LT & HT. So input energy supplied is inseparable. This comprises of input at 33 & 66 kV as the two can't be mentioned separately. Cumulated EHT sales MUs for 33kV and above are maintained and therefore bifurcation of same is not available. Same also provided in A.6 of Form-input energy sheet	
	Feeder wise AT&C losses not available.	Tata Power-DDL has ring main power distribution network. To provide uninterrupted power, back feeding points/normal operating points (NOP) are usually changed. Due to this dynamic nature of the network, connected consumers to a feeder at a particular point of time will change as the NOP changes. Tata Power-DDL is working on to implement technological solutions to capture these dynamic changes. Hence, if we compare input energy of feeder and related consumption, in many cases, input is low as compared to consumption. Vis-a- Vis,	

S. No.	Data gaps raised by Energy Auditor	Response shared by DISCOM	Status of data submission by DISCOM
		in some feeders, gap is high in positive side. Though critical parameters are monitored on monthly basis, substantial difference in the data will not be reflected on quarterly basis. Due to spread over of meter reading and billing in multiple cycle, rollover of consumers takes place from preceding quarter to current quarter and vice-a-versa. Hence, in some of the feeders the gap between input and consumption may be high on positive side and in some of the feeders it is negative. As the period of consideration widens or on rolling basis, variation gets normalized.	
	POC losses are not available	Generation at Transmission periphery- Point of connection loss (POC) and input voltage level are not available. As the power drawn is supplied through Central Transmission Utility (Power Grid) and State Transmission Utility (Delhi Transco).	

6.2.3. Summary of key responses of DISCOM management on Comments by Energy Auditor

The Auditor has identified the key issues with regards to Energy Accounting/Audit and DISCOM's management has responded to the same as summarized in the table below:

Table 42: Comments by Energy Auditor and responses of DISCOM management

S. No.	Comments by Energy Auditor	Response of DISCOM's management
	Kindly provide the filled in format as per BEE.	Provided by Tata Power-DDL
	Quarterly format as per BEE	Provided by Tata Power -DDL
	Kindly provide the identification and mapping of all of the electrical network assets	GIS Mapping
	Kindly provide the identification and mapping of high tension and low-tension consumers	GIS Mapping
	Kindly provide the details of the energy accounting data - generated from a metering system or till such time the metering system is not in place, by an agreed method of assumption as may be prescribed by	Metering is provided and home grown software is used as per tariff category.

S. No.	Comments by Energy Auditor	Response of DISCOM's management
	the State Commission	
	Whether the Metering of distribution transformers at High Voltage Distribution System up to 25KVA is done on cluster metering? If not what is the action plan	No, HVDS installation
	Energy (Electrical) Purchase report for the year 2024-25	Provided by Tata Power-DDL
	Open access consumer and their details	Provided by Tata Power-DDL
	Peak Demand of the system	Provided by Tata Power-DDL
	High loss Network segments	The segment identification is not possible due to ring main.
	Energy Conservational Schemes implemented	Provided by Tata Power-DDL
	Power Distribution Transformer	Provided by Tata Power-DDL
	Maintenance practices - Power Substation, HT lines LT lines, DTR	Provided by Tata Power-DDL
	Average Billing Rate	Category wise consumer and total billed energy of those consumer is provided by Tata Power

6.3. Revised findings based on data validation and field verification

Site visits were carried out to ascertain the meter number and the input serial number of the meters. Physical verification along with system data was verified.

At the time of field visit, feeder meters found in working conditions, Logbook is maintained on daily basis including hourly loading pattern, running feeder & energy consumption.

Field visit of Substations, DTs and commercial building was done during the audit time.

It was also observed 100% consumers are metered.

6.4. Inclusions and Exclusions

Particulars	FY 2023-24	FY 2024-25
Number of circles	5	5
Number of divisions	12	12
Number of sub-divisions	31	31
Number of feeders	19096	20003
Number of DTs	8691	8785
Number of consumers	2026104	2102837

Conclusion and Action Plan

6.5. Summary of critical analysis and way forward proposed by Energy Auditor

The primary energy-consuming areas are the distribution network, office buildings, and fleet of vehicles used for maintenance and repairs. The distribution network accounts for the majority of energy consumption, followed by office buildings and vehicles.

6.5.1. Recommendations:

1. Installation of Smart Meters.
2. Installation of Automatic Power factor controller (Capacitor Bank) at newly constructed PSS.
3. System improvement & automation.
4. Load balancing of distribution transformers.

6.5.2. Cost-Benefit Analysis:

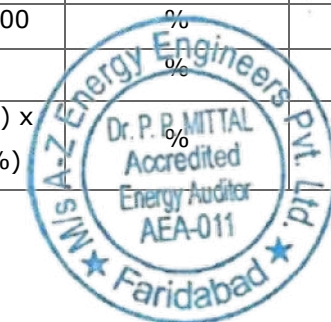
To determine the cost-effectiveness of the recommended measures, a cost-benefit analysis should be conducted. The cost of implementing the measures should be compared to the potential energy savings to determine the return on investment. This analysis will help the company prioritize the implementation of the recommended measures.

6.6. Summary of key findings – energy balance and losses

The Energy balance and losses of Tata Power-DDL for FY 2024-25 are as shown in the table below:

Table 43: Energy balance and losses

Energy Input Details	Formula	UoM	Value
Input Energy Purchase (From Generation Source)	A	MU	13094.00
Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	B	MU	11436.67
Total Energy billed (is the Net energy billed, adjusted for energy traded))	C	MU	10795.16
Transmission and Distribution (T&D) loss Details	D	MU	641.51
	$E = D/B \times 100$	%	5.61%
Collection Efficiency	F	%	100.06%
Aggregate Technical & Commercial Loss	$G = 1 - \{(1-E) \times \text{Min}(F, 100\%)\}$	%	5.55%



6.7. Recommendations and best practices

a) Energy accounting

In Accordance to clause 5(g) of Bureau of energy efficiency (Manner and intervals for conduct of Energy audit in Electrical distribution companies) Regulation issued by BEE Ministry of Power (GOI) dated 7th Oct 2021 to conduct Mandatory Annual Energy Audit & periodic Energy Accounting in Electrical Distribution Companies and as per guidelines issued by BEE vide L. No 18/BEE/DISCOM/2021/1348-94 dated 11th Feb 2022, an Energy Audit cell (EAC) is operating in Tata Power-DDL, Delhi.

There is Energy accounting cell to account for all the annual and quarterly energy accounting and audit data as per BEE regulation, vetted by internal energy manager and sent to BEE as per guidelines and Annual Energy Audit is being done by accredited energy auditor.

Energy Accounting is the first step towards identifying areas that need improvement. This will involve reviewing the current processes, systems, and data management practices.

Tata Power-DDL is laying special emphasis on Energy Accounting at all levels. Steps are being taken to account for every unit of energy supplied to consumers. The following measures are being implemented.

- i. 100% Feeder Metering at 66/33/11kV Substations.
- ii. 100% DT Metering by end of FY2024-25.
- iii. 100% Consumer metering for all categories of consumers.

b) Loss reduction

TPDDL is a private licensee that distributes power to over 9 million consumers in North and North-West Delhi. The Discom was formed in 2002 as a joint venture between Tata Power and the Government of Delhi.

Today, AT&C losses stand at 5.55% (as of FY 2024-25) which is an unprecedented reduction from an opening loss level of 53% in July 2002.

TPDDL leveraged technology to improve its operational efficiency and reduce its losses.

Tata Power-DDL has undertaken various initiatives for loss reduction. Some of the initiatives are as follows:

Renovation in Metering Infrastructure: Replacement of electromechanical meters, defective meters by electronic meters/ smart meters. Installation of AMR (Automatic Meter Reading) meters and tamper-proof static meters with facility of remote disconnection / reconnection as and when required.

Theft Control: Conducting raids in high loss areas for identifying pilferage of electricity.

Substation automation and distribution automation: Implementation of SCADA (Supervisory Control and Data Acquisition System) Control System for load management, including GIS mapping of assets.

Network reconfiguration: Electrical network designing wrt Standards and Guidelines, also considering network redundancy to increase system reliability.

Power factor improvement by automatic power factor controller.

Microgrid solutions to promote energy access to the areas where grid supply has not reached or is not feasible.

HVDS: Introduction of high voltage distribution system.

AB Cable: Replacement of bare conductor by AB (aerial bunched) Cable for shielding of possibilities of power theft.

Process re-engineering for improvement of existing customer complaint: improvement in all sorts of delay in billing, arrangement for spot billing etc. Improved process with IT application development.

Customer care services: Customer oriented management approach, like implementation of call center for 24 hours, improving customer care.

c) Energy conservation

TATA Power-DDL has always been frontrunner in choosing energy efficiency practices for building a sustainable greener tomorrow. The DISCOM has already undertaken sustainable initiative in scheme named 'Urja Arpan'.

- **Demand Side Management/ Behavioural Demand Response Program:** Demand Response (DR) is the reduction in a customer's electric load during periods of peak demand or high market price. Taking a step forward in improving system reliability, Tata Power-DDL initiated Behavioural Demand Response program to review customer's acceptability, adaptability, and its impact for balancing the demand and supply of electricity during the peak hours. Through this program customers were advised to reduce the electricity usage for a specified duration during which overall electricity demand is estimated to reach at the peak and raising awareness among individuals and organizations about responsible and efficient electricity consumption. which further helps to avoid network overloading and also defer CAPEX investment require for augmentation. The FY 2024-25 program engaged 126,500+ customers and successfully achieved a cumulative load curtailment of 452 MW.

- **Energy Efficient Lighting and Fans**

Scheme: Tata Power-DDL is offering Energy Efficient Products (EEPs) such as LED Bulbs and LED Tube Lights which have low maintenance and usage costs. These are durable and use up to 75% less energy than any other incandescent bulbs resulting in the reduction of monthly energy bills.

In addition to lighting, option for BLDC fans at discounted rates is available. BLDC fans provide high cooling efficiency with lower energy consumption and also savings in electricity bills. BLDC motors do not have brushes so they are more reliable, high life expectancy and energy savings as compared to conventional ceiling fans. With the added convenience of

remote control, BLDC fan is a financially and environmentally smart choice for customers as it consumes half electrical load as compared to ordinary fans.

Ceiling Fan Replacement - Replacement of non-star rated fans with BEE 5 Star rated fans. Consumption of 50/28 W against 70W conventional fans.

436 + items sold Replacement of conventional 70W fans with super-efficient BLDC fans (28W) at discounted price offers.

Approximately 90,210 units of energy-efficient lighting, fans, and home appliances—including washing machines—have been sold, leading to an estimated energy savings of 3.84 million units (MUs)

➤ **Launch of Clean Cooling Program with discount-based Energy efficient ACs Scheme for Customers:**

Tata Power-DDL in association with Voltas, Blue Star, General, Godrej, Hitachi is providing a discount-based scheme with or without Replacement of Old AC for all the customers.

➤ **Home Automation:**

Tata Power-DDL is offering an IoT based solution which allows to operate, schedule, monitor appliances like lighting, fans, ACs, washing machines, geysers, TVs, refrigerators, etc. with phone or with voice control via Google assistant or Alexa.

➤ **Home Automation (Sensors & Sensor Lights):**

Tata Power-DDL is offering Motion Sensors: With Motion sensors, lights and appliances can be switched on during presence of a person in the area. In absence of motion, automatically turn off the lights.

Sensor Lights: When motion is detected, sensor lights automatically illuminate at full brightness. In the absence of motion automatically dim the area lights to 20%.

➤ **Air Purifier Scheme:**

Tata Power-DDL in association with Voltas has initiated the Air Purifiers scheme at special discounted rates. The Air Purifiers are powered with 6-stage filtration process and removes up to 99.90% airborne pollutants, eliminates impurities with ease.

➤ **Washing Machine Scheme:**

Tata Power-DDL has launched a 5star rated washing machine scheme at special discounted rates in collaboration with Voltas-Beko.

➤ **AC Replacement Program-** Replacement of non- star rated AC with BEE 5 Star rated/ Inverter Technology AC by Voltas, Blue Star, Sharp & Godrej at discounted prices. More than 2028+ items sold. Energy savings of approx. 1.8 MU.

➤ **Digitization/ Adoption of digital lifestyle:** By reducing paper consumption, and exploring digital alternatives, TATA Power-DDL contributes to environmental conservation and decreased carbon footprints, thus fostering a greener and eco-friendly workplace. E-billing is being adopted for consumer billing. Supporting green initiatives: Participation in local environmental programs, such as tree planting and clean-up drives.

- **Celebration of Earth Day on 22nd April every year:** As a collective responsibility to nurture and protect Mother Earth, DISCOM is urging consumers to shut down their power for 1 hour on Earth Day.

6.8. Action plan for line loss reduction

Following energy conservation Measures (ECMs) are adopted for line loss reduction

1. Installation of Smart Meters.
2. Installation of Automatic Power factor controller.
3. System improvement & automation.
4. Improvement in metering system.
5. Replacement of conventional/ non-star rated transformer into energy efficient transformers.
6. Replacement of all conventional mechanical energy meters with static digital energy meters having less power consumption and more accuracy.
7. Laying of AB cable in theft prone area where losses are in higher side.
8. HVD system to reduce low tension line losses.
9. Replacing worm out /under sized conductors.
10. Increase in HT/LT Ratio.
11. Preventive & Periodic maintenance of line & transformer.
12. Load balancing of distribution transformers.
13. MIS Based periodic reporting of unit wise business parameters.
14. Installation of solar generation plant & solar pumps.
15. Strengthening of energy accounting infrastructure- 100% consumer metering.
16. GIS based asset mapping of all 33/11KV Substations, 11KV Lines and distribution transformers has been completed.
17. 11KV Feeder wise base line technical data i.e., length, Peak load, VR and technical loss calculated by algorithm & published in power BI.
18. Feeder wise baseline commercial loss is being collected by subtracting technical loss for total T&D loss.
19. Selection/priority of area must be made under:
 - I. Feeder having VR more than 15%.
 - II. Feeder having VR more than 9 to 15%.
 - III. High T&D loss feeder.
 - IV. High commercial loss feeder.

6.9. Action plan for monitoring and reporting

1. Energy Audit Cell as per BEE Regulation.
2. GIS Based mapping for all feeder monitoring.
3. Substation wise dashboard for all substation monitoring
4. Division dashboard for monitoring all division data.
5. Critical Parameters dashboard for KPI.
6. Customer care help line number.

6.10. Action plan for automated energy accounting

Automated energy accounting is a critical component of modern electricity distribution systems. It allows for accurate and efficient tracking of energy usage, which helps identify energy waste, reduce energy consumption, and improve billing accuracy. Mentioned below is an action plan for implementing automated energy accounting in an electricity distribution company.

Step 1: Evaluate Current Energy Accounting System

The first step is to evaluate the current energy accounting system to identify areas that need improvement. This will involve reviewing the current processes, systems, and data management practices. The evaluation should consider the following factors:

- Accuracy of billing and metering
- Timeliness of bill generation
- Data management practices
- Energy usage tracking capabilities
- Customer feedback and complaints

Step 2: Identify Automated Energy Accounting System Requirements

After evaluating the current energy accounting system, the next step is to identify the requirements for an automated energy accounting system. This will involve considering the following factors:

- Energy usage tracking capabilities
- Billing accuracy and timeliness
- Integration with existing systems
- Data management capabilities
- Scalability and flexibility

Step 3: Research and Select an Automated Energy Accounting System

Once the requirements are identified, the next step is to research and select an automated energy accounting system. This will involve reviewing available options and selecting a system that meets the identified requirements. The selected system should have the following features:

- Real-time energy usage tracking
- Automated billing and metering
- Data management and analysis capabilities

- Integration with existing systems
- User-friendly interface

Step 4: Develop Implementation Plan

After selecting an automated energy accounting system, the next step is to develop an implementation plan. This will involve determining the following:

- Timeline for implementation
- Resource requirements
- Roles and responsibilities
- Training requirements
- Data migration plan

Step 5: Implementation and Testing

Once the implementation plan is developed, the next step is to implement and test the automated energy accounting system. This will involve the following:

- Installation and configuration of the system
- Data migration from the old system to the new system
- User training
- System testing

Step 6: Rollout and Monitoring

After successful testing, the next step is to rollout the automated energy accounting system to all customers. This will involve communicating the changes to customers and ensuring a smooth transition. Once rolled out, the system should be continuously monitored to identify any issues and improve the system's performance.

In conclusion, implementing an automated energy accounting system can help Tata Power DDL improve billing accuracy, reduce energy waste, and enhance data management capabilities. The action plan outlined above provides a framework for implementing an automated energy accounting system in an electricity distribution company. By following this plan, the company can successfully implement the system and achieve its energy accounting goals.

Annexures

Annexure I - Introduction of Verification Firm

We A-Z Energy Engineers Pvt. Ltd. provides consultancy services in the areas of Energy Management while conducting Energy Audits in all segments of energy sector. For conducting detailed Energy Audits, Energy Audits under PAT (Mandatory and M&V), we have a pool of experienced BEE Accredited & Certified Energy Auditors, Electrical Engineers, Mechanical Engineers and Technicians having experience of more than 32 years. The Energy Audits are being carried out with sophisticated instruments namely Power-Analyzer, Flue Gas Analyzer, Ultra-sonic flow meter, Techo-meter, Anemometer, Hego-Meter, Digital Thermometer, Thermographic Camera's, Lux Meter, Leak detectors. Laser gun etc. etc.

Objective

- To carry out and take ahead the business of Energy Efficiency and climate change including promotion and dissemination of energy efficient product and services.
- To disseminate the culture of safe manufacturing and Services through safety audits and trainings.
- To facilitate implementation of energy efficiency projects for Demand Side Measures including optimization of energy mix for industries, railways, building sector, lighting, HVAC etc.
- To facilitate implementation of schemes, programs and policies of central and state governments or its agencies applicable for enhancing energy efficiency.
- To provide consultancy services in the field of Clean Development Mechanism and Renewable Energy Certificate projects, Carbon Markets, Demand Side Management, Energy Efficiency, Climate change and other related areas.
- To identify and impart training to build the capacity of stakeholders in the field of Energy Efficiency and safe practices in Industry.
- To act as a resource center in the field of Energy Efficiency and take up the activities of Capacity Building Training and other related activities.

Vision

- To make use of energy sustainable.
- To create and sustain markets for energy efficiency in India
- To facilitate energy efficiency improvement through private sector investments in energy efficiency.

Mission

- To assist all stakeholders in implementing energy efficiency and realizing savings.
- To create awareness regarding merits of improvement of energy efficiency and safety practices in private and public sector.

We are Accredited Energy Auditor from BEE, also empaneled by BEE for PAT M & V Audits and Mandatory Energy Audit Projects. A-Z Energy Engineers Pvt. Ltd. has been short listed by Bureau of Energy Efficiency as an Energy Service Company (ESCO), it is an ISO 9001:2015 certified company. We have completed more than 1800 nos. projects, including 75 PAT projects.

Dr. P.P. Mittal the Founder Director of A-Z Energy Engineers Pvt. Ltd. was awarded by Govt. of India in National Energy Conservation Award 2013, 2015 & 2016. MSME Ministry Govt. of India awarded "Best Services Providing Company" it was awarded by Hon'ble Prime Minister of India.

a) Name of the Firm

Name of Accredited Firm	Accredited Energy Auditor
A-Z energy Engineers Private Limited	Dr P.P Mittal (AEA 011)

b) Composition of Team

Sr.No.	Name	Qualification	Registration No	Experience (In Years)/Sector
1	Dr. P.P Mittal	Ph.D, MBA		+45 Years
2	Mr. Vipon Chanda	Sector Expert		30
3	Mr. V.P Sharma	B. Tech, EA		32 Years
4	Mr. Alok Kumar Tiwari	Energy auditor		9 Years
5	Mr. Pankaj Chauhan	Engineer		8 Years

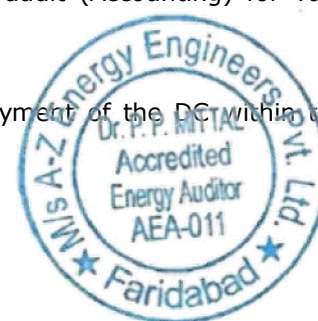
c) Registration No.

EmAEA 0024

d) Undertaking

We A-Z Energy Engineers Pvt. Ltd. hereby confirm that our AEA and any of the audit team member mentioned in this report has conducted mandatory annual energy audit (Accounting) for Tata Power-DDL, Delhi (hereafter called as DC).

We also confirm that none of our team member was in the employment of the DC within the previous four years.



Authorized Signatory


Dr. P.P Mittal

Annexure II - Minutes of Meeting with the DISCOM team

Minutes of Meeting with TATA Power Delhi Distribution Ltd., New Delhi & A-Z Energy Engineers Pvt. Ltd., New Delhi

TATA Power Delhi Distribution Ltd

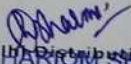
AZ Energy Engineers Pvt. Ltd.

The Audit team of A-Z Energy Engineers Pvt. Ltd., New Delhi visited EAG office of Tata Power-DDL in month of May 2025. The necessary verifications related to annual energy audit were also carried out at EAG office/ site offices of Tata Power-DDL. The Annual Energy Audit was conducted as per the provisions stipulated under BEE (Manner and Intervals for conduct of Energy Audit in electricity distribution Companies) Regulation 2021 and its amendment.

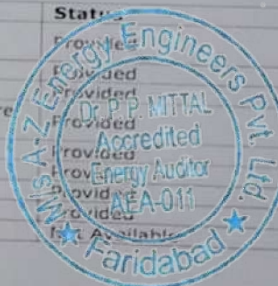
Following are the key observations during audit.

- BEE Proforma for FY 2024-25 was filled by Tata Power -DDL.
- Tata Power-DDL has provided the documents for Power purchase, Energy Input/Billed i.e. Fact sheet, ARR (Annual Revenue Requirement) (FY 24-25) Petition filed.
- Tata Power-DDL has provided details for action plan to reduce losses.
- Verified T&D losses, AT&C losses & Collection Efficiency is 5.61%, 5.55% & 100.06% respectively based on the filled in proforma and verified source documents.
- Tata Power-DDL has 100% metering available at 11/33/66 KV system, 100% metering at consumer end and 77% metering available at DT.

S. No.	Data Required for Annual Energy Audit	Status
1	Complete filled in Proforma for the year 2024-25 (Annually)	Provided
2	Quarter wise report FY 2024-25	Provided
3	Action plan to reduce the T&D and AT&C losses	Provided
4	Assets details matched with the proforma infrastructure sheets	Provided
5	Verified T&D and AT&C losses of previous years	Provided
6	Subsidy category Wise	Provided
7	DT Wise Losses	Provided
8	Feeder wise Losses	Provided
9	Voltage wise Input Energy	Not Available


Tata Power Delhi Distribution Limited
HARROW SHARMA
E. Code : 90231
AGM- Energy Audit Group
Tata Power Delhi Distribution Limited


A-Z Energy Engineers Pvt. Ltd.



Annexure III - Check List prepared by auditing Firm

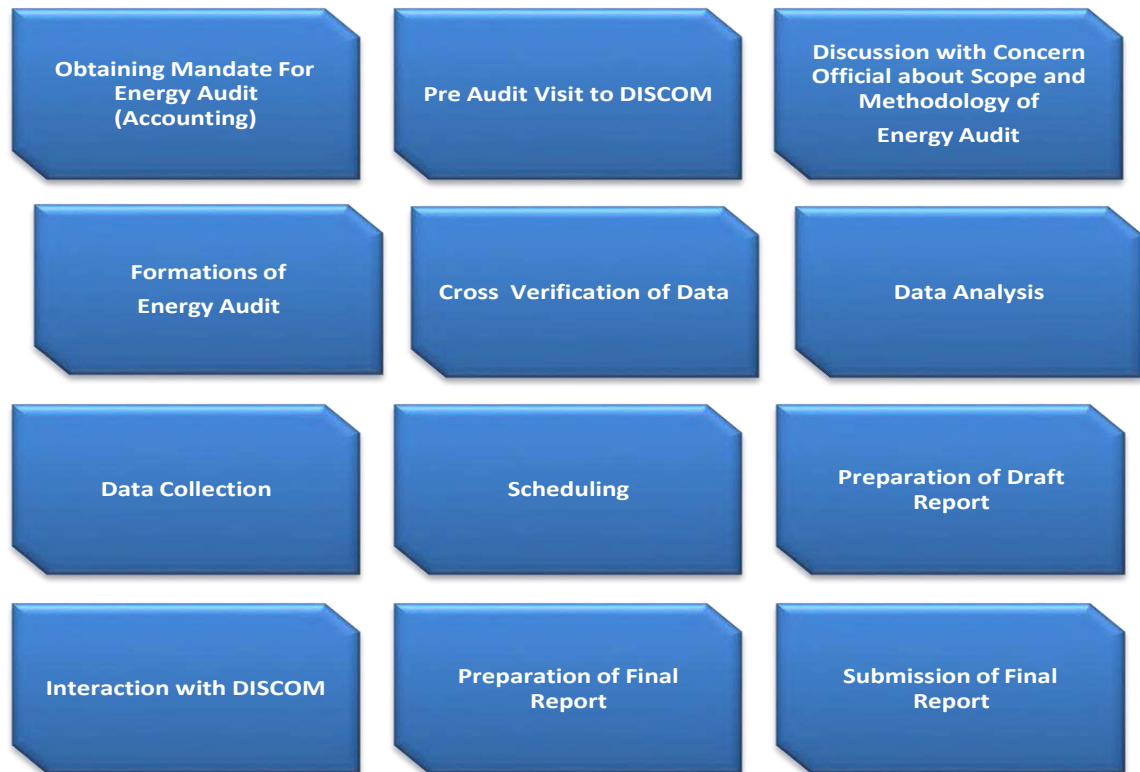
An annual energy audit checklist has been used to assess the energy efficiency of Tata Power DDL based on equipment's, Accredited Energy Audit has developed the checklist to identify opportunities for energy cost reduction and recommend solutions.

► Following information has been collected for T&D system with documentary evidence

- Month wise input and billed energy.
- T&D losses computation approach.
- Un-metered energy consumption
- Internal field audit report of input and billed energy.
- Performance of discom on T&D and AT&C losses.
- Outcome of internal audit.
- Measures taken to reduce losses.
- Zone/circle/Division/Sub-division wise loss computation.
- Reduction achieved, measures adopted for energy conservation and quantity of energy saved.
- Category wise consumers energy input and energy billed details.
- Category wise consumers total no. of connections and connected load and % connected load
- Bifurcation of Input and Billed Energy (Circle & Division Wise).
- Write up on procedure followed for technical loss analysis.
- Overloaded Feeder data

Annexure IV - Brief Approach, Scope & Methodology for audit

Scope of annual energy accounting is as per guidelines and notification from Bureau of Energy Efficiency, New Delhi dated 6th October, 2021.



Annexure V - Infrastructure Details

The infrastructure details of the DISCOM are as shown in the table below:

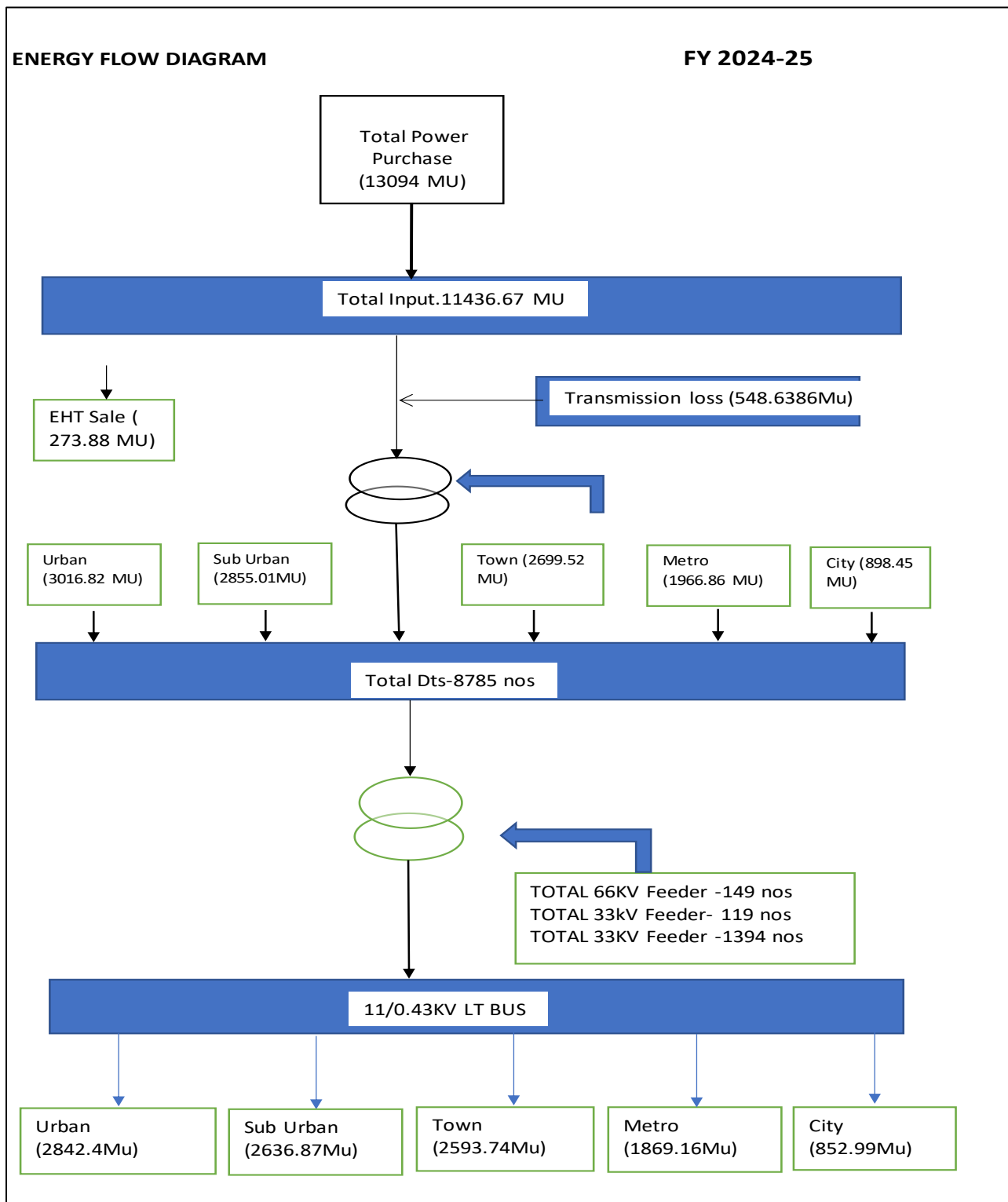
Table 44: Infrastructure details

Parameters	Total	Covered during in audit	Verified by Auditor in Sample Check	Remarks (Source of data)
Number of circles	5	5	Sample	
Number of divisions	12	12	Sample	
Number of sub-divisions	31	31	Sample	
Number of feeders	20003	20003	Sample	
Number of DTs	8785	8785	Sample	
Number of consumers	2102837	2102837	Sample	

Table 45: Metering details

Parameters	66kV and above	33kV	11/22kV	LT
Number of conventional metered consumers	0	0	0	1541229
Number of consumers with 'smart' meters	1	2	552	5,29,083
Number of consumers with 'smart prepaid' meters	0	0	0	10094
Number of consumers with 'AMR' meters	7	4	412	19,241
Number of consumers with 'non-smart prepaid' Meters	0	0	0	3190
Number of unmetered consumers	0	0	0	0
Number of total consumers	8	6	964	2102837
Number of conventionally metered Distribution Transformers	0	0	0	154
Number of DTs with communicable meters	0	0	0	6567
Number of unmetered DTs	0	0	0	2064
Number of total Transformers	0	0	0	8785
Number of metered feeders	149	119	1394	0
Number of feeders with communicable meters	149	119	1394	0
Number of unmetered feeders	0	0	0	18341
Number of total feeders	149	119	1394	18341
Line length(cktkm)	1841			
Length of Aerial Bunched Cables(cktkm)	5934.3			
Length of Underground Cables(cktkm)	6869.2			

Annexure VI - Electrical Distribution System



Annexure VII - Power Purchase Details

a) Input Purchase Power generating station for FY 2024-25

Source wise/generating station wise power purchase, contracted capacity, RPO obligation met by the DISCOM, etc.

S.No.	Name of Generation Station	Generation Capacity (In MW)	Type of Station Generation	Type of Contract	Type of Grid
1	NTPC AravaliJhajjar	613	Coal	25 Years	Inter State
2	NTPC Dadri NCTPS(Th.) Stage II	9	Coal	25 Years	Inter State
3	NTPC Kahalgaon II	48	Coal	25 Years	Inter State
4	NTPC Singrauli STPS	46	Coal	30 Years	Inter State
5	NTPC Rihand STPS-II	39	Coal	25 Years	Inter State
6	NTPC Rihand STPS-I	31	Coal	28 Years	Inter State
7	NTPC Dadri GPS	28	Gas	25 Years	Inter State
8	NTPC Auriya GPS	22	Gas	35 Years	Inter State
9	NTPC Kahalgaon I	16	Coal	Till change of allocation by MoP in lieu of Tala	Inter State
10	NTPC ANTA GPS	14	Gas	25 Years	Inter State
11	NTPC Unchahaar-II TPS	14	Coal	25 Years	Inter State
12	NTPC Unchahaar-III TPS	9	Coal	25 Years	Inter State
13	NTPC Unchahaar-I TPS	7	Coal	27 Years	Inter State
14	NTPC Farakka	7	Coal	Till change of allocation by MoP in lieu of Tala	Inter State
15	NTPC Singrauli Small Hydro	2	RE	35 Years	Inter State
16	NHPC Dulhasti	15	Hydro	35 Years	Inter State
17	NHPC Parbati III	20	Hydro	40 Years	Inter State
18	NHPC Bairasiul	6	Hydro	25 Years	Inter State
19	NHPC Tanakpur	3	Hydro	35 Years	Inter State
20	NHPC Chamera -I	13	Hydro	35 Years	Inter State
21	NHPC Chamera-II	12	Hydro	35 Years	Inter State
22	NHPC Chamera-III	9	Hydro	35 Years	Inter State
23	NHPC URI-I	16	Hydro	35 Years	Inter State
24	NHPC Uri-II	10	Hydro	40 Years	Inter State
25	NHPC Dhauliganga	11	Hydro	35 Years	Inter State
26	NHPC Sewa II	5	Hydro	35 Years	Inter State

S.No.	Name of Generation Station	Generation Capacity (In MW)	Type of Station Generation	Type of Contract	Type of Grid
27	Narora APS	14	Nuclear	15 Years	Inter State
28	RAPP 5&6	17	Nuclear	15 Years	Inter State
29	CLP Jhajjar	132	Coal	25 Years	Inter State
30	Maithon Power Limited	300	Coal	30 Years	Inter State
31	SECI Solar (Renewable)	20	RE	25 Years	Inter State
32	Tala HEP	9	Hydro	As per MoP/NRPC order	Inter State
33	CTPS 7	92	Coal	25 Years	Inter State
34	CTPS 8		Coal	25 Years	Inter State
35	MTPS 6	31	Coal	25 Years	Inter State
36	Sasan	27 MW to 136 MW	Coal	25 Years	Inter State
37	Nathpa Jhakri HPS	44	Hydro	35 Years	Inter State
38	Tehri HPP	19	Hydro	35 Years	Inter State
39	Koteshwar HEP	12	Hydro	35 Years	Inter State
40	Pragati- I	64	Gas	25 Years	Intra State
41	Pragati III	298	Gas	25 Years	Intra State
42	IPGCL GT	27	Gas	10 Years	Intra State
43	Delhi Municipal Solid Waste Solutions Ltd. (Bawana) (Renewable)	7	RE	20 Years	Intra State
44	Nanti Hydro Power Pvt. Ltd. (Renewable)	14	RE	20 Years	Inter State
45	Surya kanta Hydro energies Pvt. Ltd. (Renewable)	14	RE	20 Years	Inter State
46	Timarpur Okhla Waste management co. Ltd. (Renewable)	6	RE	20 Years	Intra State
47	Sun Edison	180	RE	20 Years	Inter State
48	Taranda	13	RE	20 Years	Inter State
49	SECI WIND	50	RE	25 Years	Inter State
50	SECI 200 MW (SBSR)*	100	RE	25 Years	Inter State
51	Tehkhand Waste to Electricity Project Ltd	9	RE	25 Years	Intra State
52	Cosmos Hydro	20	RE	20 Years	Inter State
53	Medium Term 200 MW (Kameng HEP)	200	Hydro	5 Years	Inter State
54	SECI Solar (ACME Solar)	100	RE	25-Year	Inter State

Remarks:-

1. *SBSR 200 MW: - Out of total allocation of 200 MW, only 100 MW has been commissioned till Jun 2023
2. Contract period beyond useful life may depend upon agreement/Regulatory orders.
3. NTPC >25 years few stations under litigation before various forums.

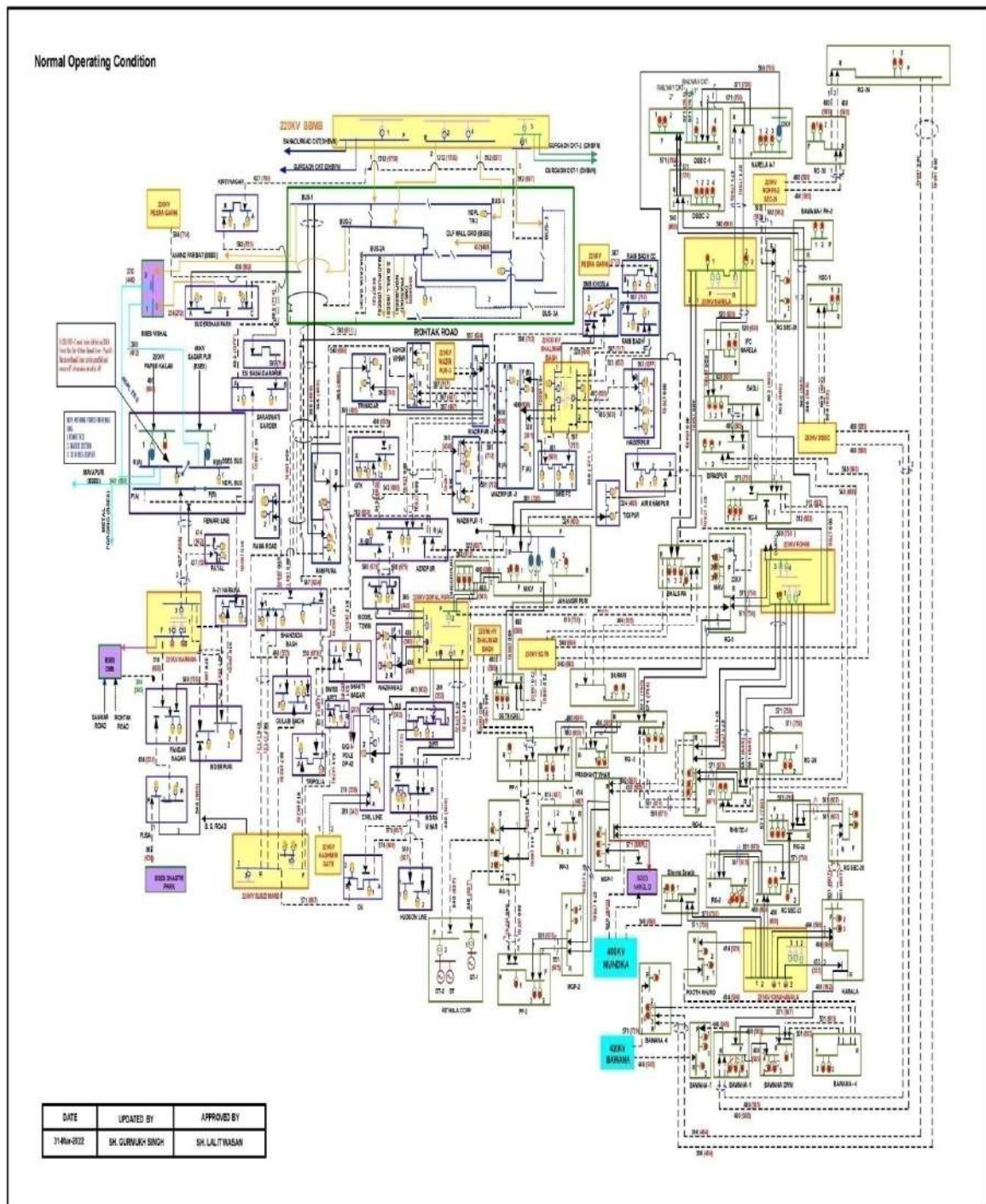
b) Embedded Power generation for FY 2024-25

S. No	Name of Generation Station	Generation Capacity (In MW)	Type of Station	Type of Contract	Type of Grid	Voltage Level (KVA)	Received at Circle (In MU)
1	Poothkhurd 54KW SPV PLANT	1	Renewable	Captive Generation	On-Grid System	11 kV	1.233872
2	Cenpeid	0.225	Renewable	Captive Generation	On-Grid System	415 V	0.229048
3	Corporate Office	0.06	Renewable	Captive Generation	On-Grid System	415 V	0.000966
4	Cennet	0.055	Renewable	Captive Generation	On-Grid System	415 V	0.029635
5	KPM	0.054	Renewable	Captive Generation	On-Grid System	415 V	0.044651
6	Narela A-7	0.05	Renewable	Captive Generation	On-Grid System	415 V	0.041502
7	BAWANA Clear Water (45 KWp)	0.049	Renewable	Captive Generation	On-Grid System	415 V	0.029363
8	DSIDC-2 Narela	0.043	Renewable	Captive Generation	On-Grid System	415 V	0.02434
9	GTK solar	0.02112	Renewable	Captive Generation	On-Grid System	415 V	0.022242
10	RG-2	0.025	Renewable	Captive Generation	On-Grid System	415 V	0.026213
11	RG-24	0.025	Renewable	Captive Generation	On-Grid System	415 V	0.027726
12	RG-5	0.024	Renewable	Captive Generation	On-Grid System	415 V	0.024754
13	RG-22	0.015	Renewable	Captive Generation	On-Grid System	415 V	0.010262
14	RG-23	0.004	Renewable	Captive Generation	On-Grid System	415 V	0.003613
15	Net Metering						4.500209

Annexure VIII - Single Line Diagram (SLD)

The SLD of the DISCOM is as shown below:

Figure 15: Single Line Diagram (SLD) of Tata Power DDL



Annexure IX - Category of service details (With Consumer and voltage-wise)

Type of consumers as per different voltage levels & number of consumers are shown in below table:

Table 46: Category of service details

Type of Consumers	Category of Consumers (EHT/HT/LT/Others)	Voltage Level	No of Consumers	Total Consumption (In MU)
Domestic			1792580	5451.54
Commercial			296695	4575.13
Water Supply				
Public Lighting				
HT Industrial				
Industrial (Small)				
HT Commercial				
Agricultural			4471	16.14
EV				
Others-3 (if any, specify in remarks)			9091	752.3504038
TOTAL			2102837	10795.16

Annexure X - Field Verification data and reports

The field inspection details are shown in the below table:

Table 47: Field inspection details







Annexure XI - List of documents verified with each parameter

The documents verified are listed in the below table:

Table 48: List of documents verified with each parameter

S. No	Data Required for Annual Energy Audit as per BEE regulation	Status	Remark
1	Complete filled in Proforma for the year 2024-2025 (Annually)	Provided	Data Attached
2	Supporting Data with Month wise breakup	Provided	NA
A	Purchase Energy		
B	Net Input in Discom		
C	Billed Energy		
D	Billed Revenue		
E	Revenue Collected		
F	Energy Export to other		
G	Transmission loss calculation Methodology	NA	NA
3	Feeder wise input, Billed Energy, T&D & AT&C losses	Sample Provided	Data Attached
4	Action plan to reduce the T&D and AT&C losses	Provided the various schemes	Action plan to reduce AT&C losses
5	Assets details matched with the proforma infrastructure sheets	Provided	Data Attached
6	DT Wise loss provided	Provided	Proforma Attached
7	Energy Flow Diagram	Data Attached	Data Attached
8	Subsidy category Wise (BEE Guideline proforma)	Format Provided	Data Attached in BEE Guideline proforma
9	High Loss area T&D & AT&C action plan to reduce losses	Provided	Data Attached
10	Power Map	Provided	Data Attached
11	Current status of Metering status at Various Voltage level of Discom	Functional & Non Functional meter details provided	100 % metered (operational)
12	Status of default meter (non-functional meters)	Provided	Data Attached

Annexure XII - Brief Description of Unit

Tata Power Delhi Distribution Limited [Tata Power-DDL] is a joint venture between Tata Power and the Government of NCT of Delhi with the majority stake being held by Tata Power Company (51%).

Tata Power-DDL is acknowledged for its consumer-friendly practices. Since privatization, the Aggregate Technical & Commercial (AT&C) losses in Tata Power-DDL areas have shown a record decline.

To ensure reliable power supply and to provide best in class service to its consumers, Tata Power-DDL has implemented several world-class technologies such as Advanced Distribution Management system or ADMS which is designed to replace the conventional SCADA-DMS-OMS system with features like real-time integration of Smart Meter Data / Distributed Generation integration and single data model from GIS , Integrated Geographical Information System (GIS) for instant services, Advanced Metering Infrastructure (AMI), Automated Demand Response (ADR), Smart Street Light Management system, Field Force Automation, Upgraded Network, Integrated Toll Free Helpline No. 19124, etc.

Tata Power-DDL is the first Indian utility to be a member of Global Intelligent Utility Network Coalition (GIUNC) which is a coalition of 14 power utilities worldwide and is working towards accelerating the development of common standards, technology solutions and processes for intelligent networks.

Tata Power-DDL provides various facilities and services to its consumers for their ease and convenience such as 24X7 Integrated Helpline, Mobile Application for both iOS and Android users, bilingual website, Multiple Payment Avenue, End to End online services for New Connection, etc.

Tata Power-DDL's contribution towards improving the ease of getting electricity connection through process simplification under 'Ease of doing Business' has contributed to raise India's ranking in 'Getting Electricity' , from 137th ranking in 2015 to an impressive 22nd ranking in 2019.

TATA Power-DDL has also added solar generation as a part of its sustainable initiatives since 2008, and has installed 14Nos. of Rooftop Grid Interactive Solar PV Generation Power Plants in its Licensed Area with a total generation capacity of 1.65 MWp, and total generation of 1.7MUs in FY 24-25. Wherein for Rooftop Solar panel installations, TATA Power-DDL has almost 4283 nos. of Consumer Solar rooftop plants. The company is already working on setting up Smart Grid with the integration of Roof Top Solar, Energy Storage, E-charging of Electric Vehicles, Home Automation etc. in its network.

Tata Power-DDL's change management experience, distributed leadership system, adoption of latest technologies; robust competence development process and innovative & open work culture are the key strategic boosters which helped in building and sustaining competitive

advantage in the changing business scenario. A journey which began 2 decades ago for empowering the consumers in Delhi now holds the potential to transform the distribution sector in India and similarly help utilities across the globe. Tata Power-DDL has a presence in India in nearly 20+ States and working with 30+ Discoms including Goa, Haryana, Uttar Pradesh, Chhattisgarh, Odisha, Ajmer etc. as well as in International cities such as Benin, Eko, Kaduna, Kano in Nigeria etc.

Tata Power-DDL is focused and committed to the road ahead and is exploring new opportunities to replicate its experience of distribution reforms both in India and abroad. It is leveraging its unique learning and skillsets solely and in collaboration with leading utilities and technology providers like GE, IBM, Enel, Omron, 3M, Panasonic, AES, Mitsubishi etc. in the areas of communications & smart grid technology, change management, consumer service delivery and business process re-engineering. Tata Power-DDL has also collaborated with leading international and national Institutions like Harvard, MIT, Ryerson University, IIT Delhi, Punjab Engineering College, Delhi University, Netaji Subhas Institute of Technology etc. to carry out research activities in energy space.

World Class Technologies, Tata Power DDL	
Advanced Distribution Management System (ADMS)	Advance Distribution Management System (ADMS) is a single integrated system which facilitates advanced monitoring, analysis, as well as control and planning, thereby enabling to enhance the reliability, safety and efficiency of the power for the consumers. This system has advanced features of reporting outages and intimating to customers upfront. This system facilitates system controller as well as maintenance team for faster restoration of supply.
Geographical information System (GIS)	Geographical Information System (GIS) is a foundational technology and single source to have repository of network, asset and consumer indexing for Tata Power-DDL. The data of this system gets integrated with ADMS, FFA, ERP, AMI, etc. for successful functioning of respective systems. This system enables delivering of results in terms of reliable & quality power along with advanced services and timely information to the consumers.
Smart Meter	Smart Meters are basic building blocks of Smart Grid. This technology encompasses Communication System (RF/ NB-IoT/ 4G based) and Data Handling Technologies (Meter Data Management System). Tata Power-DDL is implementing Smart Metering Technology (Advanced Metering Infrastructure-AMI) to bring operational efficiency in different IT and OT domains. This technology will bring transparency to consumers in terms of their consumption per month and monitoring of other critical parameters like MDI and PF on an instant basis. For Tata Power-DDL, it enables easy detection of pilferage and loss reduction. This last mile link will complete Smart Grid implementation by Tata Power-DDL.

World Class Technologies, Tata Power DDL	
Smart Street Light Management system	Tata Power-DDL jointly is working on a project for achieving a reduction in the demand of street lighting which coincides with peak load, thereby reducing the overall peak demand, improving the lux levels, improving the power factor and checking the carbon foot print as a responsibility to the society. This will translate into considerable saving to the exchequers. This system is entirely managed through a Smart Centralized Control & Monitoring System which can identify partially or completely affected streetlight circuits on a real-time basis and the type/nature of fault, thereby alerting the maintenance team without any requirement of consumer complaints for such purpose. This will enhance safety & security of general public. It can also detect pilferage from street light circuits and generate alerts.
Field Force Automation	Customer service through mobile workforce is the key to exceed the expectations of the consumer. Field Force Automation (FFA) is a system which optimizes the various tasks in hand and schedules & dispatches the nearest Crew to provide faster service to the consumers. This system not only enhances the service level but also completely tracks the allocation of workforce.

Annexure XIII - List of Parameters arrived through calculation or formulae with list of documents as source of data

Ideally, reduction of technical losses should be the parameter for evaluation of performance of Discom's sector. However, the technical losses of the Discom are not available and it involves a cumbersome process to calculate the technical losses, which varies based on various factors like loading pattern etc.

Now, only the T&D losses and AT&C losses are available as the performance parameter for achieving energy efficiency by the DISCOM.

Table 49: Formulas used to derive the parameters

Parameter	Formula	Data Source
AT&C Losses %	$\{1 - (\text{Billing Efficiency} \times \text{Collection Efficiency})\} \times 100$	Commercial
T&D Losses %	$\{1 - (\text{Total energy Billed} / \text{Total energy Input in the system})\} \times 100$	Commercial
Billing Efficiency	Total Energy Billed to Consumers (kWh) / Total Energy Input (kWh)	Commercial
Collection Efficiency	Revenue Collected (In Rupees) / Billed Amount (In Rupees)	Commercial

Annexure XIV - Detailed Formats

A. General Information

General Information				
1	Name of the DISCOM	TATA POWER DELHI DISTRIBUTION LIMITED		
2	i) Year of Establishment	2002-03		
	ii) Government/Public/Private	Joint venture		
3	DISCOM's Contact details & Address			
i	City/Town/Village	New Delhi		
ii	District	Delhi		
iii	State	Delhi	Pin	110009
iv	Telephone	011-66112202	Fax	011-27468042
4	Registered Office			
i	Company's Chief Executive Name	Dwijadas Basak		
ii	Designation	CEO		
iii	Address	NDPL House, Hudson Lines, Kingsway Camp, Delhi-09		
iv	City/Town/Village	Delhi	P.O.	GTB Nagar
v	District			
vi	State	Delhi	Pin	110009
vii	Telephone	011-66112202	Fax	011-27468042
5	Nodal Officer Details*			
i	Nodal Officer Name (Designated at DISCOM's)	H.C. Sharma		
ii	Designation	General Manager		
iii	Address	NDPL House, Hudson Lines, Kingsway Camp, Delhi-09		
iv	City/Town/Village	Delhi	P.O.	GTB Nagar
v	District			
vi	State	Delhi	Pin	110009
vii	Telephone	91-1166050595	Fax	
6	Energy Manager Details*			
i	Name	B L Gupta		
ii	Designation	AGM	Whether EA or EM	EM
iii	EA/EM Registration No.	EA-2317		
iv	Telephone		Fax	
v	Mobile	9971392990	E-mail ID	bl.gupta@tatapower-ddl.com
7	Period of Information			
	Year of (FY) information including Date and Month (Start & End)	01/Apr/2024 to 31/March/2025		

B. Summary Sheet

Performance Summary of Electricity Distribution Companies			
1	Period of Information Year of (FY) information including Date and Month (Start & End)	01/Apr/2024 to 31/March/2025	
2	Technical Details		
(a)	Energy Input Details		
(i)	Input Energy Purchase (From Generation Source)	Million kwh	13094.00
(ii)	Net input energy (at DISCOM Periphery after adjusting the transmission losses and energy traded)	Million kwh	11436.67
(iii)	Total Energy billed (is the Net energy billed, adjusted for energy traded))	Million kwh	10795.16
(b)	Transmission and Distribution (T&D) loss Details	Million kwh	641.51
	Collection Efficiency	%	5.61%
(c)	Aggregate Technical & Commercial Loss	%	100.06%
		%	5.55%

I/We undertake that the information supplied in this Document and Pro-forma is accurate to the best of my knowledge and if any of the information supplied is found to be incorrect and such information result into loss to the Central Government or State Government or any of the authority under them or any other person affected, I/we undertake to indemnify such loss.

Authorised Signatory and Seal

HC Sharma
Name of Authorised Signatory
Name of the DISCOM:
Full Address:-

Pushendra Kumar
Pushendra Kumar
HoD-EAG
E.No-94669

Signature:-
Name of AEA*: Bqn wari Lal Gupta
Registration Number: EA-2317

HAUSOM SHARMA
E. Code: 180231
AGAR- Energy Audit Group
Tata Power Delhi Distribution Limited

Seal

Dr. P. P. MITTAL
Accredited
Energy Auditor
AEA-011
Faridabad

C. Infrastructure Details

Form-Details of Input Infrastructure					
1	Parameters	Total	Covered during in audit	Verified by Auditor in Sample Check	Remarks (Source of data)
i	Number of circles	5	FY 24-25		Organizational
ii	Number of divisions	12	FY 24-25		Organizational Structure
iii	Number of sub-divisions	31	FY 24-25		Organizational Structure
iv	Number of feeders	20003	FY 24-25		GIS database
v	Number of DTs	8785	FY 24-25		GIS database
vi	Number of consumers	2102837	FY 24-25		SAP System
2	Parameters	66kV and above	33kV	11/22kV	LT
a. i.	Number of conventional metered consumers	0	0	0	1541229
ii	Number of consumers with 'smart' meters	1	2	552	5,29,083
iii	Number of consumers with 'smart prepaid' meters	0	0	0	10094
iv	Number of consumers with 'AMR' meters	7	4	412	19,241
v	Number of consumers with 'non-smart prepaid' meters	0	0	0	3190
vi	Number of unmetered consumers	0	0	0	0
vii	Number of total consumers	8	6	964	2102837
b.i.	Number of conventionally metered Distribution Transformers	0	0	0	154
ii	Number of DTs with communicable meters	0	0	0	6567
iii	Number of unmetered DTs	0	0	0	2064
iv	Number of total Transformers	0	0	0	8785
c.i.	Number of metered feeders	149	119	1394	0
ii	Number of feeders with communicable meters	149	119	1394	0
iii	Number of unmetered feeders	0	0	0	18341
iv	Number of total feeders	149	119	1394	18341
d.	Line length (ct km)		1841		
e.	Length of Aerial Bunched Cables		5934.3		
f.	Length of Underground Cables		6869.2		

3	Voltage level	Particulars	MU	Reference	Remarks (Source of data)
i	66kV and above	Long-Term Conventional		Includes input energy for franchisees	Unable to provide deta
		Medium Conventional			
		Short Term Conventional			
		Banking			
		Long-Term Renewable energy			
		Medium and Short-Term RE		Includes power from bilateral/ PX/ DEEP	
		Captive, open access input		Any power wheeled for any purchase other than sale to DISCOM. Does not include input for franchisee.	
		Sale of surplus power			
		Quantum of inter-state transmission loss		As confirmed by SLDC, RLDC etc	
		Power procured from inter-state sources	0	Based on data from Form 5	
		Power at state transmission boundary	0		
ii	33kV	Long-Term Conventional			
		Medium Conventional			
		Short Term Conventional			
		Banking			
		Long-Term Renewable energy			
		Medium and Short-Term RE			
		Captive, open access input			
		Sale of surplus power			
		Quantum of intra-state transmission loss	0		
		Power procured from intra-state sources	0		
iii		Input in DISCOM wires network	0		
iv	33 kV	Renewable Energy Procurement			
		Small capacity conventional/ biomass/ hydro plants Procurement			
		Captive, open access input			
v	11 kV	Renewable Energy Procurement			
		Small capacity conventional/ biomass/ hydro plants Procurement			
		Sales Migration Input			
vi	LT	Renewable Energy Procurement			
		Sales Migration Input			
vii		Energy Embedded within DISCOM wires network	0		
viii		Total Energy Available/ Input	0		
4	Voltage level	Energy Sales Particulars	MU	Reference	
i	LT Level	DISCOM' consumers		Include sales to consumers in franchisee areas, unmetered consumers	Majority of feeders are
		Demand from open access, captive		Non DISCOM's sales	
		Embedded generation used at LT level		Demand from embedded generation at LT level	
		Sale at LT level	0		
		Quantum of LT level losses	0		
		Energy input at LT level			
ii	11 kV Level	DISCOM' consumers		Include sales to consumers in franchisee areas, unmetered consumers	Majority of feeders are
		Demand from open access, captive		Non DISCOM's sales	
		Embedded generation at 11 kV level used		Demand from embedded generation at 11kV level	
		Sales at 11 kV level	0		
		Quantum of Losses at 11 kV	0		
		Energy input at 11 kV level			
iii	33 kV Level	DISCOM' consumers		Include sales to consumers in franchisee areas, unmetered consumers	This comprises of sales
		Demand from open access, captive		Non DISCOM's sales	
		Embedded generation at 33 kV or below level		This is DISCOM and OA demand met via energy generated at same voltage level	
		Sales at 33 kV level	0		
		Quantum of Losses at 33 kV	0		
		Energy input at 33kV Level			This comprises of input
iv	> 33 kV	DISCOM' consumers		Include sales to consumers in franchisee areas, unmetered consumers	
		Demand from open access, captive		Non DISCOM's sales	
		Cross border sale of energy			
		Sale to other DISCOMs			
		Banking			
		Energy input at > 33kV Level			
		Sales at 66kV and above (EHV)	0		
		Total Energy Requirement	0		
		Total Energy Sales	0		
Energy Accounting Summary					
5	DISCOM	Input (in MU)	Sale (in MU)	Loss (in MU)	Loss %
i	LT		9482.62		
ii	11 Kv		1027.85		
iii	33 kv				
iv	> 33 kv		273.88		

Sub-total		0	0	0	100%	0	0	0	100%	0	0	0	0	100%	0	0%	0	0	0.00%	100%
76	Total	Residential	1792580	0	1792580	85%	3882135	0	3882135	56%	11436.67	5451.541	0	5451.54123	50%	641.505	0%	3878.90451	3887.53957	100.22%
		Agricultural	4471	0	4471	0%	34519	0	34519	0%		1613539	0	161353931	0%			12.6335463	124872656	98.84%
		Commercial/Industrial-LT	295912	0	295912	14%	2418313	0	2418313	35%		3937.02	0	3937.01998	35%			6183.75495	6185.85238	100.03%
		Commercial/Industrial-HT	783	0	783	0%	333144	0	333144	5%		638.113	0	638.112997	6%			949.018358	949.606253	100.05%
		Others	9091	0	9091	0%	308459	0	308459	4%		752.3504	0	752.350404	7%			788.151642	784.640438	99.55%
77	At company level		2102837	0	2102837	100%	6976.57	0	6976.57	100%	11436.67	10795.16	0	10795.16	100%	641.505	0%	11812.463	11820.136	100.06%

** Note - It shall be mandatory to record the energy supplied separately for each category of consumers which is being provided a separate rate of subsidy in the tariff, by the state government, so that the subsidy due for the electricity distribution company is quarterly calculated by multiplying the energy supplied to each of such category of consumers by the applicable rate of subsidy notified by the state government.

Color code	Parameter
	Please enter name of circle
	Please enter circle code
0	Please enter numeric value or 0
	Formula protected

I/We undertake that the information supplied in this Document and Pro-forma is accurate to the best of my knowledge and if any of the information supplied is found to be incorrect and such information result into loss to the Central Government or State Government or any of the authority under them or any other person affected, I/we undertake to indemnify such loss.

Authorised Signatory and Seal

Signature:
Name of Energy Manager:

Name of Authorized Signatory:	Registration Number:
Name of the DISCOM:	
Full Address:-	
Seal	
	
Pushpendra Kumar HoD-EAG E. No- 94669	HARIOM SHARMA E. Code- 90231 AGM- Energy Audit Group Tata Power Delhi Distribution Limited

Banwari Lal Gupta
EA-2317

E. Form Input Energy

Form-Input energy/Details of Input energy & Infrastructure			
A. Summary of energy input & Infrastructure			
S.No	Parameters	Period from 01/Apr/2024 to 31/March/2025	Remarks (Source of data)
A.1	Input Energy purchased (MU)	13094	Power purchase Invoice
A.2	Transmission loss (%)	4%	Power purchase Invoice
A.3	Transmission loss (MU)	548.6386	
A.4	Energy sold outside the periphery (MU)	1140	Power purchase Invoice
A.5	Open access sale (MU)	96.96865	From Commercial data
A.6	EHT sale	273.88	From Regulatory data
A.7	Net input energy (received at DISCOM periphery or at distribution point)-(MU)	11527.40	
A.8	Is 100% metering available at 66/33 kV (Select yes or no from list)	Yes	
A.9	Is 100% metering available at 11 kV (Select yes or no from list)	Yes	
A.10	% of metering available at DT	77%	25 KVA and above
A.11	% of metering available at consumer end	100%	From Billing data base
A.12	No of feeders at 66kV voltage level	149	GIS Data base
A.13	No of feeders at 33kV voltage level	119	GIS Data base
A.14	No of feeders at 11kV voltage level	1394	EAG data
A.15	No of LT feeders level	18341	GIS Data base
A.16	Line length (ckt. km) at 66kV voltage level	598.5	GIS Data base
A.17	Line length (ckt. km) at 33kV voltage level	540.9	GIS Data base
A.18	Line length (ckt. km) at 11kV voltage level	5469.2	GIS Data base
A.19	Line length (km) at LT level	8035.9	GIS Data base
A.20	Length of Aerial Bunched Cables	5934.3	GIS Data base
A.21	Length of Underground Cables	6869.2	GIS Data base
A.22	HT/LT ratio	0.822385	

B. Meter reading of Input energy at Injection points																					
S.No	Zone	Circle	Voltage Level (kVA)	Division (kVA)	Sub-Division (kVA)	Feeder ID	Feeder Name	Feeder Metering Status (Metered/ un-metered / AMI/AMR)	Status of Meter (Functional / Non functional)	Metering Date of last actual meter read in if communication	Feeder Type (Aggr. Distribution / Mixed)	Status of Communication			Period from...to...				Sales		Remarks (Source of data)
												% data received through automaticity if metered	Number of hours when meter was unavailable to communication	Total Number of hours in the period	Meter S. No	CT/PT ratio	Imports (MU)	Exports (MU)			
B.1			66				Narela T X 1	Metered	Functional		Mixed	ANI				4864963	1000	328.37	0.00		
B.2			66				Narela T X 2	Metered	Functional		Mixed	ANI				4865043	1000	359.51	0.00		
B.3			66				Narela T X 3	Metered	Functional		Mixed	ANI				4865052	1000	212.78	0.00		
B.4			11				LOCAL TR Narela	Metered	Functional		Mixed	ANI				4902583	-10	-0.26	0.00		
B.5			66				Railway Ckt-1	Metered	Functional		Mixed	ANI				4864992	-625	-44.38	0.00		
B.6			66				Railway Ckt-2	Metered	Functional		Mixed	ANI				4865039	-500	-3.73	0.00		
B.7			66				Gopal Pur T X 2	Metered	Functional		Mixed	ANI				4864976	2000	21.15	0.00		
B.8			33				Gopal Pur T X 1	Metered	Functional		Mixed	ANI				4902497	1000	292.95	0.00		
B.9			33				Gopal Pur T X 3	Metered	Functional		Mixed	ANI				4864924	1000	291.19	0.00		
B.10			11				LOCAL TR Gopalpur	Metered	Functional		Mixed	ANI				4865091	-7.5	-0.20	0.00		
B.11							T X 4 (160 MVA)	Metered	Functional		Mixed	ANI				00491809	1	323.44	0.00		
B.12			220				Kasmeri Gate 220 KV DMRC 2	Metered	Functional		Mixed	ANI				4902482	500	79.79	0.00		
B.13			220				Kasmeri Gate 220 KV DMRC 1	Metered	Functional		Mixed	ANI				5128479	1000	71.14	0.00		
B.14			33				Kasmeri Gate 33KV Civil Line-1	Metered	Functional		Mixed	ANI				4864861	999.99	40.54	0.00		
B.15			11				ISBT K Gate (E/S No II Mehawir Chowk)	Metered	Functional		Mixed	ANI				4865077	183.33	5.22	0.00		
B.16			33				Kasmeri Gate 33KV Civil Line-2	Metered	Functional		Mixed	ANI				4865184	2000	26.78	0.00		
B.17			33				Kasmeri Gate 30 MVA TR	Metered	Functional		Mixed	ANI				4864797	100	28.49	0.00		
B.18			11				LOCAL TR K Gate	Metered	Functional		Mixed	ANI				4902550	-7.5	-0.25	0.00		
B.19			11				Kasmeri Gate BUS COUPLER	Metered	Functional		Mixed	ANI				4902588	-400	-0.03	0.00		
B.20			33				O/G Payal Ckt Naraina	Metered	Functional		Mixed	ANI				4865057	300	68.19	0.00		
B.21			33				O/G REVARI LINE Naraina	Metered	Functional		Mixed	ANI				4865182	4000	54.19	0.00		
B.22			33				INDER PURI Ckt-1 Naraina	Metered	Functional		Mixed	ANI				4902486	150	63.39	0.00		
B.23			33				33KV 16 MVA TR-1 Naraina	Metered	Functional		Mixed	ANI				4864880	500	33.66	0.00		
B.24			33				33KV 16 MVA TR-2 Naraina	Metered	Functional		Mixed	ANI				4864860	500	40.22	0.00		
B.25			11				Gopal Nath Bazar (-ve) Naraina	Metered	Functional		Mixed	ANI				4865146	-33.333	-10.37	0.00		
B.26			11				LOCAL TR Naraina	Metered	Functional		Mixed	ANI				4902606	-100	-9.35	0.00		
B.27			33				INDER PURI Ckt-2 Naraina	Metered	Functional		Mixed	ANI				4865117	1833.33	73.66	0.00		
B.28			33				33KV Naraina Pandav Nagar Feeder	Metered	Functional		Mixed	ANI				4864846	1000	62.35	0.00		
B.29			33				O/G 33 KV Rama Road Rohtak Road	Metered	Functional		Mixed	ANI				4865123	1250	8.29	0.00		
B.30			33				O/G 33 KV Shahjadi Bagh-2 Rohtak	Metered	Functional		Mixed	ANI				4865152	1000	5.72	0.00		
B.31			33				O/G 33 KV Rampura-1 Rohtak Road	Metered	Functional		Mixed	ANI				5295199	937.5	89.19	0.00		
B.32			33				O/G 33 KV Rampura-2 Rohtak Road	Metered	Functional		Mixed	ANI				5295126	94	88.72	0.00		
B.33			11				BUS COUPLER Rohtak Road	Metered	Functional		Mixed	ANI				4902559	300	-1.37	0.00		
B.34			33				T X 3 Rohtak Road	Metered	Functional		Mixed	ANI				5252046	625	11.14	0.00		
B.35			33				O/G 33 KV DLF Kirti Nagar Rohtak	Metered	Functional		Mixed	ANI				4864821	150	90.33	0.00		
B.36			33				T X 1 Shalimar Bagh	Metered	Functional		Mixed	ANI				4864930	1000	297.34	0.00		
B.37			66				T X 2 Shalimar Bagh	Metered	Functional		Mixed	ANI				4864917	1000	258.81	0.00		
B.38			220				DMRC Jahangirpuri	Metered	Functional		Mixed	ANI				4902501	2222.22	17.28	0.00		
B.39			11				LOCAL TR SMB	Metered	Functional		Mixed	ANI				4902561	-7.5	-0.22	0.00		
B.40			33				T X 3 Shalimar Bagh	Metered	Functional		Mixed	ANI				4864922	1000	355.58	0.00		
B.41			220				DMRC SMB RSS	Metered	Functional		Mixed	ANI				4902484	500	21.54	0.00		
B.42			66				T X 4 Shalimar Bagh	Metered	Functional		Mixed	ANI				40003535	1000	0.00	0.00		
B.43			66				T X 1 Rohini 220KV	Metered	Functional		Mixed	ANI				4864964	1000	270.21	0.00		
B.44			66				T X 2 Rohini 220KV	Metered	Functional		Mixed	ANI				4865016	1000	264.97	0.00		
B.45			66				T X 3 Rohini 220KV	Metered	Functional		Mixed	ANI				4864997	1000	396.58	0.00		
B.46			66				T X 4 Rohini 220KV	Metered	Functional		Mixed	ANI				5295171	500	322.69	0.00		
B.47			11				LOCAL TR Rohini	Metered	Functional		Mixed	ANI				4865158	-2.5	-0.35	0.00		
B.48			66				T X 1 Karjawa	Metered	Functional		Mixed	ANI				4865041	1000	296.68	0.00		
B.49			66				T X 2 Karjawa	Metered	Functional		Mixed	ANI				4902499	1000	284.69	0.00		
B.50			11				LOCAL TR Karjawa	Metered	Functional		Mixed	ANI				4865071	-266.66	-0.30	0.00		
B.51			66				66KV DMRC MUNDKA	Metered	Functional		Mixed	ANI				Y0357713	-0.4	-7.86	0.00		
B.52			66				T X 3 Karjawa	Metered	Functional		Mixed	ANI				4864788	2000	647.80	0.00		
B.53			33				T X 2 Subzimandi	Metered	Functional		Mixed	ANI				4864896	2000	170.64	0.00		
B.54			33				O/G BG Rd-1 (To BSES)	Metered	Functional		Mixed	ANI				4865119	-1393.33	-53.67	0.00		
B.55			33				O/G BG Rd-2 (To BSES)	Metered	Functional		Mixed	ANI				4864825	-133.33	-50.77	0.00		
B.56			33				T X 1 Subzimandi	Metered	Functional		Mixed	ANI				4865012	1000	367.86	0.00		
B.57			11				LOCAL TR Subzimandi	Metered	Functional		Mixed	ANI				4902604	-100	-9.12	0.00		
B.58			66				I/C 100 MVA TR, No. 1 Bawana 400	Metered	Functional		Mixed	ANI				4864847	2500	312.26	0.00		
B.59			66				66KV I/C No 1 Rohini-II 220 KV	Metered	Functional		Mixed	ANI				4864957	2500	346.37	0.00		
B.60			66				66KV I/C No 2 Rohini-II 220 KV	Metered	Functional		Mixed	ANI				5128468	1000	359.82	0.00		
B.61			33				33KV I/C No 1 Wazir pur	Metered	Functional		Mixed	ANI				4864908	1000	359.58	0.00		
B.62			33				33KV I/C No 2 Wazir pur	Metered	Functional		Mixed	ANI				4864946	1000	353.67	0.00		
B.63			33				33KV Peeragarhi Ckt. to CC raniba	Metered	Functional		Mixed	ANI				4864901	250	83.35	0.00		
B.64			33				Sudarshan Park 33KV Line-1	Metered	Functional		Mixed	ANI				4864810	200	120.12	0.00		
B.65			33				PUSA Ckt-I	Metered	Functional		Mixed	ANI				4864843	1000	24.43	0.00		
B.66			11				DCM Nuruddin Park BG Road	Metered	Functional		Mixed	ANI				4902579	500	5.93	0.00		
B.67			11				DCM chowk BG Road	Metered	Functional		Mixed	ANI				4865089	500	13.81	0.00		
B.68			11				Sadar/s/ BG Road	Metered	Functional		Mixed	ANI				4902564	500	7.37	0.00		

B.69			11			CSA colony BG Road	Metered	Functional		Mixed	AMI				4902519	499.98	4.21	0.00			
B.70			33			33 kV Pandav Nagar DMS	Metered	Functional		Mixed	AMI				5295200	100	0.00	0.00			
B.71			11			Philips DMS	Metered	Functional		Mixed	AMI				4902568	100	9.82	0.00			
B.72			66			Nangloi Ckt 2	Metered	Functional		Mixed	AMI				Y0357821	-0.8	0.00	0.00			
B.73			66			Mundka to MGP-1	Metered	Functional		Mixed	AMI				4864983	800	121.85	0.00			
B.74			66			MGP T-off to Nangloi Ckt (-ve)	Metered	Functional		Mixed	AMI				4864971	-800	0.00	0.00			
B.75			66			Mundka to Sawda Ghevra	Metered	Functional		Mixed	AMI				5295166	400	25.05	0.00			
B.76			11			Tibia College Fair Road	Metered	Functional		Mixed	AMI				4902599	1333.33	10.94	0.00			
B.77			11			EAST PARK ROAD Fair Road	Metered	Functional		Mixed	AMI				4865082	133.333	10.45	0.00			
B.78			11			Manak Pura Fair Road	Metered	Functional		Mixed	AMI				4902577	100	7.92	0.00			
B.79			11			ESI Hospital and Rameshnagar-2 V	Metered	Functional		Mixed	AMI				4902539	100	5.87	0.00			
B.80			11			Moti Nagar Tanga stand S. B. Mill	Metered	Functional		Mixed	AMI				4902548	100	0.00	0.00			
B.81			11			41 Rama Road (70 Rama Road) S. B	Metered	Functional		Mixed	AMI				4865093	100	0.00	0.00			
B.82			11			51 Rama Road S. B. Mill	Metered	Functional		Mixed	AMI				4902528	100	0.68	0.00			
B.83			11			Nazafgarh Road S. B. Mill	Metered	Functional		Mixed	AMI				4865183	133.33	5.96	0.00			
B.84			11			Moti Nagar Kiosk S. B. Mill	Metered	Functional		Mixed	AMI				4902591	1333.33	3.50	0.00			
B.85			66			sagarpur Rewari Line	Metered	Functional		Mixed	AMI				5129960	281.25	28.31	0.00			
B.86			66			Pappan Kalon Rewari Line	Metered	Functional		Mixed	AMI				4864960	1000	128.58	0.00			
B.87			11			BSES NDPL (Ex) ON BUS 1&2 Rewa	Metered	Functional		Mixed	AMI				4902529	400	0.31	0.00			
B.88			11			NDPL BSES (Ex) ON BUS 2&3 Rewa	Metered	Functional		Mixed	AMI				4902525	400	0.19	0.00			
B.89			33			33 kV Vishal -1 Rewari Line	Metered	Functional		Mixed	AMI				4902510	-400	-21.26	0.00			
B.90			33			33 kV Vishal -2 Rewari Line	Metered	Functional		Mixed	AMI				4865140	-937.5	-1.65	0.00			
B.91			33			33 kV Mayapuri Rewari Line	Metered	Functional		Mixed	AMI				4864808	-187.5	-3.36	0.00			
B.92			66			66/11 Tr. 3 Rewari Line	Metered	Functional		Mixed	AMI				4865080	2500	-10.04	0.00			
B.93			33			33/11 Tr. 3 Rewari Line	Metered	Functional		Mixed	AMI				4864796	-125	-10.38	0.00			
B.94			33			I/C from Rohtak road Sudersan Park	Metered	Functional		Mixed	AMI				4864866	937.5	54.07	0.00			
B.95			33			Vishal (Imp/Exp) Sudersan Park	Metered	Functional		Mixed	AMI				4865149	-187.5	0.24	0.00			
B.96			66			T X 2 Bawana	Metered	Functional		Mixed	AMI				4864992	1000	184.45	0.00			
B.97			66			T X 3 Bawana	Metered	Functional		Mixed	AMI				4864827	333.33	255.28	0.00			
B.98			66			T X 1 Bawana	Metered	Functional		Mixed	AMI				4864892	4000	278.05	0.00			
B.99			11			Local Tr Bawana	Metered	Functional		Mixed	AMI				4902553	-100	-0.21	0.00			
B.100			66			DELHI MSW Bawana	Metered	Functional		Mixed	AMI				4864884	1000	127.14	0.00			
B.101			66			TRANSFORMER 4 (160MVA) Bawna	Metered	Functional		Mixed	AMI				4902508	6666.66	388.43	0.00			
B.102			66			66kV Incomer 1-220kV SGTN	Metered	Functional		Mixed	AMI				XF465246	1	477.45	0.00			
B.103			66			66kV Incomer 2-220kV SGTN	Metered	Functional		Mixed	AMI				XF465248	1	474.45	0.00			
B.104			11			Vishal BSES Ramesh Nagar-1	Metered	Functional		Mixed	AMI				4902572	100	0.00	0.00			
B.105			33			DEV NGR- SEHZADA BAGH	Metered	Functional		Mixed	AMI				Q0487626	1	126.11	0.00			
B.106			33			DEV NGR-RAMA RD	Metered	Functional		Mixed	AMI				Q0487625	1	56.37	0.00			
B.107			66			GIS (160 MVA)	Metered	Functional		Mixed	AMI				Q0491811	1	230.13	0.00			
B.108			33			DEV NGR-NARAINA DTC CKT-2	Metered	Functional		Mixed	AMI				Q0487631	1	7.70	0.00			
B.109			33			DEV NGR-NARAINA DTC CKT-1	Metered	Functional		Mixed	AMI				Q0487627	1	0.09	0.00			
B.13401						Total (MU)											11527.40	0.00			
B.13402						Net input energy at DISCOM periphery (MU)														11527.40	

Color code	Parameter
	Please enter voltage level or leave blank
	Please enter feeder id and name or leave blank
	Enter meter no or leave blank
	Enter CT/PT ratio or leave blank
0	Please enter numeric value or 0
	Please select yes or no from list
	Formula protected

I/We undertake that the information supplied in this Document and Pro-forma is accurate to the best of my knowledge and if any of the information supplied is found to be incorrect and such information result into loss to the Central Government or State Government or any of the authority under them or any other person affected, I/we undertake to indemnify such loss.

Authorised Signatory and Seal

Name of Authorised Signatory
Name of the DISCOM:
Full Address:-

Seal

HARIOM SHARMA
E. Code : 90231
AGM, Energy Audit Group
Tata Power Delhi Distribution Limited

Pushpendra Kumar
HoD-EAG
E. No-94669

Signature:-
Name of Energy Manager*:
Registration Number:

Banwari Lal Gupta
EA-2317

F. Consumer & Consumption Data

[illegible]